



Abstracts from the Pharmacy Education, Science and Practice Conference 2025 Part A

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ABSTRACT



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Abstracts from the Pharmacy Education, Science and Practice Conference 2025 Part A

The abstracts for the Pharmacy, Education Science and Practice Conference 2025, sponsored and supported by the College of Pharmacy at Qatar University, are published across two journals: Part A (consisting of abstracts on the topics of Pharmacy Education and Pharmacy Practice) in the *Journal of Pharmaceutical Policy and Practice* and Part B (consisting of abstracts on the topic of Pharmaceutical Sciences) in the journal *Pharmaceutical Development and Technology*.

4

Optimal insulin initiation dose in children with newly diagnosed type 1 diabetes mellitus

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Background: Insulin dose requirements vary between patients with type 1 diabetes mellitus (T1DM). Bolus doses depend on the patient's insulin to carbohydrate ratio (ICR), for meal-time dosing, and on the correction factor (CF), for elevated blood glucose correction. In clinical practice, it usually takes several days of heuristic insulin dose adjustments, guided by blood glucose concentration measurements and carbohydrate counting before a patient's ICR and CF values can be accurately estimated.

Aims: (1) To review commonly used ICR and CF estimation models, and (2) to develop and evaluate new models for ICR and CF using retrospective patient data.

Methods: 1) A scoping literature review of online databases and search engines was conducted to identify commonly used ICR and CF formulae. 2) Linear regression analyses of retrospectively collected data was undertaken using clinical characteristic, laboratory measurements, and insulin dosing data from 70 children (≤ 18 years old) admitted to Dunedin hospitals (New Zealand) due to new onset T1DM. Regression models for ICR and CF values were developed and evaluated for bias and precision using mean error and root mean squared error, respectively.

Results: 1) According to the literature, covariates such as age and weight are commonly used to estimate ICR and CF. 2) Based on the final ICR and CF prediction models, for patients < 6 years old, total daily dose of insulin administered on first day (TDD_first) was the only significant predictor for ICR and CF ($p < 0.05$). For patients ≥ 6 years old, patient's total daily dose of insulin administered on first day, and age were found to be significant predictors for both ICR and CF ($p < 0.05$).

Conclusion: Accounting for a patient's age and insulin dose on first day allows for better estimation of true (i.e. asymptotic) ICR and CF values. This can potentially provide better glycaemic control sooner during hospital stay.

Keywords: Type 1 Diabetes, Insulin, Paediatric

7

Efficacy and safety of adding ribavirin to direct-acting antivirals (DAAs) in re-treating non-genotype1 hepatitis C- a systematic review and meta-analysis

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Background: Despite the advent of highly efficacious direct-acting antivirals (DAAs), a small percentage of individuals require re-treatment for hepatitis C.

Aim: This study aims to compare the efficacy and safety of adding ribavirin to DAAs compared to DAAs alone in treating non-genotype 1 HCV in individuals who have been previously treated.

Methods: A systematic review and meta-analysis of randomized controlled trials (RCTs) was performed. Cochrane, PubMed, SCOPUS, and preprint databases were searched for trials that compared DAAs with ribavirin to DAAs

alone in previously treated individuals with non-genotype 1 HCV infection. The primary outcome of interest was sustained virological response (SVR) at 12 weeks after treatment completion (SVR12). The quality of studies was evaluated using the Methodologic AI STandard for Epidemiological Research (MASTER) scale, heterogeneity using I^2 , and publication bias using Doi and funnel plots. Bias adjusted inverse heterogeneity models were used to compute pooled odds ratios and their 95% confidence intervals. The Grading of Recommendations, Assessment, Development, and Evaluations (GRADE) framework was used to evaluate the quality of evidence for the primary outcome.

Results: Ten RCTs with a total of 1000 participants were included. Overall, the addition of ribavirin to DAAs, compared to DAAs alone, did not show benefit in achieving SVR12 (OR 1.01, 95% CI 0.31–3.29, $I^2 = 69.8\%$), with moderate certainty GRADE evidence, and the additional ribavirin showed increased odds of adverse events (OR 2.37, 95%CI 1.88–2.98, $I^2 = 28.9\%$).

Conclusion: The moderate certainty evidence suggests that adding ribavirin to DAAs may not confer benefit in achieving SVR in previously treated individuals with non-genotype 1 HCV.

Keywords: Hepatitis C virus (HCV), Ribavirin (RBV), Direct-acting Antivirals (DAAs), Sustained virological response (SVR), Systematic review and meta-analysis

8

Hormone replacement therapy and association with depressive symptoms in menopausal women: A cross-sectional study in Qatar

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Background: Menopause is associated with a range of signs and symptoms often experienced, which can lead to psychological and physical problems. Hormonal Replacement Therapy (HRT) use may alleviate depressive symptoms in women with menopause.

Aim: By studying a population of menopausal women in Qatar, this study aims to illuminate whether HRT may play a role in alleviating the psychological burdens faced by women during this significant transitional phase of their lives.

Methods: Data was obtained from the Qatar Biobank and included a sample of 1000 women with Last Menstrual Period (LMP) of ≥ 1 year, not pregnant, and known status of HRT use, and the Patient Health Questionnaire-9 (PHQ-9). Additional variables included socio-demographics, amenorrhea duration, length of sleep, chronic disease, smoking, Body Mass Index (BMI), and vitamin B12 and D levels. Depressive symptoms were assessed using the PHQ-9 scores and women categorized into PHQ-9 score of ≥ 7 and <7 . A multivariable logistic regression model was used to determine HRT association with PHQ scores of ≥ 7 .

Results: In the multivariable logistic regression, HRT use showed 2.36 times the odds of having PHQ-9 score of ≥ 7 , with strong evidence against the model hypothesis at this sample size ($P=0.01$; 95% CI 1.26-4.40). Additionally, smokers and women with chronic diseases showed 2.67 and 2.31 times the odds of having a PHQ-9 ≥ 7 respectively, with moderate evidence against the model hypothesis at this sample size. Other factors showed little or no evidence against the model hypothesis.

Conclusion: This research highlights a positive association between HRT use and depressive symptoms in menopausal women in Qatar. It underscores the need for further exploration of the impact of HRT on menopausal women's mental health.

Keywords: Amenorrhea, Menopause, Depressive Symptoms, Hormone Replacement Therapy, Qatar Biobank

9

Comparative efficacy of subcutaneous versus intravenous interleukin 12/23 inhibitors for the remission of moderate to severe Crohn's disease: A systematic review and meta-analysis

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Background: Interleukin 12/23 (IL-12/IL-23) inhibitors are a class of monoclonal antibodies used to induce and maintain remission for Crohn's Disease (CD), a chronic inflammatory bowel disease, when patients do not respond to conventional immunomodulatory drugs or first-line monoclonal

antibodies therapies. Although monoclonal antibodies are best administered intravenously, subcutaneous administration has been trialled, with mixed results.

Aim: This meta-analysis synthesized all existing evidence to compare the efficacy and safety of subcutaneous versus intravenous IL-12/IL-23 inhibitors in achieving remission for moderate-to-severe CD.

Methods: In this systematic review and meta-analysis, we searched Cochrane, PubMed, SCOPUS, CINHALL, and preprint archives for randomized controlled trials (RCTs) that compared the efficacy and safety of subcutaneous to intravenous IL-12/IL-23 inhibitors for the remission of CD. Meta-analysis was carried out using a bias-adjusted inverse variance heterogeneity model, heterogeneity was assessed using I^2 , and publication bias using Doi plots. Evidence certainty was assessed using Grading of Recommendations, Assessment, Development, and Evaluations (GRADE).

Results: Seven RCTs, with 2179 participants with moderate to severe CD, were included. After meta-analysis, subcutaneous, compared to intravenous administration, was less efficacious for induction of remission (OR 0.77, 95% CI 0.53-1.12), with no-to-low heterogeneity ($I^2 = 0\%$, $p = 0.97$), and moderate certainty GRADE evidence. For the maintenance of remission, only two studies had analysable data, and they all showed that subcutaneous IL-12/IL-23 inhibitors were equal or better compared to intravenous administration. Further syntheses showed that subcutaneous, compared to intravenous administration of IL-12/IL-23 inhibitors had almost similar odds of adverse events (OR 0.91, 95% CI 0.63-1.32, $I^2 = 39\%$), serious adverse events (OR 0.97, 95% CI 0.61-1.53, $I^2 = 0\%$), treatment discontinuation (OR 1.06, 95% CI 0.67-1.68, $I^2 = 0\%$), and all-cause mortality (OR 1.17, 95% CI 0.27-5.20, $I^2 = 0\%$).

Conclusion: Subcutaneous administration has lower efficacy for the induction of remission but can achieve almost similar efficacy and safety in maintaining remission, with similar safety. However, more RCTs are needed to confirm these findings.

Keywords: Crohn's Disease, Interleukin inhibitors, Subcutaneous, Intravenous

11

Potentially significant drug–drug interactions: A real-world retrospective study on patients receiving psychotropic medications at Mental Health Services

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Background: Drug–drug interactions (DDIs) can lead to adverse events or altered drug effectiveness, and polypharmacy increases the likelihood of DDIs.

Aim: This study aimed to assess the prevalence and severity of clinically significant drug–drug interactions (DDIs) among patients with mental health conditions using psychotropic medications.

Methods: This was a retrospective study that included patients who had visited the outpatient mental health pharmacy at Mental Health Services in Qatar for medication dispensing. The study covered all individuals over a two-month period, totaling 1586 patients. The list of medications was cross-referenced with Lexicomp® to identify potential DDIs. Then, each identified DDIs was assessed for risk rating, severity, and reliability rating. The Statistical Package for the Social Sciences (SPSS) was used for data analysis.

Results: Out of the 1586 patient record files reviewed, it was found that 49% of studied sample did not experience any drug interactions. The remaining patients ($n = 807$, 51%) had at least one DDI. Around 44% of them had no comorbidities other than their psychiatric condition. Mental illness diagnoses: including depression (22.2%) and schizophrenia (21.9%) were the most common reported. On average, patients were taking 4.69 ± 3.36 medications, of which 2.23 ± 0.96 were psychotropics. The average number of potential psychotropic drug–drug interactions per patient was found to be 4.46 ± 5.64 . The most encountered psychotropic drug–drug interactions were those fall under Category C (86% $n = 2192$) followed by Category D (7.8%, $n = 200$).

Conclusion: The study provided insight into the prevalence of psychotropic medications DDIs among patients in Qatar. The analysis revealed significant psychotropic DDIs, particularly those leading to Torsades de Pointes (TdP) and anticholinergic intoxication, which require focused attention to mitigate potential adverse events and enhance patient safety.

Keywords: Drug–drug interactions, psychotropic medications, mental health services

12

Improving vitamin D prescribing and level monitoring for psychiatric patients at Mental Health Services in Qatar

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Background: Vitamin D deficiency has been linked to various diseases, including rickets, cancer, infections, autoimmune, cardiovascular, and metabolic diseases. Additionally, it has been found to be prevalent in psychiatric patients, impacting brain development and protection against oxidative stress.

Aim: The aim of this project was to improve the percentage of psychiatric patients with the corrected levels of vitamin D from 25% to 75% within 12 months by enhancing the process of prescribing and monitoring vitamin D.

Methodology: A multidisciplinary team was formed to implement a quality improvement (QI) project at the outpatient pharmacy in RCC2. The team identified patients during the dispensing of medications, and the outcomes measured were the percentage of adult outpatients with corrected vitamin D levels and the cumulative percentage of physicians who received the training package provided by the QI team. Data collection was done quarterly. Changes were implemented for prescribing vitamin D supplements and monitoring vitamin D levels using PDSA modules. All QI tools and charts were used as appropriate.

Results: During the 12-month period, the team reviewed the files of 665 patients to check their vitamin D status. The QI project was successful, with a 74.8% percentage of adult outpatients having corrected levels of vitamin D. Improvement was seen in each phase of the project: baseline- 25%, Phase 1–77.4%, Phase 2–85.8%, Phase 3 - 71.5 and Phase 4–60.3%.. Twenty physicians received the training package to improve prescribing practices.

Conclusion: Successful completion of the quality improvement project at RCC MHS OPD improved psychiatric patients' corrected vitamin D levels from 25% to 75% within 12 months. Targeted interventions resulted in a significant improvement in vitamin D levels for psychiatric patients within the recommended time frame.

Keywords: Vitamin D prescribing, Vitamin D monitoring, Mental Health Services

13

Deprescribing medications in geriatric care to enhance safety and quality of life: A vision 2030 aligned initiative

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Background: Polypharmacy in geriatric patients is a widespread issue associated with adverse drug reactions (ADRs), reduced quality of life, and increased healthcare costs. At King Fahad Armed Forces Hospital (KFAFH), a large number of elderly patients were on five or more medications, many of which were potentially inappropriate. This led to a need for an evidence-based deprescribing protocol to enhance patient safety and align with Saudi Arabia's Vision 2030 goal of improving healthcare quality.

Aim: The project aimed to reduce inappropriate polypharmacy among geriatric patients by 20% by deprescribing potentially inappropriate medications (PIMs) to enhance patient safety, improve quality of life, and reduce healthcare costs.

Methods: This quality improvement project used the Plan-Do-Study-Act (PDSA) methodology from January to August 2024. The outpatient pharmacy team conducted monthly comprehensive medication reviews for geriatric patients on five or more medications. Deprescribing plans were developed in collaboration with physicians and supported by patient and caregiver education. Key metrics included the average number of medications, ADR incidents, and patient satisfaction.

Results: The average number of medications per patient decreased by 37.5% (from 8.2 to 5.0), ADRs dropped by 47%, and patient satisfaction increased from 45% to 85%. There was a 20% reduction in healthcare costs per patient, and the percentage of patients on five or more medications dropped from 60% to 25%.

Conclusions: Deprescribing protocols significantly reduced polypharmacy and improved patient outcomes, demonstrating that a structured approach can enhance geriatric care. This model is sustainable, replicable, and aligns with Vision 2030s goals of improved patient safety and healthcare efficiency.

Keywords: Polypharmacy, Deprescribing, Geriatric Care, Patient Safety, Vision 2030

14

Treatment adherence and medication-related problems in patients receiving monoclonal antibodies: An experience from a secondary care hospital in the United Arab Emirates

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Background: Monoclonal antibodies (mAbs) have emerged as transformative agents in the therapeutic landscape; however, their use is associated with several challenges, including variability in clinical responses, potential for medication-related problems, such as adverse drug reactions (ADRs) and drug–drug interactions (DDIs), and treatment non-adherence.

Aim: This study aimed to evaluate treatment adherence, the prevalence of potential drug–drug interactions (pDDIs), and the occurrence of adverse drug reactions (ADRs) among patients receiving monoclonal antibody (mAb) therapies

Methods: This prospective cohort study was conducted in the inpatient and outpatient departments of a secondary hospital in the United Arab Emirates (UAE). Treatment adherence was evaluated using the 8-item Morisky Medication Adherence Scale (MMAS-8). Potential drug–drug interactions (pDDIs) were identified through the Portable Emergency and Primary Care Information Database's (PEPID) drug interactions checker, and ADRs were assessed using the Naranjo, WHO, and other standard scales.

Results: Eleven different mAbs were prescribed to the study patients. Evolocumab was the most frequently prescribed mAb (135 patients). Majority of the patients (247, 71%) had medium to high adherence with a median MMAS-8 score of 7.0. Multivariate logistic regression analysis identified ethnicity (OR:2.458, 95%CI:1.202-5.026, $P = 0.014$), education status (OR: 6.542 95% CI: 2.632-16.257, $P < 0.001$), presence of comorbid conditions (OR:0.316, 95% CI: 0.111-0.905, $P = 0.032$), type of mAb prescribed (OR:10.041, 95%CI: 3.293-30.620, $P < 0.001$), and length of the treatment (OR:1.068, 95%CI: 1.028-1.109, $P = 0.001$) as independent predictors of adherence to mAb treatment. A total of 149 pairs of interacting drugs were identified with only 1.3% of the pDDIs linked to mAbs. ADRs were identified in 3.4% (12) of the study patients with the majority being injection-site reactions (25%), mild in severity (91.7%) and linked to dupilumab use (58.3%).

Conclusion: This study highlights the high adherence to mAb treatment and the low occurrence of pDDIs and ADRs among patients receiving mAbs. It sets the groundwork for future extensive studies on mAbs within the region, providing essential insights into these important aspects of mAbs utilization.

Keywords: Medication adherence, Medication-related problems, Monoclonal antibodies

16

Efficacy of semaglutide and tirzepatide in the achievement of remission-level HbA1c in type 2 diabetes: A systematic review and dose–response meta-analysis

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Background: Glucagon-like peptide-1 (GLP-1) agonists have shown promising effects in managing type 2 diabetes, however more evidence is needed concerning their ability to achieve diabetes remission.

Aim: This meta-analysis investigated the effect of semaglutide or tirzepatide, compared to placebo, alongside standard of care treatment for achieving remission-level HbA1c.

Methods: We searched PubMed, Scopus, Embase, Cochrane CENTRAL, MedRxiv, PrePrints, ClinicalTrials.gov for randomized controlled trials (RCTs) that compared either semaglutide or tirzepatide to placebo, as add-on to standard of care. The main outcome was diabetes remission, and this was measured as remission-level HbA1c (HbA1c $\leq 6.5\%$). Secondary outcomes included achievement of controlled diabetes (HbA1c $< 7.0\%$). We computed overall odds ratios with 95% confidence intervals, using a bias-adjusted meta-analysis model. Heterogeneity was assessed using Cochran's Q test and I^2 statistics, and publication bias using the Doi plot. Dose–response analysis was also conducted to assess main outcomes.

Results: Of the 17 included RCTs, 12 were on semaglutide and five on tirzepatide. At 26–30 weeks, semaglutide showed a 10-fold increase in odds of remission-level HbA1c (OR10.71, 95%CI 5.72–20.06, $I^2 = 85.0\%$) and an eight-fold increase in odds of HbA1c $< 7.0\%$ (OR8.43, 95%CI 5.59–12.71, $I^2 = 81.8\%$). Both drugs showed a mild increase in overall adverse events (OR1.28, 95%CI 1.07–1.52, $I^2 = 50.8\%$). Dose-dependent relationships were observed between each of the drugs and remission-level HbA1c, with optimal dosages of 0.2–0.5mg for subcutaneous semaglutide (NNT = 2), 7–10mg for oral semaglutide (NNT = 3), and 5mg for tirzepatide (NNT = 2). Recommendations were of moderate GRADE certainty due to between-study heterogeneity and possibility of publication bias.

Conclusion: Semaglutide and tirzepatide are highly effective and equally safe compared to placebo in attaining remission-level HbA1c. Dose–response analysis suggested optimal dosages of 0.2–0.5mg for subcutaneous semaglutide, 7–10mg for oral semaglutide, and 5mg for tirzepatide which can guide clinical decision-making.

Keywords: Semaglutide, Tirzepatide, Diabetes, Remission

17

Retrospective comparative analysis of FDC versus SDC for *Helicobacter pylori* eradication

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Background: The eradication of *Helicobacter pylori* (*H. pylori*) is becoming more difficult despite the availability of multiple medications to treat the infection. The effectiveness of FDC (fixed dose combination) versus SDC (separate dose combination) has not yet been evaluated.

Aim: This study aimed to compare the effectiveness of fixed-dose combination (FDC) versus separate-dose combination (SDC) therapies in eradicating *Helicobacter pylori* infection among treatment-naïve patients.

Methods: This retrospective cohort study compared the effectiveness of FDC versus SDC in eradicating *H. pylori* infection. The study included patients over 18 years who were primarily diagnosed as *H. pylori*-positive between January 1st, 2020, and December 31st, 2022, and were treatment-naïve. This study was conducted in Qatar's primary governmental health system, Hamad Medical Corporation (HMC). The primary outcome was eradication failure within 1 year of treatment. Eradication was assessed using stool antigen or endoscopy. Logistic regression was used to adjust for potential confounders and to explore the factors associated with eradication failure.

Results: Of 382 patients in the study, 169 received FDC, and 213 received SDC. Most patients were female with a median age of 46 years (IQR 36–57) years. The most prevalent primary indication for *H. pylori* testing in both groups was dyspepsia. The eradication failure was comparable between the groups (33.7% in the FDC group and 34.7% in the SDC group; $p = 0.74$). In the multivariable regression, the adjusted odds ratio (aOR) of eradication failure comparing FDC to SDC was 1.04 (95% CI 0.61–1.78, $p = 0.87$). Among

other evaluated factors, only age showed an independent association with eradication failure.

Conclusion: FDC and SDC have shown similar eradication of *H. pylori* infection, with no significant difference between the two groups. Age is independently associated with eradication failure. Randomized controlled trials comparing FDC and SDC are needed to confirm these findings.

Keywords: H-Pylori, Fixed dose combination, Separate dose combination

21

Evaluation of psychological disorders in hemodialysis patients and impact of Pharmacist's educational intervention on health related quality of life

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Background: Hemodialysis patients who are having psychological disorder (anxiety and depression) could lead to worse quality of life. Pharmacist-led educational interventions can improve patient adherence, well-being, and clinical outcomes.

Aim: This study aimed to assess the association between psychological factors and health related quality of life (HRQoL) and to explore the impact of pharmacist interventions on hemodialysis patients.

Methods: A pre-post-intervention study was conducted through patients interviews. The participants were all adults (≥ 18 years) undergoing hemodialysis. Two validated tools were administered: the Hospital Anxiety and Depression Scale (HADS-A and HADS-D) to assess anxiety and depression, and the EQ-5D-5L, a standardized instrument for evaluating HRQoL.

Results: A total of 421 responses were collected from October 2023 to March 2024. Of these, 387 patients were re-interviewed after pharmacist intervention. Approximately 8.1% (43 patients) were excluded from the revisit, including 0.7% who reverted to non-dialysis status, 4.3% who had passed away, 1.4% who received kidney transplants and 1.7% who refused to participate. The study showed that a significant proportion of hemodialysis patients experienced elevated levels of

anxiety (39.3%) and depression (35.9%). Following pharmacist intervention, there was a significant reduction ($p < 0.001$) in anxiety and depression scores, which corresponded to a notable improvement in HRQoL.

Conclusion: These findings suggest that early evaluation of anxiety and depression, combined with pharmacist-led educational interventions, can significantly enhance HRQoL in hemodialysis patients. Integrating such interventions into routine care may benefit patients' psychological and overall well-being.

Keywords: End Stage Renal Disease, Health-Related Quality of Life, Anxiety, Depression, Dialysis

22

Prevalence and predictors of post-partum depression in Pakistan

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Background: Postpartum depression is a severe mental health condition that affects a significant portion of women globally, with higher rates observed in low and middle income countries (LMICs). PPD hampers a mother's and child's health but regular screening and support can reduce the risk of PPD. However, research on the prevalence and risk factors of PPD has been scarce in Pakistan.

Aim: To assess the prevalence and risk factors associated with postpartum depression (PPD) among new mothers in Rawalpindi and Islamabad.

Methods: A descriptive cross-sectional study was conducted among new mothers of twin cities over a period of three months. Three hundred and eighty two participants were recruited through convenience sampling. The Edinburgh Postnatal Depression Scale (EPDS) was employed to assess depression severity. The data was gathered by self-administering the questionnaires to the mothers of diverse age groups and backgrounds at different hospitals/clinics. Results were analysed using SPSS v 25.0.

Results: The mean postnatal period of the study population was 6.32 ± 2.724 weeks. All participants exhibited some degree of postpartum depression, with 75% experiencing severe symptoms. Moreover, the incidence of severe depression was highest in women with BMI 21-23.9 followed by 24-

25.9 and 18-20.9 while the least was observed in women with BMI above 25.9 [53 (39.0%). The incidence of severe depression was higher among women aged 26-35 years, having normal body weight, with three or more children, no daughters, and those with no miscarriage history. Lastly, occupational status and geographical location (urban vs. rural) were not significantly associated with depression severity.

Conclusion: Post-partum depression is highly prevalent among women in Pakistan. Therefore, there is an urgent need for targeted interventions and support systems for vulnerable groups, particularly women in the 26-35 age group with normal BMI. Moreover, PPD screening must be integrated into routine maternal healthcare services in Pakistan.

Keywords: Post-partum depression, Prevalence, Cross-sectional, Pakistan

23

A qualitative study exploring medicine classification and accessibility among pharmacists in Brunei Darussalam

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Background: Brunei Darussalam's medicine control governance landscape is complex, driven by legislative directives and directed by the Brunei Darussalam Medicines Control Authority (BDMCA). While the medicine dispensing system in Brunei Darussalam, across governmental and private sectors, is a critical component of the nation's healthcare, it remains under-researched, leaving gaps in understanding its distinctions, challenges, and opportunities.

Aim: This study aimed to explore the perspectives of pharmacists regarding the medicine dispensing systems in Brunei Darussalam, medicine accessibility, the classification and regulation of medicines, and the growing role of pharmacists.

Methods: A series of comprehensive one-on-one interviews were conducted with six pharmacists, exploring topics ranging from the specifics of medicine classification to the broader provision systems in place. The data was subjected to a rigorous thematic analysis, adhering to the methodology delineated by Braun & Clarke.

Results: Five themes emerged from the data: nuances of Brunei Darussalam's medicine dispensing landscape across both public and private sectors; accessibility and regulatory framework of medicines both in Brunei Darussalam and abroad; prospective implementations to enhance the current system; and the evolving role and expectations of pharmacists in the current health-care environment. Notably, a significant contribution of this study was the illumination of the differential perspectives between pharmacists operating in governmental versus private contexts.

Conclusion: This study sheds light on the previously under-explored domain of Brunei Darussalam's medicine control governance and medicine dispensing system. By drawing from the first hand experiences of practicing pharmacists, it offers actionable insights that could guide future policy developments, optimize medicine dispensation and regulation, and shape the evolving role of pharmacists in Brunei Darussalam.

Keywords: Medicine classification, Medicine accessibility, Qualitative analysis, Brunei Darussalam

24

Advancing sustainability in healthcare: Integrating circular economy principles through comprehensive ecosystem models

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Background: The healthcare sector's substantial resource consumption and waste generation contribute significantly to environmental degradation, necessitating innovative approaches like the Circular Economy (CE) to mitigate these impacts.

Aim: This study investigated the integration of CE principles into the healthcare sector, focusing on waste management, resource optimization, and the roles of key stakeholders in developing sustainable healthcare ecosystems.

Methods: The research adopts a three-phased methodology: (1) Analysis of WHO reports to identify strategic frameworks and policies for healthcare waste management and sustainability; (2) Review of academic literature on CE-driven transformations in other industries and their potential application in healthcare; and (3) Examination of case studies showcasing real-world

implementations of CE practices in healthcare settings. Insights from global reports and case studies were synthesized to assess the feasibility and impact of CE models in healthcare.

Results: The findings demonstrate that CE principles can significantly reduce healthcare's environmental impact by fostering modular, repairable, and recyclable medical devices, closed-loop supply chains, and sustainable procurement. MedTech companies, in collaboration with healthcare providers, startups, and policymakers, are pivotal in driving circular practices. However, challenges such as hazardous waste management and regulatory inconsistencies persist. Successful implementation will require coordinated stakeholder efforts and strong policy support.

Conclusion: Integrating CE principles in healthcare offers a strategic pathway toward sustainability, with coordinated stakeholder collaboration, continuous innovation, and robust regulatory frameworks being essential for overcoming existing barriers. The study underscores the importance of an approach that leverages beneficial criteria from different industries to build a resilient and sustainable healthcare system.

Keywords: Healthcare sustainability, Circular economy principles, Healthcare waste management

27

The effectiveness of interactive dashboards to enhance the quality of antibiotics prescribing in primary healthcare

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Background: Dashboards are powerful tools that track key performance indicators (KPIs) and display relevant data to healthcare professionals, helping improve the performance of outpatient pharmacies. Healthcare organizations use this data, including information on antibiotic prescribing (AP), to monitor infections and support epidemiological surveillance.

Aim: This smart dashboard aimed to promote evidence-based statistics that support drug and therapeutic committees (DTCs) in decision-making to enhance the rational use of antibiotics.

Methods: This is a quantitative approach study which was employed to extract data from Saham polyclinic, Ministry of Health (MOH) in Oman, information management system (Al-Shifa 3 plus) and formulate the (KPIs) of antibiotics

for pharmacy during the period from January to June, 2022. The data were used as a framework to determine the dashboard requirements which was built in Microsoft Excel. Nine performance analytical dashboards were developed and visualized in innovative and interactive graphs that provide descriptive statistics of twenty-one selective groups of antibiotics and five pain relievers. The study was ethically approved by the Research Ethical Review and Approval Committee (MoH/CSR/24/27854).

Results: During the study period, a total of 21,224 antibiotics were prescribed with amount of 6,885.81 Omani Rial (OMR). The most prescribed one was Amoxicillin suspension accounting for 6,854 (32.3%). The majority 12,128 (67.2%) prescriptions were prescribed in general practice clinics and emergency clinic. Among the 4,621 patients over the study period, a total culture was done as followed: 3,875 (83.467%) outpatients, 670 (14.450%) external patients and 94 (2.034%) inpatients. Out of 34 microorganisms detected, *E. coli* was the predominant (53.3%), *Klebsiella* species (9%) and *Pseudomonas* species (6%).

Conclusions: This dashboard identifies the area of intervention (General Practice clinics), improve general practitioners (GPs) prescribing attitudes will minimize the cost along with combined educational and behavioral intervention strategies.

Keywords: Dashboard, Antibiotics, Key performance indicators, Multi-drug resistance organism, Oman

28

Investigating the impact of involving students in constructing a real-world case study as a learning strategy to deepen their learning and understanding: A pilot study

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Background: Pharmacology is a challenging subject for health students due to the complexity of the course. Traditional teaching methods, such as case-based learning (CBL), have been employed to enhance student understanding. However, it may be time to explore more innovative approaches to deepen student engagement and comprehension.

Aim: This study aimed to investigate the effectiveness of engaging students in the development of real-world case studies as a learning strategy. It also compares the impact of this approach with traditional CBL methods.

Method: Fifteen students were enrolled and divided into three groups. Over a four-week period, students participated in CBL sessions and worked collaboratively to construct their own case studies. Each group identified a potential patient, a socially related elderly person or an individual with a chronic disease and gathered information on their medication history and dietary habits. A 20-minute test was administered after each learning strategy. Additionally, a Likert-scale survey and a group discussion were conducted to evaluate students' satisfaction with both approaches.

Results: Compared to the mean test score obtained after the CBL approach, test scores following the case-construction strategy showed an improvement in student performance (mean difference [MD] = 1.26, $P = 0.078$, 95% CI: -0.160 to 2.69). Although this improvement was not statistically significant, students reported higher engagement and satisfaction with the case-construction approach. The Likert-scale survey showed a score of 32.2 for the case-construction approach and 25.5 for the CBL approach. During group discussions, students criticized the preparation time, increased workload and the lack of clear guidance on essential parameters for constructing cases.

Conclusion: The case-construction approach can be effectively implemented in pharmacology courses to enhance student engagement and reinforces understanding. Balancing workload and offering clear guidelines could make this approach more feasible and effective.

Keywords: Pharmacology, Education, Case-construction, Case-based learning, Pilot implementation

30

Community pharmacists' perception and attitude about patient counseling and ongoing pharmacy education programmes in Qatar

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Background: Community pharmacists play a critical role in healthcare by providing patient counselling and ensuring proper medication use. In Qatar, pharmacists are increasingly recognised as key healthcare providers. Continuing Pharmacy Education (CPE) programmes ensure pharmacists stay updated with

the latest clinical knowledge and counseling techniques. However, there is limited research on how community pharmacists in Qatar perceive their role in patient counselling and the effectiveness of CPE programs in enhancing this practice.

Aim: This study aimed to assess the perception of community pharmacists in Qatar towards patient counselling and their participation in Continuing Pharmacy Education programs.

Methods: A cross-sectional survey was conducted among 169 licensed community pharmacists across various regions in Qatar. A structured questionnaire was distributed, covering demographic information, the frequency and confidence in patient counselling, barriers to effective counselling, participation in CPE programs, and preferences for CPE topics and formats.

Results: Preliminary results show that a significant proportion of pharmacists believe patient counselling as a crucial part of their role, with 89% rating it as "very important". However, the most frequent obstacles to successful counselling were found to be time limits (41%), patient non-compliance (24%), and language hurdles (19%). Eighty seven percent of the pharmacists attended at least one CPE session per year, indicating a modest level of participation. Over 98% of those surveyed concurred that regular CPE program participation increased their skill and confidence in patient counselling.

Conclusion: Although community pharmacists have practical obstacles in providing the best services in Qatar but have a positive attitude toward patient counselling. The quality of patient counselling at community pharmacies throughout Qatar might be substantially boosted by addressing obstacles including time limits and offering more specialized resources.

Keywords: Pharmacists, Patient counseling, Continuing Pharmacy Education (CPE), Pharmacy practice, Professional development

33

Curriculum, competency development, and assessment methods of MSc and PhD pharmacy programs: A scoping review

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Background: There is a noticeable absence of a definitive guide on how universities can effectively address the expanding challenges within pharmacy postgraduate education.

Aim: We aim to systematically review and evaluate the current landscape of postgraduate pharmacy education to: a) identify current evidence, best practices, challenges, recommendations, and solutions; and b) develop a framework to optimize postgraduate pharmacy programs.

Methods: A scoping review was conducted following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses extension for Scoping Reviews (PRISMA-ScR). Electronic databases, including PubMed, Scopus, EMBASE, ProQuest, Web of Science, and Google Scholar were utilized. The search covered studies published from January 2011 to September 2023. Following the principles of Arksey and O'Malley's framework, data charting and extraction were performed using a pre-designed data collection tool, followed by the synthesis and grouping of studies based on common themes.

Results: Of the 5542 articles found, the review included 36 eligible ones focusing on pharmacy postgraduate education (PhD and MSc), grouped into three themes: 1) courses and curriculum; 2) training and skills development; 3) assessment and mentorship methods. Utilized methodologies included descriptive analyses, questionnaires, surveys, trials, and focus groups/interviews. The studies underscored the need for competency-based curricula with regular evaluations, career planning, and diverse course offerings. Identified key skills and competencies in the studies included soft skills, communication, research, leadership and management, and critical thinking. The studies also emphasized the value of comprehensive evaluation and peer review methods. Challenges included balancing academic and real-world requirements, training, limited resources, time constraints, and faculty workload.

Conclusion: Evidence-based suggestions to improve postgraduate pharmacy education include the implementation of practice-oriented courses, value of tailored/or comprehensive assessments, focus on real-world skills, effectiveness of advanced teaching methods, and mentorship role. The proposed framework can guide program enhancement and highlight the need to improve programs holistically.

Keywords: Curriculum, Competency, Assessments, Pharmacy programs

Use of Simconverse, an AI programme, within undergraduate pharmacy teaching at De Montfort University (DMU)

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Background: With changes to the education and training for pharmacists in the UK to embed prescribing skills, simulation-led teaching and learning has been introduced into the undergraduate Pharmacy (MPharm) programme to scaffold learning and embed the necessary knowledge and skills required for prescribers.

Historically, pharmacy students have used role-play exercises with patient actors and peers to improve their communication skills. In 2024, Simconverse, an Artificial Intelligence (AI) programme was introduced to the MPharm degree.

The program delivers AI generated patients that have a pre-built brief that specifies characteristics, behaviour, medical history and outcome expectations. This allows individualised and unique experiences for students whilst ensuring an equitable and consistent approach is applied to each scenario. Post-interaction with the scenario, students are provided with a transcript of their conversation and an AI-assessed rubric, highlighting any skill or knowledge gaps. This allows students to reflect on their interactions, thus promoting the pedagogical approach of scaffolding learning, critical thinking and reflective practice.

Aim: This study aimed to assess students attitudes and perceptions on AI-based programme, Simconverse.

Methods: The Simconverse pilot involved approximately 400 students in Year 1, 2 and 3 of the degree programme. The timeline for scenario release is synchronised with the clinical teaching on the course. Completion of scenarios is mandatory and includes two attempts with feedback and a reflective account to submit. A feedback questionnaire was used to analyse students' attitudes and perceptions on AI-integrated teaching and learning.

Outcomes: Students reported increased confidence and engagement compared to learning from textbooks and traditional role plays.

Conclusion: A positive outcome to the pilot is to be followed up with rollout to Year 4 and additional scenarios created. There will also be work undertaken to look at incorporating this technology into assessments.

Keywords: Simulation-led teaching, Pharmacy education

37

Medicines use evaluation of Semaglutide at Al-Wakra Hospital

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Background: Semaglutide, a glucagon-like peptide-1 (GLP-1) receptor agonist, was introduced to Hamad Medical Corporation (HMC) formulary in 2023 as a weekly injection under the name Ozempic®, available in strengths of 0.25mg, 0.5mg, and 1mg.

Aim: To assess the initial prescribing appropriateness of semaglutide according to HMC prescribing guidelines. Moreover, changes in weight, HbA1c, and safety were evaluated.

Methods: This was a retrospective medication use evaluation of semaglutide prescribing for adult patients at Al-Wakra and Mesaieed General Hospitals. We conducted electronic chart review of a random sample of patients prescribed Semaglutide over the first 10 months of its introduction to the formulary (13/05/2023 – 15/03/2024). Evaluation criteria were; prescription for patients for the management of diabetes, or for obese diabetic patients with HbA1c >8%.

Results: Of the 268 patients identified, 100 patients were reviewed. The average age was 49 years, with 64% female. Among the 100 patients, 79 had comorbidities, often using other antidiabetic medications with semaglutide. Nearly half had previously used other GLP-1 agonists, and 72 were prescribed semaglutide for weight management. Results showed that 51% of the patients were prescribed semaglutide in accordance with HMC restrictions. Inappropriate prescribing was identified among patients receiving semaglutide for weight management, specifically in those with an initial HbA1C <8% (n = 38), and those with both BMI <30 kg/m² and HbA1C <8% (n = 11). Majority of patients (n = 69) lost weight with an average of 7kg. HbA1c reduction was observed in 41%. Twenty patients were found to have an ADR (gastrointestinal-related) attributable to semaglutide.

Conclusion: The evaluation recommends addressing inappropriate semaglutide prescribing by ensuring it is only given to qualifying patients. This can be achieved through clear prescribing guidelines, routine prescription reviews, and individualized dose titration. Additionally, improved ADR reporting and documentation are essential for enhancing patient safety and care quality.

Keywords: GLP-1 agonist, Semaglutide, Appropriateness, Prescribing

Assessment of National Essential Medicines Lists and local production of anti-cancer drugs in South Asia: A data-driven approach

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Background: Cancer poses a significant health burden in South Asia, which has the lowest health expenditure, lowest Universal Health Coverage index, and highest out-of-pocket expenditure among developing countries. Access to anti-cancer medicines is crucial and is influenced by National Essential Medicines Lists (NEMs) and local production.

Aim: This study evaluates the alignment of anti-cancer medicines in NEMs with the World Health Organization's Essential Medicines List (WHO EML), and their local production in South Asia.

Methods: Data on 67 anti-cancer medicines was extracted from WHO EML 2023, used as standard. Data on NEMs and local production was extracted from each country's regulatory authorities. Descriptive analysis, Independent T-Test, Kruskal–Wallis test, and Spearman Correlation were conducted in SPSS version 23.

Results: The median number of anti-cancer medicines listed in NEMs, and locally produced was 23 (range 0–67, IQR 32) and 6.5 (range 0–62, IQR 39) respectively. Only three countries (37.5%) included more than 50% of the 2023 WHO EML anti-cancer medicines in their NEMs. Five countries locally produce, predominantly cytotoxic agents but the import is dominant. A strong positive correlation between locally produced and NEMs' medicines ($r = 0.884$, $P = 0.004$), and a strong negative correlation between the number of locally produced anti-cancer medicines and healthcare expenditure were observed ($r = -0.732$, $P = 0.039$).

Conclusion: South Asian countries have substantial disparities in NEMs' inclusion. Local production is evident in the region indicating that regional collaboration in local production is possible and may increase access to anti-cancer medicines. This emphasizes the need for regional collaboration in healthcare, local production and targeted policies to align with the global

standards. It will target directly to access (SDG:3.8) and indirectly to cancer-driven poverty (SDG:1), inequity (SDG:10) and regional collaboration (SDG:17).

Keywords: Anti-cancer medicines, National Essential Medicines Lists, Local production, South Asia

42

Development and validation of deprescribing algorithms for patients with chronic kidney disease using consensus development methodology

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Background: Patients with chronic kidney disease (CKD) are often treated with multiple medications, leading to polypharmacy, which may be inappropriate. Deprescribing has emerged as a proactive intervention to address the risks associated with inappropriate polypharmacy. Evidence-based tools and guidelines are crucial to enhance healthcare providers' ability to deprescribe effectively. While several deprescribing tools and guidelines exist for several populations of patients, their utility in CKD settings remains limited.

Aim: This study aimed to adapt existing evidence on deprescribing in CKD to create decision-support algorithms that assist clinicians in implementing deprescribing interventions.

Methods: We employed the nominal group technique as a consensus development methodology to adapt existing tools and create comprehensive deprescribing algorithms for CKD patients. A panel of five experts in pharmacotherapy, medication safety, and clinical research was assembled. After gathering evidence on deprescribing practices for CKD patients, we conducted multiple rounds of meetings to review this evidence and develop deprescribing algorithms. Once consensus was achieved, the adapted algorithms were documented and subsequently reviewed by nephrology consultants for further validation. The algorithms were then utilized in a pilot deprescribing study to test their utility in this setting.

Results: We developed 19 evidence-based deprescribing algorithms, comprising one general algorithm and 18 drug class-specific that are frequently

prescribed in CKD care. These comprehensive algorithms were designed to: 1) assess the appropriateness of prescribed medications and 2) provide a structured deprescribing process. Validation by experts and pilot testing further refined the algorithms, underscoring their usability and convenience.

Conclusion: This study produced evidence-based deprescribing algorithms tailored for CKD patients, offering a structured approach to identify potential medicines.

Keywords: Deprescribing, Inappropriate polypharmacy, Chronic kidney disease, Guidelines, Algorithms

43

Development and evaluation of an interprofessional education program on deprescribing for older adults

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Background: Deprescribing intervention gained international recognition as a proactive approach to addressing inappropriate polypharmacy. The lack of sufficient knowledge and limited competencies of healthcare providers are among the challenges to implementing successful and effective deprescribing.

Aim: This study aimed to design and evaluate a capacity-building interprofessional education program to enable healthcare professionals to provide deprescribing interventions in various healthcare practices.

Methods: A deprescribing module was designed by mapping targeted learning outcomes to the competencies required to deliver successful deprescribing interventions. The curriculum adopted an adult-learning approach in designing and delivering topics related to the identification of target medications and the application of the deprescribing process. The modules were validated by experts locally and internationally through iterative rounds of feedback. A workshop comprising didactic and interactive sessions was delivered for healthcare providers from varied health disciplines in a tertiary hospital in Qatar. Participants' knowledge in deprescribing was assessed before and after the workshop to evaluate the success of the module in achieving its learning outcomes.

Results: Nine hours of interactive and didactic sessions were delivered to 28 healthcare professionals through a two-day workshop. The sessions covered the theoretical framework of deprescribing, the use of deprescribing tools,

and hands-on activities on several case studies to apply deprescribing process. Compared to the baseline, participant knowledge improvement was observed among all assessment questions after the workshop. The average scores of participants in the assessment improved from 53% before the workshop to 61%. Overall, 78% of the attendees were satisfied to very satisfied with the delivered topics and activities.

Conclusion: We designed, validated, and implemented interprofessional deprescribing modules to build the required competencies to successfully implement deprescribing interventions in older adults. Future studies should determine the utility of the program in healthcare and educational institutions.

Keywords: Deprescribing, Inappropriate polypharmacy, Older adults, Adult learning, Interprofessional

45

Embedding experiential learning into the curriculum: Pharmacy students supporting the delivery of the seasonal flu vaccination programme in the UK

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Background: The 2021 General Pharmaceutical Council (GPhC) Standards for the initial education and training of pharmacists focus on training that supports a more integrated health and social care system, with pharmacists supporting patients to access the care they need in the right setting at right time. To ensure that pharmacists have the correct skills and attributes to deliver this care, undergraduate curriculums need to change and provide more opportunities for hands-on experiential learning UK pharmacy students must also show how they can participate in the promotion and protection of public health.

In the UK, community pharmacies provide vaccination services and are central to delivering the National Influenza Immunisation Programme.

Aim: This programme/project aimed to embed pharmacy students into vaccine programmes while achieving key objectives within their education and training.

Method: Following taught sessions regarding how pharmacists lead on pharmaceutical public health, including the provision of influenza vaccination services, third year (Level 6) pharmacy students were trained to be vaccinators and deployed to support flu vaccination services in community

pharmacies across Leicestershire. In October–November 2024, students attended a local community pharmacy for 2 days. Under supervision, students were able to administer flu vaccinations under the UK National Protocol and support the team with associated record keeping. Additionally, students were expected to proactively support with identifying patients eligible for vaccination and encourage uptake.

Results: This pilot demonstrated that pharmacy students were able to effectively support with the administration of influenza vaccinations under supervision and provide valuable support in public health services in a community pharmacy setting.

Conclusion: In their evaluation of the placement, supervisors commented on the professionalism of students and that the students' confidence. Future evaluation of patient experience and pedagogical outcomes will support development of other related public health services in the future.

Keywords: Experiential learning, Pharmacy education, Seasonal flu vaccination

46

Best timing to improve medical students' attitudes towards physician–pharmacist collaboration by using an educational podcast

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Background: Newly graduated physicians play a crucial role in prescribing medications and are expected to do so safely. However, studies show they make twice as many prescribing errors compared to their senior colleagues, underscoring the need for optimized collaboration with pharmacists. Many medical students and early-career doctors are unaware of pharmacists' expertise and the value they bring to patient care.

Aim: This study investigates the optimal timing for educating medical students about the role of pharmacists using an educational podcast, to improve interprofessional collaboration.

Methods: We recruited second-year bachelor's and final-year master's medical students to participate in this study. Participants completed an initial questionnaire based on the Scale of Attitudes Towards Physician-Pharmacist

Collaboration (SATP2C). Afterwards, they were asked to listen to a podcast about the daily practice and contributions of community pharmacy and complete a follow-up questionnaire consisting of the SATP2C and additional open-ended questions about their learning experience.

Results: So far, 96 students have been recruited. Recruitment will continue until December 2024. All participants completed the pre-survey, with 27 participants completing the post-survey. Preliminary results indicate that the podcast positively impacts students' attitude and perceptions towards pharmacists. The average SATP2C-score for bachelor students before the intervention was 53.2 (SD = 6.1) and after 57.8 (SD = 2.6), showing an improvement of 4.6 points ($p > 0.05$). For master students, the average score improves from 52.6 (SD = 4.3) to 53.5 (SD = 5.6), a 0.9-points increase ($p > 0.05$). These results suggest a bigger impact on bachelor students. Participants report gaining more insights into pharmacists' roles and value the ability to easily obtain advice, which strengthens collaboration and patient care.

Conclusion: Preliminary results suggest that early education about pharmacists influences more positive attitudes towards collaboration.

Keywords: Interprofessional collaboration, Curriculum, Medical students, Pharmacy education

47

Development of an innovative clinical pharmacy service in a urology surgical unit: A new initiative from Qatar

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Background: In surgical settings, pharmacists play a crucial role in medication management, particularly for complex perioperative needs like pain control, anticoagulation, and antimicrobial prophylaxis. Their integration into surgical teams, including urology, is essential for ensuring comprehensive medication oversight.

Aim: To assess the impact of a clinical pharmacy initiative in a surgical urology unit by examining the nature, significance, medications involved, and acceptance of pharmacist interventions.

Methods: A cross-sectional study was conducted at the Ambulatory Care Center (ACC), Doha, Qatar. Data related to the clinical pharmacist

interventions and associated rationale were classified according to the nature of the intervention using an adapted classification system. Assessment of the severity followed the National Patient Safety Agency (NPSA) Risk Matrix. Linear regression, Kruskal–Wallis, and post-hoc analyses were performed to determine the association between patient-related and medication-related characteristics on pharmacist interventions.

Results: A total of 3284 interventions (1486 patients) were analyzed. Most patients ($n = 1105$; 74.4%) had 1–2 interventions. Age and gender showed a positive linear correlation with the number of interventions per patient ($p < 0.01$). Majority of interventions were related to pharmacological strategy ($n = 1858$; 56.6%) and quantity of drug ($n = 821$; 25%). Additional drug therapy ($n = 748$; 22.78%) was the most common subcategory followed by optimum dose/frequency ($n = 691$; 21.04%) and discontinuation of medications ($n = 352$, 10.72%). Anti-infectives were the most identified drug category ($n = 798$, 55.1%). Most interventions (59.4%) were of moderate significance, with older patients more likely to receive these compared to minor interventions ($p = 0.032$). Prescriber acceptance was high, exceeding 90%, and increased by 6.6% from 2021 to 2023.

Conclusion: This study found that clinical pharmacy services in urology surgery effectively enhanced therapeutic optimization and medication safety, with pharmacists also addressing untreated health issues. Although the urology patient population presents unique challenges, the core principles of clinical pharmacy practice remain consistent.

Keywords: Clinical pharmacy, Surgical pharmacy, Pharmacist interventions, Urology, Surgery

48

Analysis of clinical pharmacists' interventions in a rehabilitation setting

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Background: Drug-related problems (DRPs) are prevalent in hospitalized patients, particularly in rehabilitation settings, where complex medication regimens increase the risk of harm. Clinical pharmacists are essential in identifying and preventing DRPs, yet their specific role and impact in rehabilitation have not been well-studied

Aim: To elucidate the role of clinical pharmacists in a rehabilitation setting by examining the type, severity, medications involved, and the level of acceptance of pharmacist interventions.

Methods: A cross-sectional study was conducted at Qatar Rehabilitation Institute (QRI) in Doha, Qatar. Data on clinical pharmacists' interventions and their rationales were categorized based on the type of intervention using a validated classification system. The severity of these interventions was assessed using the National Patient Safety Agency (NPSA) Risk Matrix. Linear regression and chi-square analyses were employed to explore the relationships between patient-related and medication-related characteristics and the pharmacist interventions.

Results: A total of 3,807 clinical pharmacists' interventions involving 815 patients were collected and analyzed. Findings indicated that most interventions addressed pharmacological strategy ($n = 1670$, 43.9%) and drug quantity ($n = 1166$, 30.7%). The most frequently reported intervention subtypes included dose optimization ($n = 749$, 19.7%), additional drug therapy ($n = 673$, 17.7%), and medication discontinuation ($n = 476$, 12.5%). Cardiovascular agents were involved in 37.1% of the interventions, followed by endocrine medications (17.1%) and centrally acting agents (11.7%). A significant proportion of interventions were deemed to have moderate severity (79.8%). Statistical analysis revealed a positive linear correlation between age and the number of interventions per patient ($p < 0.001$); with no significant difference in the severity of interventions between adult and elderly patients ($p = 0.09$).

Conclusion: This study emphasizes the varied roles of clinical pharmacists in rehabilitation, where the complexity of patients demands both standard practices and tailored approaches. It calls for further research to evaluate the impact of these interventions on key clinical outcomes.

Keywords: Pharmacist interventions, Rehabilitation, Medication management

49

Clinical outcomes of implementing Diabetic Ketoacidosis PowerPlan

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Background: Diabetic Ketoacidosis (DKA) is a life-threatening complication of diabetes mellitus that requires prompt prescribing of insulin bolus/infusion and multiple fluid/electrolyte regimens. Ordering multiple regimens simultaneously may result in prescribing errors which may affect the clinical outcomes.

Aim: The purpose of this study is to assess the effectiveness of implementing an electronic order set for patients with DKA on the time to resolution of DKA when compared to individual prescribing.

Methods: A retrospective cohort study was conducted for adult patients with DKA admitted to Al-Wakra Hospital. Eligible cases were identified via electronic medical records, with DKA resolution defined by two of the following criteria: serum bicarbonate ≥ 15 mEq/L, venous pH > 7.3 , or anion gap closure ≤ 12 mEq/L. Other outcomes included hospital length of stay (LOS), DKA recurrence, and mortality. Wilcoxon rank-sum and chi-square tests were used for data analysis.

Results: Of the 485 identified DKA patients, 135 received the DKA PowerPlan (PP), while 450 did not. Most patients were males, and approximately one-third of the cohort required intensive care unit (ICU) admission. Additionally, there was no difference in age, pH, bicarbonate, anion gap and need for ICU admission between the two groups at baseline. The time to DKA resolution was significantly shorter in the PP group compared to the non-PP group (10.8 hours vs. 12.9 hours; $p = 0.022$). Mortality [1 (0.7%) vs. 8 (1.9%), $p = 0.46$] and LOS [3.2 days vs. 3.7 days, $p = 0.20$] were lower in the PP group, though not statistically significant. No difference in the DKA recurrence rates between the two groups.

Conclusions: The DKA PowerPlan significantly reduced the time to DKA resolution, highlighting its potential to enhance management efficiency, though mortality, length of stay, and recurrence rates showed no significant differences.

Keywords: Diabetic Ketoacidosis, PowerPlan, Resolution, Order set

50

Role of the PHCC pharmacist in the utilization of near-expiry medications

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Background: The volume of expired medicines is evident through the annual statistics provided by the Pharmacy Department as well as the monthly inventory from Jan to July 2022. In the year 2022, the initiative worked effective for reducing the financial loss due to near expiry medication for tablets. Therefore, the initiative was spread to Vaccines. The baseline financial loss of near expiry medication for vaccines starting from January 2023 to June 2023 was 23,634QR.

Aim: To reduce the financial loss due to expired medication (cold chain) by 40% from baseline (QAR 23,634) from first of July 2023 to end of December 2023 at Al Daayen Health center.

Methods: The percentage of utilization and percentage of reduction in financial loss of near-expiry medication was assessed. The team adopted reallocation as their change idea in assistance with the Cerner system and WhatsApp. The project was conducted at the Pharmacy department in collaboration with the Supply chain at the Al Daayen Health Center, Primary Health Care Corporation Doha Qatar. We have adopted brainstorming, fishbone diagram and process maps as our QI tools.

Results: In July 2023, a total of 279 doses of the Boostrix vaccine were received, with 96% utilization and only 10 doses lost. In August 2023, 170 pens of Humalog Mix insulin were supplied; however, only 45% were utilized, resulting in the loss of 93 pens. By April 2024, 100% utilization of available stock was achieved. At the end of the PDSA cycle, the total value of near-expiry medication loss was reduced to QAR 2,962, while medications worth QAR 17,078 were successfully utilized through reallocation among health centers.

Conclusions: The implementation of targeted inventory management and reallocation strategies significantly improved medication utilization rates and minimized wastage. By the end of the PDSA cycle, near-expiry medication losses were substantially reduced, demonstrating the effectiveness of proactive redistribution among health centers in optimizing resource use and supporting cost-effective healthcare delivery.

Keywords: Near expiry medication, Pharmacist role, Utilization

52

Early detection and management of anemia in pre-dialysis chronic kidney disease patients in Qatar: A real-world, retrospective cohort study

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Background: Anemia is a common complication in patients with chronic kidney disease (CKD) that is not always detected and treated mainly in pre-dialysis patients. There is no sufficient data to reflect the real-life detection of anemia in patients with CKD.

Aim: This study aims to assess the early detection and management rates of anaemia in pre-dialysis CKD patients and the medications in CKD-induced anaemia treatment.

Methods: This is a retrospective cohort study that included pre-dialysis CKD patients with evident anaemia recruited from Hamad Medical Corporation, Doha, Qatar. Electronic medical records were employed to gather data between September 2019 - April 2023. Descriptive and inferential statistic demographics, explain the patient demographics, and baseline clinical parameters, rate of anemia detection severity, treatment modalities, and outcomes.

Results: Among the 2,051 patients included, 65.3% received anemia treatment with the most prevalent treatment being iron supplementation $\pm$ Vitamin B (54%). Severe anemia was more prevalent among stage 5 CKD patients and treatment rates were significantly higher in nephrology clinics compared to other settings ($p < 0.032$). Patients receiving Iron + Vit B $\pm$ erythropoietin-stimulating agents showed the highest improvement in eGFR levels after six months ($p < 0.001$). Moreover, patients receiving Vit B showed the most significant improvement in haemoglobin levels after six months ($p < 0.001$).

Conclusion: Early detection and treatment of anemia in CKD patients remain suboptimal, particularly in the early stages of the disease. More systematic management and increasing the involvement of nephrology specialists in patient care might improve outcomes in pre-dialysis CKD patients.

Keywords: Anemia, Chronic kidney disease, Anemia detection

54

Scrambler Therapy versus guideline treatments in managing chronic neuropathic pain: A sensitivity analysis of cancer and non-cancer patients

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Background: Chronic neuropathic pain (CNP) is a prevalent condition worldwide and has cancer and non-cancer origins. Targeted research can help understand the nature of therapeutic options and develop effective management strategies for alleviating CNP stemming from both origins. Current pharmacological treatments often lack the sustainability needed to optimize the quality of life in these two groups. Recently, Scrambler Therapy (ST) has shown sustainable pain reduction across diverse populations. However, the

difference in this sustainability between cancer and non-cancer pain is yet to be determined.

Aim: To assess the sustainability of reduction of CNP achieved by ST compared to standard therapeutic options in patients with cancer and non-cancer aetiology.

Methods: We conducted a dose–response meta-analysis (DRMA) using the robust error meta regression model to evaluate the sustainability of pain reduction achieved by ST compared to standard therapeutic options. A sensitivity analysis was performed to examine the impact of different underlying causes of CNP on pain reduction patterns, specifically comparing cancer-induced (N = 3) and non-cancer-induced (N = 5) CNP.

Results: Overall, the DRMA results demonstrate that ST achieves a greater reduction in pain scores than the comparison group. The results of the sensitivity analysis show that the cancer group experienced a peak reduction in mean pain scores of 3 points by day 15, after which their scores returned to baseline levels by day 70. In contrast, the non-cancer group exhibited a more sustained reduction in pain, achieving a 3-point decrease that was maintained throughout the follow-up period of 90 days.

Conclusions: Scrambler therapy offers sustained pain relief for more than 90 days, offering a new treatment choice in pain management of CNP due to non-cancer aetiology.

Keywords: Scrambler therapy, Chronic Neuropathic Pain, Dose–response Meta-analysis

57

Clinical outcomes of every 48-hour vs. 72-hour fentanyl patches in pain management: A retrospective comparative study across DAAM and Enaya

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Background: Fentanyl patches are a widely used transdermal opioid therapy for the management of moderate-severe pain, particularly in patients with chronic pain conditions or cancer. These patches provide a consistent release of fentanyl over a designated period, with two common formulations being the 48-hour (Q48H) and 72-hour (Q72H) patches.

Aim: To compare the effectiveness, compliance, and safety of the Q48H and Q72H fentanyl patches in clinical practice, with the goal of optimizing pain management strategies.

Methods: Data on all fentanyl patches administered between January 2023 and August 2024 were extracted from the Cerner database across DAAM and Enaya, Rumaila Hospital.

Results: 19 patients (mean age ≥ 65 years, 73% Male, 58% Cancer) were administered either of the two frequencies. 37% of the patients were on 25 mcg/hr Q72H patch. Among patients with moderate pain (4-6 on Abbey Scale), those using the Q72H patch showed a 42% reduction to mild pain (1-3 on Abbey Scale), while patients on the Q48H patch experienced 100% improvement to mild pain. Among patients experiencing extreme pain (>10 on Abbey Scale), both patches demonstrated a 100% reduction in pain, bringing it down to moderate or mild levels. Patients on Q72H patch had lower incidence of side effects compared to Q48H patch respectively (Sedation 29.4% vs 100%; Constipation 47.1% vs 50%). Compliance rates were at 100% for both groups.

Conclusion: Patients with higher pain scale will benefit from Q48H patch compared to Q72H patch for improved pain management. Although the Q48H patch has a pronounced side effect profile, the overall benefit-risk ratio is more favourable. The findings could assist clinicians in selecting the optimal fentanyl patch formulation, improving pain management and informing healthcare policies to enhance treatment protocols and cost-effectiveness.

Key words: Fentanyl, Transdermal Opioid Therapy, Pain Management, Pain Scale, Adherence

59

Empowering early career pharmacists: Unleashing non-clinical competencies through foundation pharmacy residency

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Background: Pharmacy residency programs traditionally prioritise clinical skills development. However, non-clinical competencies, such as leadership, conducting education, and innovation are now emerging as pivotal factors in propelling pharmacists toward excellence in practice. The extent to which these non-clinical skills are effectively fostered by residency programs remains unclear.

Aim: This study aims to explore how residency programs propel the development of crucial non-clinical competencies such as leadership, conducting education, and innovation.

Methods: Participants pharmacists who completed a foundation residency program and their preceptors from a tertiary teaching hospital took part in semi-structured interviews. Thematic analysis, employing an inductive approach and aided by NVivo software, was used to identify recurrent themes in the interview responses.

Results: Competency development was shaped by four key themes: system-dependent facilitators, system-dependent barriers, individual resident attitudes, and pharmacy department influences. The structure of the residency program was perceived to strongly support competency development in conducting education. The impact on the leadership and innovation competencies development was comparatively lesser.

Conclusion: Pharmacy residency is perceived as effective in supporting non-clinical competency development when there is a clear structured framework with objectives and guidance for pre-defined activities and tasks known to support competency development. Ambiguity and a lack of standardized guidance in developing specific competencies were identified as factors that diminish their relevance for both residents and preceptors. To enhance residency programs, it is essential to establish clear frameworks, with predefined objectives and activities known to support competency development and supplement them with the necessary skills building courses where appropriate.

Keywords: Pharmacy residency, Competencies, Early career pharmacists

60

Safer care through intensive medications reconciliations (IMR)

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Background: Intensive medication reconciliation (IMR) is the systematic and comprehensive review of all the medications a patient is taking to ensure that any medications being added, changed, or discontinued are carefully evaluated. In October 2023, a retrospective audit observed that among 157 DM2 patients with 4 or more medications prescribed, only 15 had completed IMR with a baseline compliance to IMR as 9.5.

Aim: To improve percentage of intensive medications reconciliation for type 2 diabetes patients with 4 or more prescribed medications from 9.5% to 15% at Al Daayen HC from 1st of November 2023 to 31st January 2024.

Methods: Percentage of Intensive medication reconciliation was assessed on a weekly basis through an audit. The team adopted Patient awareness leaflets and boards, which was developed in conjunction with the patients. Patient satisfaction was also assessed at the end of the PDSA cycle. The project was conducted at the Pharmacy department in collaboration with physicians at the Al Daayen health Center, Primary Health Care Corporation Doha Qatar.

Results: The utilization improved to an average of 21% at the end of PDSA Cycle in January 2024. The percentage of IMR Utilization was monitored for sustainability of the process with patient engagement and the compliance is an average of 28% at the end of August 2024. Ninety percent of the patients were aware of the IMR service availability, while 86% had utilized the service. Seventy seven percent of the patients were satisfied with pharmacist education while 23% were moderately satisfied.

Conclusion: The glycemic control for diabetic patients has improved by 2% during the last 2 quarters of 2024 with more compliance to the medications. The project gave insight on the use of different patient engagement platforms, such as engaging them in developing the education material and improving staff skills in delivering the education as well.

Keywords: Intensive medications reconciliation, Medication safety, Medication reconciliation

61

Preliminary assessment of a risk prediction tool for multidrug-resistant organism infections in critically ill patients in a tertiary care hospital in the united Arab emirates

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Background: Hospital-acquired infections caused by multi-drug-resistant organisms (MDRO) increase healthcare burden, patient morbidity, and mortality. Internationally available risk assessment tools for MDRO lack generalizability to different populations. This preliminary study aimed to investigate risk factors for MDRO infection in the UAE population and assess the feasibility of performing the study on a larger scale.

Methods: This is a retrospective case-control study. Patient data from the ICU were collected from a tertiary care academic hospital in Sharjah through the period of March to May 2022. A total of 39 patients were screened by the principal investigator through random sampling. Patients admitted to ICU for more than 24 hours for a hospital-acquired infection were included. Risk factors for MDRO infections were identified from the patient data. Student t-test was used for parametric variables, Kruskal-Wallis test for non-parametric variables, and Chi-square test for nominal variables, all used wherever applicable at $\alpha = 0.05$.

Results: Cases were divided into MDRO ($n = 11$), Non-MDRO ($n = 10$), and no-growth ($n = 18$) groups. Out of 39 patients, 61.5% ($n = 24$) were females with a mean age of 80.26 years (± 12.49 SD). Recurrent infection was lower in the no-growth group versus others ($p = 0.045$). A reported history of bacterial growth was identified to put patients at a 2.83-fold risk of getting MDRO infection (95%CI 1.489–5.39), and those who required enteral tube feeding had a 2.22 higher risk (95%CI 1.369–3.608). Moreover, the de-escalation of antibiotics was statistically significant ($p = <0.001$) and higher in the MDRO group versus non-MDRO with 81.8% ($n = 9$) & 60% ($n = 6$), respectively.

Conclusion: History of bacterial growth and the use of enteral feeding are associated with a higher risk of developing MDRO infection. Based on this pilot study results, modifications to the data collection method were performed before starting a larger-scale investigative study to identify all risk factors harbouring MDRO infection in UAE.

Keywords: Risk prediction tool, Multidrug-resistant organism infections, Critically ill patients

62

Quantitative and qualitative assessments of medication reconciliation module for hospital pharmacists in the United Arab Emirates: Lessons learned

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Background: Authors of this research developed the first Medical Reconciliation (MedRec) module for pharmacists in the United Arab Emirates.

Aim: This study aimed to assess the potential impact of the MedRec module quantitatively and qualitatively.

Methods: In this mixed-method study, a pre–post single-arm assessment was done at baseline (pre-intervention), and after six months (post-intervention) on the knowledge, perception, and barriers towards MedRec practice among hospital pharmacists ($n = 35$) who underwent the module. The assessment tool was a content and face-validated, reliable (Cronbach alpha = 0.781) questionnaire. Pre–post score comparisons were made using the Wilcoxon signed ranks test (at $\alpha = 0.05$). The qualitative assessment included in-depth interviews ($n = 8$) and performing thematic analysis using AtlasTi.

Results: At baseline, the median scores were: knowledge 8 (IQR 6–10)/12, perception 28 (IQR 25–30)/35, and barriers 26 (IQR 23–29)/30. Post-intervention, the median knowledge score increased to 12 (IQR 10.75–12, $p < 0.001$), perception score to 35 (IQR 31.75–35, $p < 0.001$) with no significant change in barriers score [26.5 (IQR 22.75–29), $p = 0.640$]. The module enhanced participants' knowledge of the definition of MedRec ($p = 0.039$) and the types of medication discrepancies ($p = 0.022$), perceptions toward the need for consulting pharmacists when medical staff considers discontinuing or prescribing medications ($p = 0.013$). Post-intervention, the perceived time barrier to practicing MedRec was reduced ($p = 0.003$). Qualitative feedback provided an in-depth assessment of the module. *Just narrate the findings from qualitative part in one or two sentences rather putting a quote.*

Conclusion: The study findings suggest that individual-level interventions can improve only knowledge and perceptions of MedRec while minimizing the practice barriers requires interventions at higher levels.

Keywords: Mixed-methods research, Medication reconciliation, United Arab Emirates

63

Exploring pharmacy students' perceptions on curriculum enhancement in a pharmacology course: A qualitative study

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Background: Pharmacology as a subject links basic and clinical science, therefore it is crucial for pharmacy students to have a strong foundational understanding of pharmacology early in their education. The evolvement of pharmacy curricula worldwide has led to the implementation of vertically

integrated pharmacy programs to facilitate the applicability of theoretical knowledge to practice. To address this need, the foundations of pharmacology course at the College of Pharmacy in Qatar University was redesigned with revised learning objectives and content.

Aim: To explore pharmacy students' perceptions of integrating a foundations of pharmacology course into a vertically aligned curriculum, supported pharmacology knowledge development. Additionally, we aimed to explore the course impact on students learning process and preparedness for future pharmacology courses.

Methods: Focus group interviews were conducted with twenty-four pre-registration pharmacy students to explore their perceptions on how the revised course supported their foundational pharmacology knowledge. Data were analyzed by inductive thematic analysis.

Results: Four key themes emerged from the data 1. Optimal timing for course introduction 2. Students' readiness to apply course content 3. Teaching strategies and course execution 4. Opportunities for course enhancement. Overall, students had positive perceptions on the revised curriculum, which significantly enhanced their understanding of complex pharmacological concepts. The students appreciated learning foundational pharmacology principles in their first professional year, as it allowed them to progressively build on their pharmacology knowledge and transfer this knowledge to other pharmacy subjects.

Conclusion: This study highlights the significance of introducing the foundations of pharmacology early in the pharmacy education curriculum and recognized the importance of content repetition for knowledge retention. Further studies are required to evaluate how the pharmacology course supports students' competency development in pharmacology knowledge, application, and understanding.

Keywords: Curriculum, Pharmacology education, Qualitative analysis

66

Evaluation of drug-related problems by clinical pharmacists in an Intensive Care Unit: A Systemic Review

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Background: Drug therapy for critical care patients usually requires specialized care management, and complex medication regimens. In critical

healthcare settings, clinical pharmacy services have been expanded significantly. Therefore, incorporating clinical pharmacists into the clinical team can be an ideal approach to preventing drug-related problems (DRPs) and improving drug safety and efficacy.

Aim: This study aims to evaluate DRPs identified by clinical pharmacists in the ICU.

Methods: Three English databases were searched (Embase, The Cochrane Library, and PubMed) for observational and interventional studies reporting drug-related problems identified by clinical pharmacists in critical care settings. Two of the researchers independently conducted literature screening, quality evaluation, and data extraction and qualitatively analysed the results.

Results: A total of 4,222 DRPs were reported by eleven selected studies. The most frequently reported DRP category was dose selection, indicated by either over-dose or underdosing, as identified by nine studies. The second most common category was drug selection, with 1,805 reported DRPs. The most common medication category with identified DRPs was anti-infectives. Most clinical pharmacist interventions involved dosage adjustments and therapeutic drug monitoring, with an acceptance rate ranging from 73% to 100%.

Conclusion: Dose and drug selection were the most reported DRPs in critical care settings. This directed the focus of clinical pharmacist interventions towards dosage adjustment and therapeutic drug monitoring. It highlights the vital role of pharmacists utilizing PK/PD knowledge in optimizing medication therapy to enhance patient outcomes.

Keywords: Drug-related problems, Intensive care unit, Systematic review

69

Clinical audit to limit protease inhibitor-based regimen use among adult living with human immunodeficiency virus (HIV)

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Background: Protease inhibitor (PI)-based regimen have demonstrated virologic efficacy, durability, and high genetic barrier to resistance. They are helpful for patients with integrase strand transfer inhibitor resistance. However, long term toxicities and major drug interactions with medications metabolized by cytochrome P-450 would require close monitoring.

Aim: to evaluate and optimize the use of boosted PI among adults at the HIV clinic in Qatar in accordance with international guidelines. By limiting the use

of PIs for patients with resistance or intolerance to other classes of antiretrovirals, thus reducing risk of toxicities from PI use such as diarrhea, dyslipidemia, diabetes, and cardiovascular risk. Additionally, switching from boosted PI-based to INSTI-based regimen where applicable. Ultimately, to sustain viral suppression and improve adherence.

Methods: A retrospective cohort review was conducted of adults with HIV at the Communicable Disease Center who were receiving darunavir/cobicistat (DRV/COBI) based regimen, between Jan 1st 2022 and Jan 1st 2023. Data was collected from patients' records. Then, Audit results were discussed in a multidisciplinary team.

Results: Thirty patients on DRV/COBI were screened, 83% males and with median diagnosis of 13.5 years. Seven patients were excluded as they were switched to INSTI-based regimen. A total of 23 patients were reviewed, mostly (96%) were virologically suppressed with CD4 counts >200 cells/mcl. The MDT concluded 10/23 patients (43%) would benefit from switching to INSTI-based regimen. Therefore, we switched 6/10 patients to (BIC/F/TAF), 1/10 patient to (DTG/3TC) in which they remained virally suppressed. Protease inhibitors were maintained in 13/23 patients (57%) as they had two class resistance mutations or poor compliance to the previous regimens.

Conclusion: We recommend switching 10 patients to single pill integrase strand transfer inhibitors-based regimen. This drug use audit is expected to enhance the care of HIV patients by reducing drug interactions, drug toxicities and pill burden.

Keywords: Protease inhibitors, Darunavir-Cobicistat, HIV, Antiretrovirals, Compliance

70

Efficacy and safety of iron treatment strategies in athletes with iron deficiency and iron deficiency Anemia in Qatar

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Background: Iron deficiency (ID) and iron deficiency anemia (IDA) among athletes can affect performance and recovery; however, the efficacy of iron supplementation in athletes remains understudied.

Aim: To assess the efficacy and safety of iron therapeutic regimens among athletes with ID or IDA.

Methods: Retrospectively, we included athletes with ID or IDA who received at least one month of iron treatment (ferrous sulphate tablets, iron polymaltose syrup, or iron sucrose injections), between March 2020 and March 2022, with valid baseline and follow-up laboratory assessments. Primary outcomes were ferritin and hemoglobin changes; secondary outcomes included other laboratory parameters, side effects, and adherence statistics. The study was approved by Aspetar Ethics Committee (IRB X202301053).

Results: The study included 275 athletes with ID or IDA (81.8% male, mean age 17.5 ± 6.7 years) across various sports, mainly football (41.8%) and athletics (19.6%). Among them, 37.1% had IDA, and 62.9% had ID. Ferrous sulfate tablet (94.2%) was the primary treatment, with most receiving 60–65mg elemental iron once daily (75.6%). After 2–5 months of follow-up, ferritin levels rose from 20.3 ± 1.1 $\mu\text{g/L}$ to 32.5 ± 1.3 $\mu\text{g/L}$ (60.1%, $p < 0.001$), while haemoglobin increased from 12.7 ± 0.1 g/dL to 13.1 ± 0.1 g/dL (3.6%, $p < 0.001$). Greater increases in ferritin and haemoglobin were observed among power and skill sport athletes compared to other sports, and among females compared to males ($p < 0.001$). Higher serum Vitamin D (>30 ng/mL) was correlated with greater ferritin gains (40.2%, $p = 0.006$) than severe vitamin D deficiency (<10 ng/mL). There were no significant cases of complications or adverse events.

Conclusion: Various iron dosing strategies are effective and safe for improving iron status in athletes with ID or IDA. The optimal dosing strategy in each sports category should be further explored.

Keywords: Iron deficiency, Anemia, Athletes

71

Development and evaluation of an inpatient multidisciplinary team deprescribing program for older patients in a tertiary care hospital in Qatar: A preliminary analysis

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Background: Deprescribing is a recognized structured intervention that effectively prevents adverse events associated with inappropriate polypharmacy. The literature has abundant examples of successful deprescribing interventions in outpatient settings. However, inpatient

deprescribing is not as well-established. In this study, we aim to design and assess the impact of an inpatient deprescribing multidisciplinary team (IDMT) intervention for older patients in a tertiary hospital in Qatar. Aim?

Methods: This research is a proof-of-concept, pre-post, single-site study. Inpatient deprescribing is performed by a trained multidisciplinary team using a structured and an integrated workflow. Patients included are medically stable ≥ 65 years who are admitted to Qatar Rehabilitation Institute inpatient unit and on polypharmacy (≥ 10 medications). Eligible patients undergo standardized stages of deprescribing workflow which includes assessment, formulation, implementation, active monitoring of deprescribing plan, and discharge and post-discharge follow-up. Outcome measures include treatment burden (assessed by TBQ score), health-related quality of life (assessed by EQ-5D-5L tool), Medication Regimen Complexity Index (MRCI), Charlson's comorbidity index, cost benefit and cost-savings.

Results: The study commenced in mid-January 2024 and is presently ongoing. Given the early phase of recruitment, very limited preliminary findings are currently available, with more comprehensive results anticipated by the end of April 2025. The average age of the participants is 72 years. All patients had a medical history encompassing stroke, type II diabetes, and dyslipidemia. Baseline data reveal average initial medication count of 15, Charlson's comorbidity score of 8, a TBQ score of 34, and an MRCI score of 62.

Conclusion: The implementation of IDMT program is anticipated to improve health outcomes in older adults with polypharmacy and are expected that it would result in cost saving.

Keywords: Deprescribing, Older patients, Hospital pharmacy

72

Exploring the perceptions of stakeholders on the facilitators and barriers to implementing clinical pharmacy services in private hospitals in Qatar: A preliminary study

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Background: Clinical pharmacists are essential to medication safety, contributing to error prevention and optimized therapy. In Qatar, their role is

well established in public hospitals but remains limited in private hospitals, leading to missed opportunities for improving patient safety in the private sector.

Aim: The study aims is to identify the barriers and facilitators regarding clinical pharmacists' integration across various private hospitals in Qatar.

Methods: Three face-to-face semi-structured interview discussions, using an interview guide that was drawn upon the CFIR, were conducted with 3 hospital pharmacy directors to explore the barriers and facilitators for the implementation of clinical pharmacy services (CPS) in private hospitals in Qatar. Interview discussions were audio-recorded, transcribed, and analyzed using manual sorting and coding of the transcripts.

Results: Key major domains emerged from the study include pharmacists' professional identity, the barriers and facilitators for implantation, perceptions toward CPS, and strategies for implementation. The results revealed an ongoing barrier by Qatar's private hospitals to establishing clinical pharmacy service identities as difficulty in finding candidates with the desired level of expertise, financial constraints, limited government support, and a lack of demand for CPS within private hospitals. Facilitators of the involvement of clinical pharmacy practice identified include effort, willingness to implement, and reliable administrative support.

Conclusion: The study identified key barriers to the integration of clinical pharmacy services in Qatar's private hospitals, including workforce limitations, financial challenges, and insufficient institutional and governmental support. However, facilitators such as administrative willingness and commitment to implementation indicate a positive outlook for future integration. Addressing these barriers through targeted strategies could support the expansion of clinical pharmacy services in the private healthcare sector.

Keywords: Barriers, Facilitators, Clinical pharmacy services, Qatar, Private hospitals

73

Pharmacists' perspectives on depression care: A scoping review

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Background: Pharmacists are well placed to support people with depression, complementing the roles of other healthcare professionals. However, there is

paucity of literature regarding pharmacists' attitudes, beliefs, barriers, and enablers on depression care.

Aim: To explore pharmacists' perspectives in depression care worldwide, with a particular focus on their beliefs, current attitudes, and the barriers and key enablers to providing effective depression care.

Method: The scoping review was conducted according to the Joanna Briggs Institute (JBI) methodology for scoping review aligned with the Arksey and O'Malley methodology and reported by the Preferred Reporting Items for Systematic Reviews Extension for Scoping Reviews (PRISMA-ScR) guidelines. The databases searched included Medline, Scopus, ProQuest, and Cochrane databases. Data were presented in tabular and narrative formats, with key themes analyzed to capture pharmacists' perspectives on depression care. Any discrepancies were settled by consensus.

Results: This review included 14 studies, 5 qualitative and 9 quantitative papers about pharmacists' attitudes and beliefs in supporting depression care. Pharmacists' attitudes and belief themes were further divided into five emerging subthemes, including pharmacists' attitudes toward engagement and support in depression care, the negative attitudes toward delivering depression care, the accessibility of pharmacy care, the expanded role of pharmacists in depression care, and pharmacists' perceived role limitations and misconceptions in depression care. Barriers included competing priorities, interprofessional boundaries, and limited training and resources, while enablers encompassed specialized mental health training and remuneration.

Conclusion: The findings of this review indicated that pharmacists generally hold favourable attitudes and beliefs toward depression care. Training programs, remuneration, and standardized guidelines are needed to further improve the unfavorable pharmacists' attitudes and overcome limiting beliefs.

Keywords: Pharmacist, Depression, Perspective, Attitude, Beliefs

74

Senior medical students' and recent graduates' perspectives on prescribing competencies in undergraduate medical curriculum at Qatar University

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Background: Safe and effective prescribing is a core competency for medical graduates and a critical component of patient care. Medical schools have a responsibility to ensure that their curricula adequately prepare students to prescribe competently and confidently. However, limited data exist on how medical students and recent graduates perceive their preparedness to prescribe, as well as the effectiveness of the curriculum in achieving prescribing-related learning outcomes.

Aim: This study aimed to determine the perceptions of senior medical students and recent graduates from College of Medicine (CMED) at Qatar University regarding their undergraduate education and training related to prescribing, their confidence in achieving the intended learning outcomes specified in the curriculum and their preparedness to prescribe in practice.

Methods: This was a cross-sectional web-based survey with qualitative components, utilizing a 22-item pre-validated questionnaire developed through a multi-phase iterative process. The survey was administered to all senior medical students (year 4–6) and recent graduates of CMED at Qatar University.

Results: One hundred sixty-seven medical students and graduates responded to the survey (response rate ~45%). Overall, the participants expressed positive attitude towards curricular content related to prescribing that aimed to prepare them for practice, including pharmacology and therapeutics, history-taking, physical examination, diagnostic investigations, application of evidence-based medicine, and other related skills. Over 60% of the participants believed that the prescribing components of the undergraduate curriculum made them competent in providing safe and effective prescriptions in practice. Notably, there was an association between seniority and perceived curricular effectiveness, highlighting the impact of clinical experience on participants' views. Recommendations for curriculum enhancement focused on practical integration, contextualized education, and a holistic approach to both theoretical and clinical learning.

Conclusion: The study revealed positive opinions regarding curricular content related to prescribing and identified areas for improvement in medical curriculum. Continuous monitoring, evaluation and refinement based on participants' feedback are crucial for ensuring the preparedness of medical graduates for safe and effective prescribing in real-world clinical practice.

Keywords: Medical Education, Curriculum Evaluation, Contextualized Education, Curriculum Enhancement

75

Improving compliance of the narcotic and controlled medications maintenance checklist in the heart hospital-a quality improvement project

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Background: Controlled and narcotic medications constitute valuable therapeutic treatment for patients across the healthcare facilities. Proper management of narcotic and controlled medications is a critical aspect of our hospital, ensuring regulatory adherence, prevention of misuse and patient safety. Baseline data of the Heart Hospital units showed clear lack of awareness among staff, and compliance to maintenance checklist (MCL) falls far short of what is expected.

Aim: This quality improvement project aimed at increasing the percentage of Heart Hospital units' compliance with all standards of the MCL of narcotic and controlled medications from 0% in October 2022 to a 100% by June 2023.

Method: A quality improvement methodology was adopted. Three Plan-Do-Study-Act (PDSA) cycles started in October 2022 were used to test the success of the interventions. Checklists are commonly used in healthcare and are important tools in improving performance. Hamad Medical Corporation has a MCL of 15 standards as part of improving the narcotic and controlled medications management. A run chart was used to display the monthly percentage compliance to MCL. Cause and effect diagram disclosed the unforeseen causes of non-compliance to the MCL standards.

Result: All units stored narcotic and controlled medications were reported non-compliant (0%) to the MCL (PDSA 1). In the second PDSA, raised staff awareness and introduction of one-to-one educational session. Compliance increased to 67%. In the third PDSA, compliance continued to improve reaching 92%. The target 100% achieved in May 2023 where, recognition, accountability and appreciation were implemented.

Conclusion: Compliance to standardized MCL is an asset to maintain narcotic and controlled medications in our hospital. It provides a system to prevent, detect and respond to any discrepancies. We managed to improve compliance to a 100% and continuously sustaining the target.

Keywords: Controlled medications, Narcotic drugs, Patient safety, Checklist

Improve emergency cart medications management to error proof in the heart hospital-a quality improvement project

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Background: Successful cardiopulmonary resuscitations (CPR) relies on availability of emergency carts that provide required equipment and recommended medications. Missing, dis-organized or inconsistent medication drawers in emergency carts or faulty equipment can significantly compromise resuscitation process. We identified gaps in maintaining emergency medications in emergency carts in the Heart Hospital (HH).

Aim: The aim of this project was to improve compliance to emergency cart medications availability, uniformity and security in the HH from 58% in August 2021 to a 100% by August 2022.

Method: A baseline data was collected in August 2021. This was followed by a quality improvement project using a Plan-Do-Study-Act (PDSA) cycles over a 12-months period. Interventions to fill in the gaps included a total of 5 PDSA cycles. This involved development of comprehensive checklist; standardized medications layout and labels across the hospital; staff education and awareness about the processes; development of the picture portfolio for stocked medications and risk assessment for medication administration.

Result: Overall, compliance improved from 58% to a 100% with huge sustainability obstacles such as frequent changes of medications brands, and supply of short expiration dates of emergency medications.

Conclusion: The successful improvement over the 12-month period can prove that the employed interventions were effective. Sustainability was maintained for a long period post completion despite the challenges that were faced. Our next step is focusing on reducing emergency cart medication wastage.

Keywords: Emergency medications, Code blue, Cardiopulmonary resuscitation

Improving reporting of adverse drug reactions by the operational pharmacists in the heart hospital

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Background: Adverse drug reactions (ADRs) exert a substantial burden on the healthcare system. Pharmacists possess the essential expertise to identify and report ADRs. Nevertheless, within the operational pharmacy practice, there has been a noticeable under-reporting of ADRs among operational pharmacists. This under-reporting is of critical concern and presents an area for improvement.

Aim: This quality improvement project aimed at increasing the percentage of ADR reported by operational pharmacists in the Heart Hospital from 1% in December 2021 to 10% by December 2022

Method: The team adopted application of 'model for improvement', including use of run charts, process mapping and Plan-Do-Study-Act (PDSA) approach. To identify the barriers to ADR reporting among operational pharmacists, data were collected through a combination of surveys and focus group discussions. The data were analyzed using pareto analysis and a fishbone diagram. An online educational intervention was conducted twice. Written educational materials and reminder cards were developed. A non-monetary incentive approach was used to encourage staff.

Result: ADR reporting improved following a series of PDSA cycles where modifications were made based on data and feedback from staff. The most significant barriers were identified, including lack of knowledge and training, absence of reminders and feedback along with high workload. By the end of Year 2022, the percentage of ADR reports had increased to 16%. This improvement was continuously sustained throughout the entire year 2023 to available data of the year 2024. The awareness and education of pharmacists significantly improved, having a profound impact on enhancing Hamad Medical Corporation (HMC) pharmacovigilance program.

Conclusion: Educational interventions, feedback and reminders improved percentage of ADR reported by operational pharmacists. This improved reporting and maintenance of a continuous intervention are recently extended to other healthcare professionals, including physicians and nursing.

Keywords: Adverse drug reactions, Adverse event, Pharmacovigilance, Pharmacy profession

A reflection on a transforming experience with value improvement program and a call for national and international pharmacy experience

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Background: In response to the growing need for high-value care, the Value Management approach, originally developed by the Institute for Healthcare Improvement (IHI), was adapted and redesigned by the Heart Hospital (HH) in collaboration with IHI and the Hamad Healthcare Quality Institute. This collaboration led to the launch of the Value Improvement Program (VIP) in 2018, aimed at enhancing care quality while optimizing resource utilization

Aim: The aim of this reflection was to explore the HH pharmacy department's experience on implementing the value improvement methodology to improve our care services and also to enhance our staff skills and capabilities.

Method: Three tools were utilized to implement the VIP. These included a displayed visual management board, Pharmacy box score with listed projects along with their data collected and a monthly scheduled communication huddles that suits the pharmacy setting.

Result: This methodology has yielded positive outcomes in performance and cost domains. Among the included projects and compared to the baseline data on hand prior to implementation, improvement in adverse drug reactions (ADRs) increased by 16%; near-expire medications cost saving increased to 66%; compliance to narcotic and controlled drugs maintenance checklist has improved to a 100%.

Conclusion: Throughout this journey, conducting team huddles has brought about numerous benefits for our pharmacy team, fostering a culture of collaboration and enhanced patient care. Emphasis on continuous improvement and cost-effectiveness has prompted us to critically assess our resources allocation and proactively identify areas for improvement. Pharmacy staff was upskilled at an individual level to ensure improvement methodologies are commonly understood and become widespread at all levels of practice.

Keywords: Value improvement, Quality improvement, Visual board

Hospital pharmacists' perspectives on integrating artificial intelligence (AI) into continuing professional development (CPD): A qualitative study

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Background: Continuing professional development (CPD) is essential for developing skilled healthcare professionals. Recently, artificial intelligence (AI) has emerged as a revolutionary force across various sectors, including healthcare. However, its adoption in the education and training of healthcare professionals remains limited. Utilizing the great potential of AI could fundamentally reshape our approach to CPD.

Aim: This qualitative study aimed to explore hospital pharmacists' perspectives and challenges on integrating AI into CPD.

Method: A qualitative descriptive methodology was employed, involving semi-structured face-to-face interviews. Twelve hospital pharmacists from different hospitals in Saudi Arabia were selected through purposive sampling for the interviews. The recorded interviews were transcribed and analyzed using thematic analysis.

Results: A total of three themes were extracted. Overall, Pharmacists had a positive perception of AI in CPD; they believe that CPD is poised to undergo significant transformation in the foreseeable future, propelled by modern concepts such as AI and robotics. AI systems can adapt to individual needs, providing personalized learning experiences, assessing strengths and weaknesses, providing real-time feedback, and recommending tailored modules based on individual learning patterns and knowledge gaps. This incorporation of AI can enhance adaptability, efficiency, and engagement, ultimately empowering pharmacists to excel in their profession. Furthermore, integrating AI into CPD programs presents an opportunity to overcome several limitations of the current CPD system, such as lacking indicators or evaluations to assess pharmacist development, and time constraints. However, pharmacists' lack of awareness of proper tools for utilizing and implementing AI hinders their ability to fully leverage AI in CPD.

Conclusions: The study highlights the need for healthcare organizations to assess the current state of AI in education and incorporate AI into CPD programs to optimize continuous development within pharmacy's profession.

Keywords: Continuing professional development, Artificial intelligence, Education; Hospital pharmacists, Qualitative study

Artificial intelligence (AI) in pharmacy: A scoping review

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Background: Artificial Intelligence (AI) is rapidly transforming pharmacists from their traditional dispensing roles to becoming an integral part of the decision-making multidisciplinary healthcare team, providing safe and effective patient care. AI can be used in providing a wide range of services including pharmacogenomics, drug information, medication safety, antimicrobial stewardship and continuous education etc.

Aim: The current review aims to explore the implications of AI in pharmacy, examine various tools supporting accurate and evidence-based clinical decision making and identify key facilitators and barriers to implementing AI in pharmacy.

Methodology: A systematic search around implications of AI in pharmacy was conducted across major databases including MEDLINE and CINAHL from inception until October 2024. The search included literature pertaining to AI utilization in pharmacy and relevant topics. All study designs were included. Two researchers independently extracted the data.

Results: The interim analysis revealed that 24 out of 112 studies met the inclusion criteria. Machine learning, robotics, clinical decision support, and virtual assistance were the most used AI techniques. AI applications predominantly focused on drug information, detecting adverse drug reactions, automated dispensing, tools to adhere medication adherence and dose optimization etc. More than 40% of the studies have also reported the use of AI in detecting and mitigating medication errors and adverse events and promoting medication safety practices through education. Key facilitators include enhanced patient care, data integration, operational and clinical efficacy, while barriers involve implementation cost, data privacy, ethical and regulatory challenges.

Conclusion: AI has the potential to revolutionize pharmacy practice, improving patient care and outcomes. Pharmacist must embrace AI and collaborate with researchers to develop and implement effective AI-driven solutions.

Keywords: Artificial intelligence, Pharmacy, Patient care

83

Evaluating the use of virtual simulation training to support pharmacy students' competency development in conducting dispensing tasks

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Background: The incorporation of digital health technologies in undergraduate teaching, such as the MyDispense program, provides a simulated opportunity to support students' competency development in dispensing tasks.

Aim: This study aimed to evaluate the impact of using MyDispense, an online pharmacy simulation platform, on pharmacy students' self-reported reaction, learning, and accuracy in performing dispensing tasks.

Methodology: 16 MyDispense cases simulating real-practice scenarios from prescription dispensing and verification to patient counseling were created for 55 professional year 1 pharmacy students as part of a professional skills course. A pre-post survey, nestled on Kirkpatrick's Model of Training Evaluation, was conducted on participants in the cohort to evaluate how the incorporation of MyDispense supported their development of dispensing tasks. A virtual formative assessment was also done using MyDispense to compare students' self-reported accuracy with faculty assessment scores to evaluate how MyDispense can support students' knowledge and metacognitive abilities in dispensing.

Results: The study showed an increase in students' confidence after using MyDispense, which led to an increase in the "learning" level of Kirkpatrick's Model. However, small changes were observed in other levels and sublevels of Kirkpatrick's Model. A notable discrepancy was observed between students' self-assessment scores and faculty assessment scores, with students overestimating their performances. Correlation analysis showed a weak relationship between students' self-assessment scores and the learning level of Kirkpatrick's Model.

Conclusion: This study highlights the positive impact of MyDispense on pharmacy students' confidence in dispensing. However, virtual simulation training may be best incorporated when students have exposure to experiential training placements, to maximize the learning outcomes and knowledge in dispensing processes.

Keywords: Competency-development, Dispensing, MyDispense, Professional skills, Virtual simulation

Exploration of professionalism and its tenets among health professions education students: A mixed-methods study from Qatar University health sector

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Background: Professionalism, defined as judicious use of communication, knowledge, skills, reasoning, values, and reflection, for benefits of individuals and communities, is recommended as an explicit component of health education curricula, and is demarcated by distinct tenets. Professionalism among health professions education students at Qatar University (QU) has not been previously explored.

Aim: The current study aimed at assessment of professionalism and its tenets among QU Health Sector students and exploring associated factors.

Methods: A mixed-methods design was used, following ethical approval from QU Institutional Review Board (Ref: QU-IRB 1977-EA/23). For the quantitative phase, data were collected anonymously from students at the five QU Health Sector colleges, using the Pharmacy Professionalism Instrument (PPI), which assesses overall professionalism score, and scores of each of six tenets: Excellence, Respect, Altruism, Duty, Accountability, and Honor/Integrity. The qualitative phase involved five focus groups that explored students' perspectives on professionalism and opportunities for its fostering. Data from each phase were analyzed separately then integrated.

Results: A total of 536 survey responses were analyzed. The mean PPI score indicated fair levels of professionalism and of Excellence, Altruism, Duty and Honor/Integrity tenets, while Accountability and Respect exhibited excellent levels. Scores statistically increased with progress in year of study for the overall professionalism score, Excellence, and Respect. Focus groups confirmed students' views on tenets of excellence, respect, altruism, accountability, and Honor/Integrity. Yet, a slight discrepancy in the latter tenet emerged regarding necessity of reporting errors, with some students expressing concerns about potential negative consequences.

Conclusions: This is the first assessment of professionalism among QU Health Sector students, revealing favorable tenets developing across the years of study. Amidst evolution in both education and sociocultural norms, health education curricula should uphold the cultivation and emphasis of

professionalism concepts, and establish an environment of recognition and nurturing of professionalism.

Keywords: Assessment, Professionalism, Mixed methods

87

Cost-effectiveness of gene-test-guided HIV therapy in drug-resistant patients – A systematic review of the literature

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Background: HIV drug resistance has become a burden on the healthcare system and a financial concern for all parties. The development of mutations in HIV results in resistance to antiretroviral therapy (ART), necessitating the use of HIV genotypic drug resistance testing to guide treatment.

Aim: This study aims to systematically review and evaluate the cost-effectiveness of HIV drug resistance testing at the time of diagnosis, compared with no testing, in guiding treatment decisions for patients with HIV drug resistance.

Methods: A systematic literature search of PubMed, EMBASE, and EconLit, was done to identify all the economic evaluations related to the cost-effectiveness of the HIV genotypic drug resistance test compared to the clinical judgement. The Quality of Health Economic Studies tool was used for quality assessment.

Results: The search identified 8 articles where genotype-guided treatment was compared to clinical judgment. Six studies showed that genotype testing is cost-effective compared to clinical judgement based on 3 different drug classes which were NNRTI, NRTI, and PI, while 2 studies showed that it is not cost-effective for INSTI drug-based regimen. All studies evaluated cost/ QALY except for 2 studies where the cost was per year gained. All included articles are of good quality.

Conclusion: Based on current economic evaluations in the literature genotypic drug resistance testing may not be considered cost-effective in contexts where INSTIs are the first-line treatment. On the other hand, it is cost-effective in settings where NNRTIs, NRTIs, and PIs are the first-line regimens.

Keywords: HIV, Drug-resistance, NNRTI, NRTI, PI, INSTI, Gene-test-guided, Cost-effective

Reducing NSAID prescriptions in patients receiving ACE inhibitors/ARBs and diuretics: A quality improvement initiative

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Background: Nonsteroidal anti-inflammatory drugs (NSAIDs) are widely prescribed, but their concurrent use with angiotensin-converting enzyme inhibitors (ACEIs) or angiotensin-II receptor blockers (ARBs) and diuretics can significantly increase the risk of acute kidney injury (AKI). This “triple whammy” of medications presents a major safety concern in hypertensive patients. At Abu Nakhla Health Center, a quality improvement (QI) initiative was launched to address the frequent prescribing of NSAIDs to patients receiving ACE inhibitors/ARBs and diuretics.

Aim: This project aimed to reduce NSAID prescriptions by 50% among patients receiving ACE inhibitors/ARBs and diuretics within three months (May to July 2024) by raising awareness among healthcare providers and educating patients on the associated risks.

Methods: A multidisciplinary team implemented several interventions, including provider education on safe prescribing practices and patient counselling at the pharmacy. Data was collected monthly through audits from the health center’s electronic medical records (Cerner) to monitor the prescription rates of NSAIDs in the target population. The effectiveness of these interventions was evaluated through the reduction in NSAID prescriptions.

Results: Over the three-month period, NSAID prescriptions in patients on ACE inhibitors/ARBs and diuretics decreased from 31% in May to 10% in July. The success of the project was attributed to improved provider adherence to prescribing guidelines and increased patient awareness of NSAID risks.

Conclusion: This QI initiative effectively reduced the prescribing of NSAIDs in high-risk patients, enhancing patient safety and preventing medication-related AKI. The project highlights the importance of targeted education and can serve as a model for similar initiatives in other healthcare settings.

Keywords: NSAIDs, ACE inhibitors, ARBs, Diuretics, Acute kidney injury

89

Adverse reactions to intravenous carboxymaltose: A closer look

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Background: Intravenous carboxymaltose is widely used for iron deficiency anaemia treatment. Despite its effectiveness, adverse drug reactions (ADRs) pose significant safety concerns, requiring thorough evaluation to optimize clinical practice.

Aim: To analyze ADRs related to intravenous carboxymaltose at AWH from 2020 to 2023, focusing on patient demographics, reaction types, infusion rates, and timing of side effects.

Methods: A retrospective review was conducted using ADR data collected by the Medication Safety Officer. Parameters analyzed included demographics, types of ADRs, infusion durations, timing of reactions, and reporting sources. Descriptive statistics were used to identify trends and inform recommendations.

Results: A total of 81 ADR cases were reported, with 98.8% occurring in females and 72.8% in patients aged 21–40 years. The most common reactions were allergic and skin-related (79%), followed by respiratory symptoms such as shortness of breath (19.8%). Moderate infusion rates (over one hour) were used in 75.3% of cases and correlated with fewer severe reactions compared to rapid infusions. Notably, 55.6% of ADRs occurred after infusion completion, highlighting the need for extended monitoring. Nurses reported most ADRs (86.4%), followed by clinical pharmacists (13.6%).

Conclusion: The study identified females aged 21–40 as the most affected demographic, with allergic and skin reactions as the predominant ADRs. Moderate infusion rates were associated with a lower incidence of severe reactions. Findings emphasize the need for gender-specific studies, extended post-infusion monitoring, and a review of infusion protocols to improve safety and clinical outcomes.

Keywords: Adverse drug reactions, Intravenous carboxymaltose, Infusion protocols, Medication safety, Patient monitoring

Optimizing medication reconciliation practices in the obstetrics and Gynecology department: A quality improvement initiative

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Background: Medication reconciliation is a critical process to prevent medication errors and enhance patient safety. Challenges in compliance and accurate documentation within the Obstetrics and Gynecology (OBS/GYN) Department at AWH prompted the need for targeted improvement efforts.

Aim: To improve compliance with medication reconciliation protocols in the OBS/GYN department and achieve a compliance rate of 95% by the end of 2023.

Methods: This quality improvement initiative identified key barriers, including misconceptions about the purpose of reconciliation, time constraints, and unclear responsibilities for documenting home medications. Interventions included interactive in-person presentations, daily follow-ups, and assigning accountable personnel for documentation. Monthly audits were conducted to assess adherence to protocols and measure progress.

Results: Interventions led to a marked improvement in compliance rates, progressing from 75.00% in January 2023 to 95.24% by July 2023. These results highlight the effectiveness of targeted strategies in optimizing medication reconciliation practices and enhancing physician adherence.

Conclusions: This initiative successfully improved compliance with medication reconciliation protocols in the OBS/GYN Department at AWH. The outcomes demonstrate increased physician engagement, enhanced understanding of reconciliation's importance, and improved accuracy of medication histories. Sustained monitoring, continuous education, and active engagement are essential to maintain and further these gains. This project underscores the significance of quality improvement efforts in promoting patient safety and healthcare efficiency.

Keywords: Medication reconciliation, Quality improvement, Patient safety, Compliance, Obstetrics and gynecology

91

Prevalence, causative microorganisms, and antibiotic sensitivity of UTIs in pregnant women: A retrospective study at AWH

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Background: Urinary tract infections (UTIs) during pregnancy present significant risks to maternal and fetal health due to physiological and hormonal changes. Identifying prevalence, causative pathogens, and antibiotic resistance patterns is critical to guiding treatment and improving outcomes.

Aim: To assess the prevalence, common microorganisms, and antibiotic sensitivity patterns of UTIs in pregnant women attending Al Wakra Hospital (AWH).

Methods: A retrospective review of 485 urine cultures from pregnant women was conducted over two years at AWH. Data analyzed included culture indications, identified microorganisms, and their antibiotic sensitivity profiles.

Results: The prevalence of positive urine cultures was 11.75%. Routine screening accounted for 60.62% of culture indications, followed by evaluations for chorioamnionitis (2.06%), preterm labor (1.65%), and fever (2.27%). Predominant microorganisms included *Escherichia coli* (2.45%), *Streptococcus agalactiae* (9.78%), and *Klebsiella pneumoniae* (1.22%), alongside other pathogens such as *Candida* and *Pseudomonas aeruginosa*, reflecting pathogen diversity.

Antibiotic sensitivity analysis showed strong bacterial response to penicillin, Augmentin, Nitrofurantoin, and Ampicillin (66.67%). However, certain bacteria, including extended-spectrum beta-lactamase (ESBL) producers and *Pseudomonas*, exhibited reduced sensitivity (17.54% and 15.79%, respectively), highlighting challenges in antibiotic efficacy.

Conclusions: This study highlights the considerable burden of UTIs in pregnant women, with routine screening proving essential for early detection. The diversity of pathogens and variable antibiotic sensitivity emphasize the complexity of UTI management in pregnancy. Tailored treatment strategies based on sensitivity profiles are vital for optimizing outcomes and minimizing adverse maternal and fetal effects.

Keywords: Urinary tract infections, Pregnancy, Antibiotic resistance, Prevalence, Maternal health

Improving influenza vaccination rates in pregnancy: A clinical pharmacist-led pilot study

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Background: Influenza immunization during pregnancy is essential for maternal and fetal health, yet vaccination rates remain suboptimal. National and international guidelines strongly advocate for influenza vaccination in antenatal care. This pilot study evaluated the feasibility and impact of a clinical pharmacist-led outpatient vaccination clinic to address this gap and promote vaccination adherence among pregnant women during the flu season.

Aim: To assess the feasibility of a pharmacist-led clinic to improve influenza vaccination compliance and to evaluate pregnant women's knowledge, attitudes, and vaccination barriers.

Methods: Over a two-week period, clinical pharmacists approached 88 pregnant women during their antenatal visits at an outpatient clinic. The pharmacists provided counselling, recorded patient responses, and identified barriers to vaccination. A survey captured participants' knowledge and attitudes toward influenza vaccination during pregnancy. Reasons for vaccine refusal or delay were documented to guide future interventions.

Results: Among 88 approached patients, 50 (56.8%) agreed to receive the influenza vaccine. Of the 38 who did not, 27 (71.1%) had already been vaccinated, while others cited reasons such as deferring to a spouse's opinion (2 patients), language barriers (1 patient), or allergy concerns (1 patient). Seven patients outright declined. Survey findings revealed that 44 women had not been vaccinated during prior pregnancies, with varying opinions on vaccine effectiveness and side effects.

Conclusion: This pilot study demonstrated that pharmacist-led antenatal vaccination clinics can significantly improve influenza immunization rates among pregnant women. The intervention highlights the critical role of clinical pharmacists in addressing vaccination barriers and promoting awareness. Further research should explore the scalability, sustainability, and long-term impact of such initiatives on public health outcomes for pregnant women and their infants.

Keywords: Influenza vaccination, Pregnancy, Clinical pharmacists, Antenatal care, Immunization barriers

93

Efficacy and safety of dinoprostone formulations in cervical ripening and labor induction: A comparative study

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Background: Cervical ripening and labor induction are critical in managing post-term pregnancies, maternal or fetal complications, and other obstetric indications. Dinoprostone, a synthetic prostaglandin E2 analogue, is widely used in various formulations, including inserts, gels, and tablets. However, limited research directly compares the efficacy and safety of these formulations, necessitating a detailed evaluation to guide clinical decision-making.

Aim: To compare the effectiveness and safety of dinoprostone inserts, gels, and tablets for cervical ripening and labor induction, and to analyze variations in patient characteristics and outcomes across these formulations.

Methods: This retrospective study reviewed 181 women who underwent labour induction at our hospital using dinoprostone insert ($n = 62$), gel ($n = 76$), or tablet ($n = 43$). Patient demographics, including age, BMI, parity, gestational age, initial Bishop score, and induction indications, were analyzed. Changes in Bishop scores and labor outcomes were assessed using descriptive statistics and comparative analysis.

Results: Significant differences were observed in final Bishop scores among the three groups, with the insert group achieving the highest mean score (8.32 ± 2.24), followed by the gel (7.2 ± 1.97) and tablet (6.72 ± 1.25) groups. The tablet group demonstrated the shortest median time to vaginal delivery (16 hours). Vaginal delivery was the predominant mode of birth in all groups. While fetal distress and maternal complications were comparable, tachysystole was more frequent in the insert group.

Conclusions: All three dinoprostone formulations proved effective and safe for cervical ripening and labor induction. Inserts yielded higher Bishop scores but were associated with increased tachysystole, whereas tablets facilitated quicker vaginal delivery. These findings enable clinicians to tailor formulation choice based on individual patient needs. Prospective studies are warranted to validate these results and optimize clinical guidelines.

Keywords: Dinoprostone, Cervical ripening, Labor induction, Obstetrics, Bishop score

94

Effectiveness of carbetocin in preventing post-partum hemorrhage in high-risk women: A prospective clinical audit

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Background: Post-partum hemorrhage (PPH) poses a significant risk to maternal health and places a considerable financial burden on healthcare systems. High-risk women require targeted preventive measures to mitigate PPH effectively. Carbetocin, a uterotonic agent, shows potential in reducing PPH incidence and improving maternal outcomes.

Aim: To evaluate the efficacy of Carbetocin in reducing PPH rates among high-risk women through a prospective clinical audit.

Methods: This audit was conducted in the Obstetrics and Gynaecology Department of a secondary care hospital in Qatar from April to September 2022. High-risk women were identified based on admission risk assessments. Compliance with Carbetocin use and PPH prevalence were analyzed, with comparisons made between Carbetocin and non-Carbetocin (oxytocin) groups. Data from 2823 deliveries were assessed to examine compliance rates, PPH incidences, and maternal outcomes.

Results: Among the study population, 11.41% (322) were identified as high-risk. Carbetocin use compliance improved from 80.36% to 96% during the audit period ($p < 0.05$). PPH rates were significantly lower in the Carbetocin group (1.64%) compared to the oxytocin group (13.64%; $p = 0.0037$). This highlights Carbetocin's effectiveness in reducing maternal morbidity associated with PPH.

Conclusions: The audit underscores Carbetocin's role as a preventive measure in high-risk women, demonstrating reduced PPH rates and enhanced compliance. Regular clinical audits are crucial for optimizing maternal care and ensuring the safe application of evidence-based interventions.

Keywords: Post-partum hemorrhage, Carbetocin, Maternal health, Clinical audit, High-risk pregnancy

Assessment of patients' adherence to oral anticoagulants in Eastern Saudi Arabia: A cross-sectional study

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Background: Oral anticoagulants (OACs) are essential for treatment and prevention of cardiovascular diseases. Despite their importance, adherence to OACs' therapy remains significantly low in the Middle East and Gulf region.

Aim: To measure adherence to OACs among patients who are taking these medications in Alhassa region, Eastern Saudi Arabia.

Methods: The study employed a cross-sectional design to evaluate medication adherence in patients aged 18 and older prescribed warfarin, dabigatran, rivaroxaban, or apixaban at the Prince Sultan Cardiac Centre (PSCC) in Alhassa, Saudi Arabia. Participants provided informed consent, and those who declined were excluded. Data collected included patient demographics, medical conditions, OAC types, and experiences with OACs. Statistical analysis was conducted using SPSS (version 26), with numerical data expressed as means and standard deviations, categorical data as frequencies and ratios, Chi-square tests for comparisons, and multivariate logistic regression for significant factors ($P < 0.05$). Odds ratios (OR) and 95% confidence intervals (CI) were calculated. The study received approval from the PSCC research committee and the University of Birmingham.

Results: The study included 312 participants, comprising 163 females (52.2%) and 149 males (47.8%). Age of participants ranged between 18 and 94 years. OACs were primarily prescribed for valve disease (47.8%) and atrial fibrillation (46.2%). Other indications included stroke (1.9%), deep vein thrombosis (2.6%), and pulmonary embolism (1.6%). Adherence rates to OACs were (34%) among warfarin users and (22%) for those on direct oral anticoagulants (DOACs). Overall adherence to OACs was (29.8%). Moreover, the results indicated that adherence to OACs was significantly impacted by factors including age, gender, duration of treatment, knowledge about OACs, and the frequency of missed doses.

Conclusions: Additional measures are needed to improve adherence to OACs in the study area to ensure better therapeutic outcomes and reduce the risk of complications associated with suboptimal adherence.

Keywords: Oral anticoagulants, Pulmonary embolism, Adherence, Atrial fibrillation

Exploring viewpoints of health care professionals on patients adherence to oral anti coagulants in Eastern Saudi Arabia

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Background: Staying consistent with oral anticoagulants (OACs) is crucial for achieving effective treatment outcomes. Understanding the perspectives of healthcare professionals (HCPs) plays a vital role in improving adherence to these medications. Their insights can help shape policies that maximize therapeutic benefits while reducing the risks linked to non-adherence.

Aims: To explore healthcare professionals (HCPs)'s perspectives on patient adherence to oral anticoagulants (OACs), with a focus on patient education to improve adherence, factors influencing OAC prescribing, and the key determinants of non-adherence.

Methods: The study gathered insights through semi-structured interviews, with the data analyzed using thematic analysis. It took place in Hofuf, within the Alhassa region of Saudi Arabia's Eastern Province. Healthcare professionals from the Prince Sultan Cardiac Centre (PSCC) were randomly selected to take part. The research received approval from the PSCC research committee and the University of Birmingham. All participants provided informed consent before participating.

Results: HCPs reported varying levels of adherence to OACs, noting that adherence was generally higher for direct oral anticoagulants (DOACs) compared to vitamin K antagonists (VKAs). Factors such as education level, forgetfulness, the type of OAC prescribed, and polypharmacy were identified as key contributors to non-adherence. HCPs highlighted that adherence scales are rarely used in routine practice. They confirmed that both VKAs and DOACs are readily available and that patient counselling on OAC use is provided. However, the role of hospital pharmacists is primarily limited to dispensing medications and offering basic counselling, with no systematic use of adherence measurement tools in day-to-day practice.

Conclusion: Future research is recommended to investigate patients' perspectives on improving adherence to OACs. Furthermore, the role of pharmacists could be expanded to include monitoring INR levels.

Keywords: Oral anticoagulants, Pulmonary embolism, Adherence, Atrial fibrillation

103

A 12-year scientometric analysis of research productivity among clinical pharmacy and practice faculty members at Qatar University

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Background: Scientometric analysis is crucial for academic institutions to enhance research activities and productivity by assessing institutional goals, identifying priorities, addressing gaps, and recognizing faculty contributions. Given the advancements in the clinical pharmacy and practice field and Qatar's ever-evolving research infrastructure, it becomes essential to assess the contributions of its clinical pharmacy and practice department to the field and the global research landscape.

Aim: To conduct a scientometric analysis of the research output from a clinical pharmacy and practice department at a Middle Eastern university, focusing on indicators related to both the quantity and quality of contributions.

Methods: Research articles published from 2012 to 2024 were retrieved using Web of Science (via InCites), Scopus (via SciVal), and Google Scholar. The analysis focused on total publications, journal impact, Q ranking, subject area analysis, and collaboration networks.

Results: The 31 faculty members produced 1,266 articles indexed in Web of Science, 1,270 in Scopus, and 1,737 in Google Scholar, with an average of 5 articles annually per faculty member and 41 publications per faculty member over their tenure. Active publication years accounted for 85% of faculty tenures, with a steady annual growth that peaked between 2021 and 2023. Most publications were in Q1 or Q2 journals, with the *International Journal of Clinical Pharmacy and Research in Social and Administrative Pharmacy* being the most commonly targeted outlets. The research output covered diverse subject areas, focusing on pharmacy practice, health education, medication safety, and chronic disease management.

Conclusion: This analysis provides valuable insights into the research quality of the clinical pharmacy and practice faculty, highlighting its progression, and enabling benchmarking against regional and global trends in clinical pharmacy and practice research. These findings assist researchers, academic institutions, and policymakers to advance research initiatives, foster collaborations, and shape informed policies.

Keywords: Research productivity, Scientometric analysis, oductivity

Impact of pharmacist interventions on therapeutic inertia among patients with diabetes in primary care setting: A systematic review

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Background: Despite well-established diabetes prevention and management strategies, therapeutic inertia, defined as the failure to modify therapy despite inadequate glycemic control, remains a global challenge. There are conflicting results in the literature regarding the impact of pharmacist-led interventions in addressing therapeutic inertia within primary care settings.

Aim: This systematic review aimed to evaluate the effectiveness of pharmacist-led interventions in addressing diabetes-related therapeutic inertia in primary care settings, compared to usual care (UC).

Methods: A systematic search of MEDLINE, Embase and Web of Science, supplemented by Google Scholar was conducted from database inception to October 2024. Comparative studies evaluating pharmacist-led interventions targeting diabetes therapeutic inertia in primary care settings were included. Risk of bias was assessed using Risk of Bias 2 and Risk of Bias in Non-randomized Studies of Intervention tools. Furthermore, the interventions were appraised using the Descriptive Elements of Pharmacist Intervention Characterization Tool (DEPICT).

Results: Twenty-two studies across 112 centers involving 29,593 patients were included in this review, consisting of 4 randomized trials, 5 quasi-experimental studies and 13 cohort studies. Pharmacist interventions varied, but most commonly involved medication counseling (77.3%), and recommendations for therapeutic modifications (72.3%). Sixteen studies reported therapeutic inertia-related outcomes, with the intervention arm achieving HbA_{1c} targets earlier (7.9 vs. 12.4 months) and a higher proportion of patients reaching glycemic targets (37% vs. 24%) compared to UC. The intervention group also showed a greater mean HbA_{1c} reduction from baseline (−1.5% vs. −0.5%) compared to UC, with similar hypoglycemia rates between the two groups (5.1 vs. 4.9 episodes over 12 months).

Conclusion: This review highlights the positive impact of pharmacist interventions in reducing diabetes therapeutic inertia and improving glycemic control in primary care settings. Further research with robust designs and

detailed intervention descriptions is needed to better define the role of pharmacists in managing therapeutic inertia.

Keywords: Pharmacist interventions, Therapeutic inertia, Diabetes

105

Therapeutic inertia quantification in diabetes care: How do we measure it?

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Background: Therapeutic inertia, defined as the failure to intensify or de-intensify therapy despite suboptimal glycemic control, is a significant challenge in diabetes management with adverse health consequences and a significant burden on healthcare system resources. Despite that, current research lacks harmonized and effective ways to measure therapeutic inertia, complicating the efforts to address this issue effectively in diabetes care.

Aim: To summarize the approaches used to quantify therapeutic inertia in diabetes care, with the goal of improving clinical management and patient outcomes.

Methods: A search was conducted to identify relevant articles through a MEDLINE (PubMed), Embase, and Web of Science databases from inception until August 2024, employing keywords such as 'therapeutic inertia', 'quantification', and 'diabetes'. Articles focusing on quantifying therapeutic inertia in diabetes care were included. Quantification methods were summarized in terms of formula, scoring, validation, advantages, and limitations.

Results: Current methods for quantifying therapeutic inertia include duration between HbA_{1c} measurements above target levels and/or treatment intensification, proportion of patients with elevated HbA_{1c} levels undergoing treatment intensification within a specified timeframe, glycemic burden, and other metrics. However, these methods have limitations, leading to the development of therapeutic inertia indices such as the norm-based method (NBM), standard-based method (SBM), and American Diabetes Association (ADA) method. These indices aim to describe healthcare providers' practices and link process measures and key glycemic outcomes. Comparative evaluations have shown SBM to be superior in predicting clinical outcomes over time.

Conclusion: Standardizing therapeutic inertia measurement approaches and developing comprehensive interventions are needed to enhance diabetes

management. Future research should assess quantification and interpretation of therapeutic inertia, establish consensus on measurement methods and thresholds, and encourage collaboration among researchers, clinicians, and policymakers to effectively measure and tackle therapeutic inertia.

Keywords: Therapeutic inertia, Diabetes, Quantification

107

Evaluation the awareness and perception toward artificial intelligence (AI) among pharmacy students

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Background: Artificial Intelligence (AI) has significantly changed clinical decision-making and diagnosis procedures. This advancement has made AI possible to therapy monitoring and improve patient outcomes. The gap between the education and AI revolution has grown significantly, consequently educational initiatives should focus on.

Aim: This study aimed to evaluate Artificial Intelligence awareness and perception among pharmacy students.

Methods: This was a cross-sectional study that used a validated Likert scale system that measured AI awareness and perception among pharmacy students at National University, Oman. The data was analyzed by using SPSS version 25. The null hypothesis assumed that pharmacy students have high awareness and positive perception towards AI, while the alternative hypothesis expected that pharmacy students have low awareness and negative perception towards AI.

Results: The results of the internal consistency coefficient of the items of awareness and perception toward AI were high (Cronbach's Alpha > 0.85). The awareness toward AI found that more than 50% of pharmacy students are familiar with the uses of AI and know its importance in scientific research (58.3%). However, more than half of the participants didn't know that AI can support therapeutic diagnosis (50.6%) and the importance of AI in pharmacy education (57%). It was also observed that pharmacy students have a positive perception of the importance of AI in facilitating pharmacy and healthcare professionals' access to information (84.2%) and patients' access to the service (80.8%), whereas 80.8% of the pharmacists considered that AI will raise new social challenges and new ethical challenges (79.5%) in the pharmacy field.

Conclusion: The awareness of AI was low although the perception was positive. This supports the recommendation to establish and improve the knowledge among students toward the role of AI in the health care system and pharmacy education.

Keywords: AI, Pharmacy education, Perception, Education

109

Patients' and primary carers' views on clozapine treatment for schizophrenia: A cross-sectional study in Qatar

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Background: Subjective views of clozapine treatment among patients with schizophrenia in the Middle East and North African MENA Region have not previously been assessed. Globally, few studies have assessed the views of carers on clozapine treatment.

Aim: We aimed to assess the views of patients on clozapine in terms of benefits, side effects, and education received from the clinical team.

Methods: A cross-sectional survey, using a clinician-administered structured interview, of patients with schizophrenia/schizoaffective disorder prescribed clozapine in Qatar, and their primary carers. The participants were asked to rate clozapine against prior their antipsychotic treatment in terms of specific benefits and side effects.

Results: Forty-two patients and 33 carers participated in the study. Of the patients, two thirds were male, approximately half were Qatari with mean age was 33.9 years. Patients and carers rated clozapine as superior to prior antipsychotic treatment on all 7 listed potential benefits. The greatest perceived benefit was improved mood. Patients rated clozapine as less likely to cause extrapyramidal side effects but more likely to cause 18 other potential side effects compared to prior antipsychotics, with the greatest difference being for nocturnal salivation, increased appetite, and constipation. Nearly half of patients (48%) and two thirds of carers (64%) stated that they would have preferred to start clozapine earlier in their illness. Sixty percent of patients and 37% of carers regarded the information that they had received from health professionals on clozapine as inadequate. Generally, there were significantly positive correlation between patients and carers regarding the overall side effects

of clozapine. Likewise, improvement in hearing voices paranoid thoughts correlated with improved quality of life.

Conclusion: Patients and carers appreciate the benefits of clozapine despite its side effects being problematic. The results support clozapine being offered earlier in treatment and services providing more information on clozapine to patients and carers.

Keywords: Clozapine, Patient views, Schizophrenia, Mental health

110

A comprehensive real-world safety profile analysis of the FDA Adverse Event Reporting System (FAERS) data for Lapatinib

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Background: The clinical application of lapatinib, a dual tyrosine kinase inhibitor for breast cancer, has been constrained due to its adverse drug events (ADEs). Additionally, there remains a gap in understanding its real-world safety, particularly concerning rare ADEs that could be serious and/or life-threatening.

Aim: To perform data mining of FAERS to provide an updated and comprehensive analysis of lapatinib's safety, identify and quantify strong safety signals associated with lapatinib that may be serious or life-threatening, and explore the serious outcomes of these ADEs.

Methods: FAERS data from March 2007 to July 2024 were retrieved using OpenVigil software to analyze lapatinib-related ADEs. ADEs were categorized into preferred terms (PTs) and system organ classes (SOCs) via the Medical Dictionary for Regulatory Activities (MedDRA). Descriptive and disproportionality analysis via detection algorithms were conducted to summarize report characteristics and quantify signals.

Results: Out of the 25,506,744 FAERS reports collected, 18,407 PTs identified lapatinib as the primary suspect, resulting in the analysis of 10,959 ADEs signals. Females (77.9%) and those aged 18–64 (45.38%) were predominantly reported. A total of 16 SOCs were affected, with 155 lapatinib-related PTs identified, 115 of which were significantly disproportionate, including 57 potentially novel PTs. Most signals were of medium intensity. Although gastrointestinal (GI), and dermatological disorders were the most commonly reported, skin and nail ADEs showcased a stronger association with lapatinib, with diarrhea

being the only strong significant GI signal detected. Cardiac events were less commonly linked with lapatinib. Some newer significant signals found include cheilitis, erysipelas, and cholestatic jaundice, warranting further investigations. The most common outcomes reported were initial or prolonged hospitalization, followed by death and life-threatening events.

Conclusions: This study underscores the importance of monitoring for lapatinib-related ADEs in clinical settings to enhance the quality of life and minimize the risk of life-threatening outcomes.

Keywords: Lapatinib, FDA, Adverse drug reactions, Patient safety

111

Psychological impact of acne on depression levels among Lebanese adolescents: A cross-sectional study

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Background: *Acne vulgaris* is a widespread chronic inflammatory skin condition with a high prevalence among adolescents affecting approximately 9.4% of the global population. Treatment strategies typically address a variety of factors associated with pathogenesis as bacterial growth, proliferation, sebaceous gland hyperactivity leading to excessive sebum production and clogged pores. However, in many cases treatment protocols neglect the psychological impact of acne, particularly the development of depression

Aim: This study aimed to assess the range of factors influencing depression levels among Lebanese adolescents with acne, using the Patient Health Questionnaire-9 (PHQ-9).

Methods: A cross-sectional observational study was conducted among 284 Lebanese patients. The questionnaire was validated and approved by the Lebanese International University board. Each patient responses were summed to yield a score which will be categorized into minimal depression (1-4), mild depression (5-9), moderate depression (10-14), moderately severe depression (15-19), and severe depression (20-27). The higher the score the higher the risk of having more severe depression.

Results: The study involved 284 Lebanese adolescents who met the inclusion criteria. The mean age of participants was 16.31 years (± 2.81), and the mean age of acne onset was 12.05 years (± 2.86). The average PHQ-9 score for

depression was 10.64, indicating moderate depression levels. Depression scores were higher among females (mean score = 11.49) compared to males (mean score = 10.57). Moreover, it is evident to highlight those participants with cystic acne reported significantly higher depression scores (mean score = 12.49).

Conclusion: The study demonstrates a strong association between acne and elevated depression levels among Lebanese adolescents. This emphasizes the need for integrating mental health support into acne treatment regimens. Future studies should be directed toward assessing the relationships between acne and dermatological conditions as well as the effectiveness of integrated treatment approaches.

Keywords: Acne, Depression, Adolescents, Lebanon

114

The specialized competency framework for community pharmacists in Lebanon: A validation and evaluation study

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Background: Community pharmacists in Lebanon play a key role in patient care and public health, yet a nationally validated competency framework tailored to their practice has been lacking. As pharmacy practice evolves, defining clear competencies is essential to support professional development, enhance practice standards, and guide continuing education.

Aim: This study aimed at developing and validating the Lebanese specialized competencies framework for community pharmacists (SCF-CP) as a tool to transform community practice and support the professional development and career progression of community pharmacists.

Methods: Content validity was assessed and improved through a team of experts. After a thorough literature review and utilizing the Delphi

technique, six domains were defined in the framework, with their respective competencies and behaviors. A cross-sectional study was then carried out using an online questionnaire. The snowball technique was applied to reach community pharmacists across all the Lebanese governorates.

Results: The final sample included 512 community pharmacists. The construct validity of the framework was confirmed by factor analysis. The Kaiser–Meyer–Olkin measures of sampling adequacy were satisfactory for all models ranging from 0.500 to 0.956 with a significant Bartlett’s test of sphericity ($P < 0.001$). The internal consistency of all competency domains was confirmed by Cronbach’s alpha, with values ranging from 0.803 to 0.953. All competencies were significantly correlated with their respective domains ($P < 0.001$), and all domains were significantly correlated with each other and with the framework ($P < 0.001$). The participants declared being competent in all domains relating to fundamental skills, safe and rational use of medicines, pharmacy management, professional skills, public health fundamentals, and emergency preparedness and response, with some exceptions, such as compounding, management, and emergency preparedness. A higher declared competency level was associated with having more experience and receiving more than 50 patients per day.

Conclusion: Findings demonstrate that the SCF-CP framework is a valid and reliable tool. This framework could help assess the minimum competencies that community pharmacists should possess or acquire and direct initial and continuing education for better practice.

Keywords: Community pharmacy, Lebanon, Pharmacist, Specialized competency

115

Instructional educational games in pharmacy experiential education: A quasi-experimental assessment of learning outcomes, students’ engagement and motivation

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Background: Active learning strategies, such as educational games, have gained attention in health professions education for their potential to enhance student engagement, motivation, and knowledge retention. In pharmacy education, incorporating gamification into practice-based learning may support the achievement of intended learning outcomes and competency development. However, evidence on its effectiveness in pharmacy practice experience courses remains limited.

Aim: This study aimed to determine the impact of implementing instructional educational games on attaining the intended learning outcomes mapped with the competencies of a pharmacy practice experience course, and to assess students' attitudes towards motivation and engagement in this active learning activity.

Methods: This was a quasi-experimental study that utilized a pre-test-post-test for the research groups. Different gaming platforms were used to create different questions that help students memorize medications' brand names, and acquire advanced community knowledge and skills. The attainment of the intended learning outcomes was assessed and compared between the experimental and control groups.

Results: A total of 233 students were enrolled in the study. The experimental group had a significantly higher total post-test average compared to the control group (Beta = 7.695, 95% CI = 4.964–10.425, $P < 0.001$). The experimental group had also significantly higher averages of competency domains related to foundational knowledge (Beta = 1.471, 95% CI = 0.723–2.219, $P < 0.001$), pharmaceutical care (Beta = 1.650, 95% CI = 0.673–2.627, $P < 0.001$), essentials to practice and care (Beta = 1.838, 95% CI = 0.626–3.050, $P < 0.003$), and approach to practice and care (Beta = 2.736, 95% CI = 1.384–4.088, $P < 0.001$) averages. The experimental group reflected positive attitudes toward gamification engagement and motivation, with greater than 60% of the students recommend engage educational games to be part of the course.

Conclusion: Incorporation of educational games into pharmacy practice experiences resulted in better learning outcomes. This kind of active learning appears to be acceptable and motivational for students, and is recommended for further research in didactic courses in the pharmacy curriculum.

Keywords: Educational games, Instructional games, Experiential education, Learning outcomes, Competencies

116

Cost-effectiveness analysis of genotype-guided optimization of major depression treatment in Qatar

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Background: Pharmacogenetic testing (PGx) enhances the effectiveness and safety of antidepressant treatment for moderate to severe major depressive disorder (MDD) by identifying genetic differences that affect how medications are metabolized, allowing for tailored treatment plans.

Aim: This study evaluated the cost-effectiveness PGx-guided strategy compared to standard of care (SoC) in managing MDD.

Methods: We structured a two-stage decision tree for a short-term follow-up of six weeks, as well as a lifetime Markov model with three-month cycles. The analysis compared the current SoC with a PGx-guided therapy strategy using multi-gene panel for adult people with MDD. Clinical outcomes and utility values were sourced from published studies, while healthcare cost data was obtained from Hamad Medical Corporation, Qatar. The short-term incremental cost-effectiveness ratio (ICER) was evaluated based on treatment response both with and without side effects and relapse. The long-term ICER focused on quality-adjusted life years gained (QALYs) and years of life saved (YLS).

Results: Implementing PGx-guided therapy led to cost savings of Qatari Riyal 2,289 (95% confidence interval (CI): -22,654 to 26,340). In the short term, it achieved higher response rates without side effects and relapse (mean difference 0.10, 95% CI: 0.09-0.15) and higher response rates with or without side effects and without relapse (mean difference 0.05, 95% CI 0.04-0.06). Over the long term, PGx testing resulted in an average of 0.13 YLS and 0.06 QALYs gained per person, alongside cost savings of QAR 46,215 (95% CI: -15,744 to 101,758). Sensitivity analyses confirmed the stability of the model's findings.

Conclusion: The implementation of PGx testing to guide antidepressant therapy not only improved health outcomes for the population but also significantly reduced costs in Qatar.

Keywords: Cost-effectiveness analysis, Genotype-guided optimization, Major depression, Qatar

Dihydropyrimidine dehydrogenase deficiency (DPYD) genotyping-guided fluoropyrimidine-based adjuvant chemotherapy for breast cancer. An economic evaluation

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Background: The risk of severe adverse drug reactions (ADRs) is increased for those with dihydropyrimidine dehydrogenase deficiency (DPYD).

Aim: The objective of this study was to examine the cost-effectiveness of offering DPYD pharmacogenetic-guided care, where genetic testing informs personalized dosing versus the current standard of care (SoC) for local or metastatic breast cancer patients in Qatar.

Methods: We developed a two-stage decision analysis, with an analytic tree model over a 6-month, followed by a life-table Markov model over a lifetime horizon. We compared the SoC with the alternate strategy of DPYD genetic screening in patients living in Qatar with local/metastatic breast cancer and eligible for adjuvant fluoropyrimidine therapy. Clinical outcomes and utilities were obtained from published studies, while health-care costs were estimated from Hamad Medical Corporation, Qatar. The short-term outcome included the incremental cost-effectiveness ratio (ICER), defined as cost per success (survival without grade III/IV ADRs) at six months. The long-term outcome was the ICER, defined as cost per quality-adjusted life year (QALY) gained, with a 3% annual discount rate. The study adopted a public healthcare perspective. Sensitivity analyses were conducted to explore the impact of key input parameters on the robustness of the model.

Results: In the short-term model, at its base case, DPYD genomic screening was dominant over SoC with a mean cost-saving of QAR 84,585 (95% confidence interval (CI), 45,270–151,657). This cost saving reflects the overall economic benefits associated with the implementation of DPYD genomic screening. In the long-term model, compared to the current SoC, DPYD genetic screening would result in an ICER of QAR 21,107 (95% CI –59,382–145,664) per QALY gained.

Conclusion: Based on our model, implementing DPYD genetic screening to detect DPYD mutations in breast cancer patients before therapy initiation seems to be a cost-saving and cost-effective strategy in Qatar.

Keywords: Dihydropyrimidine dehydrogenase deficiency, Adjuvant chemotherapy, Genotype guided therapy, Economic evaluation

118

The next decade of cardiovascular disease burden in Qatar, a Gulf Cooperation Council Country: Projections from 2024 to 2033

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Background: Cardiovascular diseases (CVD) are a great public health challenge in Qatar, with significant impacts on long-term population health and societal costs.

Aim: We aimed to forecast the health and economic burden of the CVD in Qatar from 2024 to 2033, from both healthcare and societal perspective.

Methods: A validated two-stage dynamic model was developed. The model spans a 10-year period and focuses on individuals aged 40-79. The CVD incidents (i.e. myocardial infarction [MI], stroke) were estimated using the 2013 Pooled Cohort Equation, while recurrent events were obtained from the global REACH registry. The model outcomes included fatal and non-fatal MI and stroke, years-of-life lived (YLL), quality-adjusted life years (QALYs), total direct costs, and total productivity loss costs. Utility and cost inputs were derived from published sources. Outcomes were discounted at a rate of 3% per annum. Calibration and validation were performed to ensure model accuracy. A multivariate sensitivity analysis was also conducted.

Results: By 2033, there will be 271,260 non-fatal MI events (95% confidence interval [CI] 271,249-271,277), 258,892 non-fatal strokes (95%CI 258,858-259,094), and 20,413 CVD deaths (95%CI 20,405-20,429). The cumulative YLLs and QALYs were 13,806,845 (95% CI 13,802,149-13,811,541) and 10,655,665 (95%CI 10,652,720-10,658,611), respectively. The direct costs were QAR71.14 (95%CI QAR70.62-71.66) billion, and the productivity loss costs were estimated to surpass QAR108.12 (95%CI QAR106.88-109.36) billion. The exchange rates used were based on 2024 values (1QAR = 0.27US\$).

Conclusions: This study offers valuable insights into the projected burden of CVD in Qatar, highlighting the need for effective preventive strategies to reduce risk.

Keywords: Cardiovascular disease, Qatar, Disease burden

119

Unraveling the future productivity burden of cardiovascular disease in Qatar: Investigating the modifiable risk factors control in type 2 diabetes

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Background: Insufficient risk factor control can lead to a loss of millions of productivity-adjusted life years (PALYs).

Aim: We aimed to assess the productivity burden of cardiovascular disease (CVD) in type 2 diabetes (T2D) and examine the potential advantages of enhancing the control of modifiable CVD risk factors in Qatar.

Methods: Models were developed to quantify the productivity burden, in terms of PALYs, of CVD in Qataris with T2D, aged 40–65 years, from 2024 to 2033. The financial value of PALYs was determined based on the gross domestic product (GDP) per full-time worker (i.e. US\$80,573). The base-case model estimated the productivity burden of CVD, and interventional scenarios were simulated to assess potential gains resulting from improved control of modifiable risk factors, including reduced incidence of T2D, lower systolic blood pressure (SBP), decreased number of smokers, and improved total cholesterol. All costs and outcomes were discounted at an annual rate of 3%.

Results: The base-case analysis projected that CVD in T2D would result in an estimated 2,096,536 PALYs (95% confidence interval, 1,689,272–2,182,939), contributing US\$225.46 (95%CI, 1,689,272–2,182,939) billion to the country's GDP. However, implementing interventions to decrease the T2D incidence, lower SBP, reduce the number of smokers, and improve the total cholesterol could potentially yield gains of 200,408, 198,173, 194,725, and 113,462 PALYs, respectively. These improvements would also lead to economic gains of US\$20.01 billion, US\$20.17 billion, US\$19.78, and US\$12.79 billion, respectively.

Conclusions: Implementing interventions that prioritize risk factor control and prevention of CVD can help enhance overall productivity in the country.

Keywords: Cardiovascular disease, Type 2 diabetes, Risk factors, Qatar

122

Association of plasma levels of Sestrin2 with adiposity and metabolic function indices in healthy and diabetic subjects from Qatar Biobank

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Background: Despite the accumulating evidence from cellular and animal studies, the role of circulating Sestrin2 (SESN2), a stress-inducible antioxidant protein, in human cardiometabolic health remains largely unexplored.

Aim: The current study aimed to investigate the association between circulating SESN2 and cardiometabolic risk factors in healthy and diabetic individuals.

Methods: This cross-sectional study leveraging data and plasma samples from the Qatar Biobank investigated the relationship between plasma SESN2 levels and various cardiometabolic indices in 326 healthy and 518 diabetic subjects. The study protocol was approved by the Institutional Review Boards (IRB) of Qatar Biobank (QBB) (#Ex-2021-QF-QBB-RES-ACC-00049-0173) and Qatar University (#QU-IRB 1624-E/21).

Results: The study found that SESN2 levels were significantly lower in diabetic individuals compared to healthy controls (5.49 vs 8.25, $p < 0.001$). In the healthy cohort, higher SESN2 levels were associated with a favorable metabolic profile, indicated by lower odd ratios (OR) of high HbA1c (OR: 0.33), HOMA-IR (OR: 0.58), visceral adiposity index (OR: 0.46), lipid accumulation product (OR: 0.49), atherogenic index of plasma (OR: 0.42) and metabolic syndrome (OR: 0.23). Conversely, in the diabetic cohort, higher SESN2 levels were paradoxically linked to increased triglycerides (OR: 1.57), the product of triglyceride glucose and waist circumference (OR: 1.8), body fat (OR: 1.72), waist size (OR: 1.82), waist-to-hip ratio (OR: 1.96) and metabolic syndrome (OR: 1.48).

Conclusions: These findings suggest that SESN2 may play a complex and context-dependent role in metabolic regulation, potentially serving as a protective factor in healthy individuals but contributing to metabolic dysfunction in the context of established diabetes. Further research is needed to elucidate the underlying mechanisms and implications for targeted interventions.

Keywords: Adiposity, Biomarker, Diabetes mellitus, Metabolic syndrome, Sestrin2

Prescription pattern of ocular hypotensive therapy and knowledge of patients with advanced glaucoma

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Background: Glaucoma is a chronic progressive condition and the leading cause of irreversible blindness globally. Improved patient knowledge enhances adherence to therapy and reduces complications.

Aim: To determine the prescription pattern of anti-glaucoma medications and patients' knowledge about glaucoma and its therapy.

Methods: A retrospective cross-sectional study, which included glaucoma patients (n = 40), was conducted at Magrabi Private Hospital in Riyadh from January to May 2024. The study was approved by the IRB committee at KSU (**E-24-8581**). Medical files were examined to retrieve current medications, and ophthalmic data, including intraocular pressure (IOP), Optical Coherence Tomography (OCT), cup-to-disc ratio, and visual field test. Patients were interviewed at the recent visit and answered Gray's quiz.

Results: Most participants were males (80%) and non-Saudi (60%), with a mean age of 50 ± 14.3 years. Of them, 42% reported a family history of glaucoma, and 32% had primary open-angle glaucoma. About 80% had an increased cup-to-disc ratio and retinal thinning. Carbonic anhydrase inhibitors were the most prescribed medication (62.5%), either as monotherapy or combined with beta-blockers or alpha agonists. Fixed combinations of two and three medications were prescribed for 37.5% and 10% of patients, respectively, while 52.5% received monotherapy. The median knowledge score was 12. Nine (22.5%) of patients did not know what glaucoma is, while six patients (15%) did not know which eye part can become damaged by glaucoma. Seventeen (42.5%) were unaware of what part of vision glaucoma affects the most, and 40% patients were unaware of the hereditary nature of glaucoma. Fifteen patients (37.5%) did not know the name of their prescribed drops. Gray's score was independent of drug class, combination type, and administration frequency.

Conclusion: Patients with advanced glaucoma require educational support to improve their understanding of the illness, increase self-efficacy, and effectively adhere to the treatment protocol.

Keywords: Prescription patterns, Advanced glaucoma, Ocular hypotensive therapy

125

Enhancing practical learning in pharmacy education: A kinesthetic approach using clinical pharmacist menu software

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Background: A key learning area for pharmacy students is using appropriate medicines and applying knowledge to optimise patient care via medication therapy. Case-based learning, often involving paper-based care plans, helps students identify, resolve, and prevent drug-related problems. However, students often lack real-world experience and face challenges in decision-making, such as identifying drug-related problems, selecting appropriate medications, and determining appropriate dosages, resulting in low performance.

Aim: This study aimed to enhance the students learning through Clinical Pharmacist Menu Software and to evaluate its acceptability, usability, and effectiveness.

Methods: The CPM software was designed to incorporate CDS tools, utilizing a product from Oracle Co., USA. A prospective, questionnaire-based comparative study was conducted among 20 undergraduate pharmacy students. To measure the changes in students' knowledge and decision-making skills for appropriate medication use, the students were initially asked to formulate a care plan using traditional paper-based methods. Later, they were asked to formulate a care plan using the CPM software. The feedback of all students was recorded after each activity through a validated questionnaire. CDS systems were analyzed for their potential to improve student learning and their effectiveness in identifying and resolving medication errors

Results: The findings of the study indicated that CPM was able to support student learning in optimizing the medication therapy, showing 48% improvement in response to questionnaire $p < 0.0001$ and that the students found using CPM was clear, usable, easy to follow, and helps in enhancing knowledge. The number of clinical pharmacy interventions using the CPM software was higher than that of the PCP (7% vs. 4.5%).

Conclusion: CPM has the potential to significantly improve student kinesthetic learning, knowledge, and decision-making skills in medication therapy. CPM could be a valuable tool for enhancing pharmacy students' ability to optimize medication therapy

Keywords: Pharmacy education, Kinesthetic learning, Medication therapy management, CDS tool

The association between plasma Sestrin2 levels and the risk of acute ischemic stroke: A case cohort study

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Background: The effective management of acute ischemic stroke (AIS) requires sensitive biomarkers for rapid and accurate diagnosis and prognosis. Oxidative stress is a key factor in AIS pathogenesis, underscoring the need for biomarkers linked to cellular stress responses. Sestrin2, a stress-inducible anti-oxidant protein, is upregulated in response to oxidative stress and may have a protective role.

Aim: This study evaluated the association between plasma Sestrin2 levels, total antioxidant capacity (TAC), and nitric oxide (NO) with the risk of AIS.

Methods: The study included 186 AIS patients and 30 healthy controls. Circulating levels of Sestrin2, TAC, and NO were measured within 48 hours of onset, and subsequently at 5 days, 30 days, and 1-year post-acute events.

Results: At the onset of acute events, plasma levels of Sestrin2, TAC, and NO were significantly lower in AIS patients compared to healthy controls: Sestrin2 (8.383 ± 7.39 vs 1.434 ± 3.57 ; $p < 0.001$), TAC (2.279 ± 0.326 vs 1.88 ± 0.42 ; $p < 0.001$), and nitrite/nitrate (65.951 ± 44.07 vs 53.5 ± 47.9 ; $p = 0.045$). Sestrin2 levels remained lower in AIS patients at 5- and 30-days following the acute events compared to controls. In contrast, plasma NO levels increased significantly on day 5 compared to onset levels ($z = -2.471$, $p = 0.013$). Multiple logistic regression showed that being male, diabetic, having a high National Institutes of Health Stroke Scale (NIHSS) score, and the presence of small vessel disease (SVD) all increase the risk of AIS while being from the Middle East decreases this risk. Patients in the second tertile for Sestrin2, third tertile for TAC, or second tertile for NO had reduced AIS risk, with adjusted odds ratios of 0.123 ($p < 0.001$), 0.327 ($p = 0.008$), and 0.382 ($p = 0.24$), respectively.

Conclusions: These findings suggest that Sestrin2 plays a protective role against AIS and its recurrence.

Keywords: Sestrin2, Stroke, Ischemia, Nitric oxide

129

Safety of high-dose versus standard-dose influenza vaccination on long term residents admitted in Rumailah hospital: A retrospective study

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Background: Influenza infection is a major cause of illness and death globally, contributing to a significant health burden. Older adults and individuals with frailty or chronic diseases often experience suboptimal immune responses to standard flu vaccines, leading to the development of high-dose inactivated influenza vaccine (HD-IIV), which is intended to boost immunity.

Aim: This study aimed to evaluate the safety, tolerability, and adverse event profile of high-dose versus standard-dose inactivated influenza vaccines among long-term care residents.

Methods: A retrospective study was conducted to assess adverse events, tolerability, and safety profile of the influenza vaccines in long term care residents. Additionally, the study sought to identify gaps in the prescribing process to provide optimum care for the residents. We identified 160 residents who received either HD-IIV or standard-dose inactivated influenza vaccine (SD-IIV), using Cerner system and the data was collected through a standardized collection sheet.

Results: Out of 148 residents reviewed between October to December 2024, 64% were males aged ≥ 60 years (57%) with multiple comorbidities. Prescription compliance in accordance with Ministry of Public Health standards of Qatar, was 91% vs. 88% between HD-IIV and SD-IIV groups. Within 7 days of vaccination, 21% and 10% of residents in the HD-IIV and SD-IIV groups, respectively, experienced at least one side effect. 53% of residents in the HD-IIV group experienced pain, compared to 50% in the SD-IIV group. Previously immunized residents experienced fewer side effects than those who were not (16% vs. 20%). Diabetic residents receiving HD-IIV had a higher incidence of side effects (24%) compared to those with coronary artery disease (21%).

Conclusion: Compared to SD-IIV, HD-IIV appeared to have a higher rate of side effects among long-term residents. Moreover, the findings could help refine vaccination strategies to ensure better protection and improving overall health outcomes.

Keywords: Influenza vaccine , Long-term care, Adverse events, Safety, High-dose inactivated vaccine (HD-IIV)

Impact of medicines price deregulation on access to oral antidiabetic medicines in Pakistan's private pharmacies: A WHO/HAI-based study

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Background: Pakistan, with the highest global prevalence of diabetes, critically requires equitable access to affordable anti-diabetic medicines. In early 2024, the prices for medicines not on the National Essential Medicines List (NEML) were deregulated necessitating the evaluation of this policy's effect on medicine access.

Aim: To evaluate access to oral diabetes medicines and to analyze the effect of the recent price deregulation policy on their prices and affordability in Pakistan.

Methods: A WHO/HAI methodology-based survey was conducted in 30 private pharmacies across 6 cities, to obtain prices and availability data of 30 Oral anti-diabetics (OADs) including lowest-priced generic (LPG), highest-priced generic (HPG), Originator brand (OB), categorized by NEML enlistment and formulation types. Published and surveyed pre-post-deregulation prices were compared. Difference-in-differences (DiD) analysis was employed to determine the policy's effect on prices, with NEML medications as the control group. Affordability for the lowest-paid employee to purchase the monthly standard treatment, and percentage availability of medicines were calculated and compared using non-parametric statistical tests.

Results: The median MUPs of OADs rose from PKR 18.29 to PKR 20.70 for LPGs, PKR 24.29 to PKR 32.32 for HPGs, and PKR 37.02 to PKR 39.41 for OBs post-deregulation. The DiD analysis confirmed significant price increments among OBs ($p = 0.004$), HPGs ($p = 0.02$), and LPGs ($p = 0.01$). The DiD-interaction term showed a price rise for non-NEML LPG ($0.77, p = 0.35$) but a relative price reduction for non-NEML HPGs ($-0.20, p = 0.95$) and OBs ($-10.85, p = 0.05$) post-deregulation. The prices of single active ingredient OADs and fixed-dose combinations were increased significantly. The affordability reduced by 19% ($p < 0.01$). Most of the surveyed medicines had availability from 47% to 97%, post-deregulation.

Conclusion: The study highlights the protective role of NEML enlistment against price increases. The diverse market responses to price deregulation

underscore the need for careful market price tracking and a critical policy adaptation

Keywords: Medicines price deregulation, Medications access, Oral antidiabetic medicines, Pakistan

135

Comparative analysis of adverse events following COVID-19 vaccination: Insights from a retrospective study in Qatar

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Background: The COVID-19 pandemic has profoundly affected global health, with millions of cases and deaths worldwide. Vaccination is critical in controlling virus transmission and reducing severe outcomes. However, the rapid development and deployment of COVID-19 vaccines necessitate robust systems to monitor vaccine safety, particularly adverse events.

Aim: This study aimed to compare the incidence and types of adverse events reported with different COVID-19 vaccines (AstraZeneca, Moderna, Pfizer, and Pfizer Paediatric) across doses to identify patterns in local and systemic reactions, including severe events like anaphylaxis.

Methods: A retrospective cross-sectional analysis was conducted using Electronic Health Records from Qatar's Primary Health Care Corporation, including individuals aged six months and older. Data on adverse events were collected via the "COVID-19 Post Vaccine Assessment Form," documenting local and systemic symptoms. Statistical analyses using IBM SPSS evaluated relationships between vaccine type, dose, and adverse event incidence.

Results: Among 121,100 patients, 28,700 (22%) reported at least one adverse event. Moderna had the highest symptom prevalence after the second dose (33%), while AstraZeneca showed the highest rates after the third dose (96.9%), with injection site pain being most common (55%). Anaphylaxis was most frequent following Pfizer Paediatrics' first dose (0.9%), while the lowest symptom rates occurred with Pfizer Paediatrics' third dose (0.5%). Systemic symptoms, including fever and fatigue, were frequently reported across vaccines.

Conclusion: Significant differences in adverse event profiles exist among COVID-19 vaccines and doses. Moderna and AstraZeneca were linked to higher local and systemic symptom rates, especially AstraZeneca's third dose. Pfizer Paediatric demonstrated fewer adverse events overall, though severe reactions like anaphylaxis were notable. These findings underscore the importance of continuous post-vaccination monitoring to enhance vaccine safety and guide administration strategies.

Keywords: Adverse events, COVID-19 vaccination, Qatar

137

An evaluation of pre–post antimicrobial stewardship intervention studies utilizing the Kirkpatrick model across healthcare education levels: A scoping review

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Background: Antimicrobial stewardship (AMS) education is crucial for fostering rational antibiotic use among healthcare professionals (HCPs) and students. However, the effectiveness of AMS educational interventions in achieving behavioral changes and clinical outcomes remains unclear. This study systematically evaluates AMS education interventions using the Kirkpatrick Model to assess their impact across reaction, learning, behavior, and results levels.

Aim: To examine the effectiveness of AMS educational interventions by mapping their outcomes to the Kirkpatrick model and identifying gaps and future directions for optimizing AMS education.

Methods: A scoping review was conducted using the Joanna Briggs Institute (JBI) framework. Nine databases were systematically searched for studies published between January 2010 and December 2024. Eligible studies employed a pre–post-intervention design targeting healthcare professionals or students. Outcomes were categorized according to the four Kirkpatrick levels: reaction, learning, behavior, and results.

Results: A total of 63 studies were included, mapping 192 outcomes across the Kirkpatrick levels: reaction (23.96%), learning (26.56%), behavior (26.04%), and results (23.44%). Most studies (76%) targeted HCPs, emphasizing behavior (Level 3) and clinical results (Level 4), whereas undergraduate students (24%) primarily focused on knowledge acquisition (Level 2). Didactic

methods (59%) were effective for knowledge transfer but had limited behavioral and clinical impact. Online modules (27%) provided flexibility but required interactive elements for sustained behavioral changes. Multi-faceted approaches (14%) demonstrated the highest potential for comprehensive outcomes but were underutilized.

Conclusions: AMS educational interventions effectively enhance knowledge and engagement but require improvements in behavioral and clinical outcomes. Multi-faceted approaches integrating didactic, online, and interactive strategies are needed to bridge educational gaps. Future research should focus on designing hybrid interventions, incorporating rigorous study designs, and evaluating long-term impacts on patient outcomes to strengthen AMS education and combat antimicrobial resistance.

Keywords: Antimicrobial stewardship, Education, Kirkpatrick Model

139

Evaluating home use patterns and adherence to chronic cardiovascular medications

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Background: Chronic cardiovascular diseases are among the leading causes of morbidity and mortality globally. Effective management of these conditions heavily relies on the consistent and proper use of prescribed medications at home.

Aim: This study aimed to evaluate the patterns of chronic cardiovascular medication use and self-reported adherence among patients managing their conditions at home.

Methods: A cross-sectional study was conducted, involving participants diagnosed with chronic cardiovascular conditions and prescribed at least one long-term cardiovascular medication. Data were collected using a structured questionnaire and adherence assessment using ARMS, a validated self-reported scale.

Results: Patterns of medication use indicated a high prevalence of monotherapy (55%), with a minority on combination therapy regimens. Antihypertensives (40%) and statins (30%) being the most commonly used medications. The study found that 78% of participants adhered to their medication regimens. Commonly reported barriers to adherence included forgetfulness (35%), side effects (20%), and financial constraints (15%). Polypharmacy (≥ 3

medications) was observed in 20% of cases, associated with lower adherence rates (55%).

Conclusion: The findings highlight the need for targeted interventions to address non-adherence among patients with chronic cardiovascular diseases.

Keywords: Cardiovascular medications, Chronic cardiovascular disease management, Home, Medication adherence, Patient barriers

140

Assessment of antibiotic prescribing practices in the management of neonatal sepsis

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Background: Neonatal sepsis remains a leading cause of morbidity and mortality in newborns worldwide. Prompt and appropriate antibiotic therapy is critical for successful management. However, variability in prescribing practices can lead to suboptimal outcomes, including antimicrobial resistance and treatment failure.

Aim: This study aimed to evaluate antibiotic prescribing practices for the management of neonatal sepsis.

Method: A retrospective study was conducted in the neonatal intensive care unit of a private institute in the UAE. The medical records of neonates diagnosed with sepsis and admitted over a 4-year period were reviewed. Data collected included clinical and demographic factors, as well as neonatal parameters such as sex, gestational age, birth weight, clinical condition at admission, duration of hospital stay, and antibiotic exposure. Maternal factors such as maternal age, obstetric history, perinatal complications, and risk factors like fever or prolonged rupture of membranes were also analyzed. Sepsis-related data included pathogen identification, antibiotic susceptibility patterns, and risk factors for sepsis. Statistical analysis was performed to identify patterns of antibiotic prescribing.

Results: Among the 25 cases reviewed, empirical antibiotic therapy was initiated in all cases, with 88% receiving treatment within the first hour of diagnosis. The combination of ampicillin and gentamicin was the most commonly prescribed regimen, used in 72% of cases. Broad-spectrum antibiotics, such as meropenem, were used in 16% of cases, primarily in neonates with severe clinical presentations. All antibiotic regimens were adjusted based

on culture and sensitivity results when available. The average course length of treatment was 7–10 days.

Conclusion: Standardized protocols and continuous medical education are essential to improve prescribing practices and ensure better clinical outcomes.

Keywords: Antibiotic Prescribing Practices, Guideline Adherence, Infection Control, Neonatal Sepsis

143

Evaluation of antibiotic prescribing patterns in intensive care units

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Background: Antibiotics play a crucial role in managing infections in critically ill patients admitted to intensive care units (ICUs). Understanding prescribing patterns is essential to optimize therapy and mitigate risks such as antimicrobial resistance and treatment failures.

Aim: This study aimed to evaluate antibiotic prescribing patterns in ICU settings.

Method: A retrospective cross-sectional study was conducted at a tertiary care facility in the UAE. Medical records of ICU patients admitted between January 2023 and December 2024 were reviewed. Data collected included patient demographics, diagnoses, types of antibiotics prescribed, routes of administration, and durations of therapy. Statistical analysis was performed to identify trends in prescribing patterns.

Results: A total of 58 antibiotics were prescribed for 20 ICU patients, with an average of 2.9 antibiotics per patient. Empirical antibiotics were initiated for all patients upon ICU admission. Piperacillin-tazobactam was the most commonly prescribed antibiotic, accounting for 45% of all prescriptions, followed by meropenem at 30%. The average duration of antibiotic therapy was 7 days.

Conclusion: The prescribing patterns indicate frequent use of multiple antibiotics, highlighting the need for routine evaluations of therapy to ensure appropriateness and optimize outcomes.

Keywords: Antibiotic Prescribing Patterns, Antimicrobial Resistance, Infection Management, Intensive Care Units

145

Implementing smart infusion pumps with drug libraries: Enhancing medication safety at cardiac care facility

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Background: Intravenous drug administration requires meticulous precision to prevent medication errors that can lead to severe complications or fatalities. Intravenous infusion errors remain a critical challenge, where medication errors contribute to approximately 400,000 to 500,000 annual deaths. Contributing factors include insufficient training, lack of standardized protocols, and outdated technology.

Aim: To implement a comprehensive drug library in smart infusion pumps to enhance patient safety and optimize medication administration protocols.

Method: A quality improvement project was initiated in June 2023 using a Plan-Do-Study-Act (PDSA) cycle to analyse and improve infusion pump use. Key interventions included staff education, protocol standardization, and phased implementation of smart infusion pumps. Critical care units piloted the system on June 1, 2024.

Results: Post-implementation, no medication errors were reported in intensive care units, reflecting improved safety and efficiency. The initiative addressed knowledge gaps, established standardized drug dilution libraries, and enhanced protocol adherence through continuous staff training and audits. High nursing turnover and outdated equipment were mitigated by systematic upgrades and integration of smart infusion technologies.

Conclusion: Following ICU success, plans are to expand the project to include pre-mixed syringes prepared in clean room environments. This approach aims to reduce contamination, prevent IV-related infections, and minimize Look-Alike, Sound-Alike medication errors, further improving medication safety. The implementation of smart infusion pumps with drug libraries at represents a significant advancement in clinical safety and quality care. Aligning with ISMP Best Practice 8, the project standardized drug administration, reduced errors, and enhanced workflow efficiency.

Keywords: Smart infusion pumps, Medication safety, ICU, Medication error

146

Evaluating perceptions of feedback: Insights from undergraduate students

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Background: Regular evaluation of student academic needs is crucial for adapting improved teaching methods. Feedback plays a vital role, especially in skill-based fields like medicine. Research suggests a need for a more robust approach to studying how feedback processes work.

Aim: This study investigated how undergraduate medical and pharmacy students perceive feedback, the factors influencing their engagement, and the impact of these perceptions on their learning.

Methods: A mixed-methods approach was employed, combining surveys and semi-structured interviews. Two hundred and fifty undergraduate students from all academic years at medical and pharmacy colleges were included in the study after obtaining informed consent. A structured questionnaire covering various dimensions of feedback, including timeliness, constructiveness, clarity, and the perceived impact of feedback on academic and clinical performance, was administered. Semi-structured interviews were conducted with students until data saturation to gather more in-depth insights into their experiences with feedback. Survey data were analyzed using descriptive statistics. Interview data were coded and thematically analyzed.

Results: Findings indicate that most students acknowledge the importance of feedback. Seventy percent of the students agreed that feedback plays a crucial role in their learning. While 69% of students reported receiving regular feedback on all assessments and practical work, 10% felt it could be more structured and focused. Key factors influencing students' perception of feedback included timeliness, clarity, a supportive academic environment, specificity, and constructive criticism. In contrast, vagueness, generalizability, inconsistent feedback from instructors, and perceived biases were identified as key barriers to effective feedback.

Conclusion: This study highlights the critical role of feedback in medical and pharmacy education, emphasizing the need for a more student-centered approach. Findings underscore the importance of timely, clear, and constructive feedback, delivered in a supportive environment, to maximize its impact on student learning and professional development.

Keywords: Feedback Perception, Medical Education, Pharmacy Education, Academic environment, Academic performance

147

Navigating optimal antibiotic practices: Prospective audit and feedback on Piperacillin/Tazobactam utilization in a high-risk cardiac care setting in Karachi, Pakistan

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Background: Antimicrobial resistance (AMR) poses a significant threat to global health, especially in patient care. The misuse of antibiotics has contributed to the rise in antibiotic-resistant bacterial infections. This has intensified the emphasis on the responsible use of antibiotics. Antimicrobial stewardship programs, since their inception, have been highly successful in optimizing antibiotic use.

Aim: The purpose of the study was to assess the appropriate use of broad-spectrum antibiotic, Piperacillin Tazobactam from Nov 2023 – Feb 2024 through the WHO antimicrobial stewardship tool of “Prospective audit and feedback” and assess the appropriateness of therapy in compliance with references of IDSA (Infectious Disease Society of America) guidelines.

Method: We conducted a prospective study spanning four months i.e. Nov 2023- Feb 2024 in a specialized cardiac care hospital. The focus of the study was to conduct a systematic audit of the use of piperacillin-tazobactam. In this hospital, antibiotics were selected for review every quarter bases to evaluate antibiotic appropriateness. The appropriateness of antibiotics was assessed using the WHO “Prospective-audit and feedback” tool for antimicrobial stewardship and the IDSA guideline, which recommends trained professionals (physicians/pharmacists) to review therapy and provide recommendations when it is suboptimal.

Results: In this prospective audit analysis, 518 patients are included, all have received piperacillin/tazobactam. Out of these 518 patients, N = 215(88.8%) treated under Cardiothoracic are declared compliant as per guidelines, N = 156 (60.2%) in cardiology, N = 5(50%) in pulmonary, N = 2 (100%) in nephrology and N = 4(80%) in neurology under care patients compliant as per appropriateness criteria.

Conclusion: This study offers valuable perceptions into the usage and appropriateness of broad-spectrum antimicrobials. Effective AMS implementation can help mitigate the rise of resistant organisms and enhance patient safety.

Keywords: Prospective Audit, Antibiotic stewardship, Feedback mechanism, Appropriateness

148

Assessment of the new unfractionated heparin [UFH] updated institutional protocol awareness among emergency department nurses

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Background: Unfractionated heparin (UFH) is one of the high-alert medications that has a greater potential to cause patients harm if used improperly. Determining how UFH is administered and ensuring that nurses have enough understanding of this high-alert drug are thought to be crucial for promoting a culture of anticoagulation safety. Our institution recently established a new aPTT target based on a modification to the reagent used in our laboratory. In order to maintain a safe atmosphere for the greatest patient care, we must ensure that all healthcare professionals are aware of the latest modifications. The emergency department (ED), which has 226 patient stations, is regarded as the biggest in the area.

Aim: To assess ED nurses' knowledge of the updated UFH institutional protocol to identify gaps or areas for improvement and formulate an educational plan accordingly.

Method: A cross-sectional survey was conducted for nurses to gauge their understanding of the newly updated UFH institutional protocol. Based on the survey results, educational lectures were delivered to ED nurses to improve their knowledge of UFH institutional protocol. A follow-up survey was administered to assess knowledge of ED nurses.

Results: Among the 199 nurses surveyed pre-intervention, 63% were aware of the aPTT target for DVT and PE, while only 47% knew the target for NSTEMI and AFIB. Following the intervention ($n = 222$), awareness increased to 91% and 92%, respectively.

Regarding knowledge of the maximum UFH loading dose, 43% of participants pre-intervention were aware of the correct dose for PE and DVT,

compared to 89% for NSTEMI and AFIB. Post-intervention, awareness rose to 81% for PE and DVT, and 96% for NSTEMI and AFIB.

In terms of dose calculation, 61% of respondents accurately computed the UFH loading dose for NSTEMI and AFIB pre-intervention, and 67% correctly calculated the initial infusion rate. After the intervention, these figures improved to 85% and 79%, respectively.

Conclusion: Targeted educational interventions significantly enhanced nurses' knowledge of UFH dosing and monitoring parameters. Post-intervention improvements in awareness and dose calculation accuracy highlight the effectiveness of standardized training in promoting safe and consistent anticoagulation practices in clinical settings

Keywords: Unfractionated heparin, Emergency department, Nurses

150

Clinical pharmacist-driven medication therapy management clinic in a mental health helpline: One-year insights from Qatar

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Background: Medication therapy management (MTM) services are aimed at enhancing patients' comprehension of appropriate drug usage and promoting medication therapy adherence, resulting in favorable clinical, economic, and humanistic outcomes. The MTM in mental health services under HMC which is led by clinical pharmacists recently participated in the national mental health helpline (NMHH) services.

Aim: To identify and describe the outcome of the clinical pharmacists' interventions for patients seeking input from MTM clinic and NMHH

Method: This was a retrospective review of registry of patients receiving face-to-face and virtual care at the MTM and NMHH services over a 12-month period from Dec-2023 to Dec-2024. Consultation reasons, source of referrals and number of patients for whom a psychiatrist was consulted were analyzed.

Results: With the establishment of NMHH services, our face-to-face MTM clinics shifted focus primarily to Clozapine and Lithium cases requiring follow-ups and lab monitoring, ensuring close care for 120 patients on Clozapine and 103 on Lithium. Simultaneously, our virtual NMHH services recorded an impressive 451 calls over the year, catering to a predominantly adult (97%) and female

(65%) demographic, with 26% being Qatari nationals. Remarkably, 23% of callers were new to Mental Health Services (MHS).

Key reasons for referrals to Helpline Clinical Pharmacy services included medication education (47%), managing side effects (26%), and ensuring safety during pregnancy/breastfeeding (9%). Outcomes demonstrated significant impact: 43% of consultations improved adherence, 25% resolved adverse reactions, and 13% optimized medication effectiveness.

Conclusion: MTM and NMHH services contributed positively to the overall care of patients diagnosed with mental health problems. Results encourage the expansion of the service. Increasing collaboration with helpline coordinators and psychiatrists could enhance the safety, efficiency, and effectiveness of patient care. Future studies shall focus on the impact of MTM clinics on reduction in hospital readmission and other clinical and economical outcomes.

Keywords: Medication therapy management, Clinical pharmacy, Education

151

Cost-effectiveness analysis of preemptive Dihydropyrimidine Dehydrogenase genetic screening before fluoropyrimidine-based adjuvant chemotherapy in colorectal cancer in Qatar

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Background: Colorectal Cancer (CRC) is one of the most commonly diagnosed cancers globally, accounting for more than 930,000 deaths in 2020. Fluoropyrimidine-based chemotherapy is the gold-standard treatment for CRC patients. Around 80% of fluoropyrimidine-based chemotherapies are catabolized by Dihydropyrimidine Dehydrogenase (DPYD) enzyme, which is encoded by the DPYD gene. A genetic deficiency in the DPYD enzyme affects around 3–8% of the general population. Administering full dose fluoropyrimidine-based chemotherapies in CRC patients with DPYD deficiency can lead to life-threatening toxicities. Therefore, personalized treatment approaches are warranted.

Aim: To evaluate the cost-effectiveness of DPYD screening prior to adjuvant chemotherapy in CRC patients in Qatar.

Methods: A decision-tree with a 6-month horizon and a Markov model over a lifetime horizon were built to achieve the aim of this study. The analysis was conducted from the Qatari healthcare provider perspective. The primary outcomes were Incremental Cost-Effectiveness Ratio (ICER) expressed as Qatari Riyals (QAR) per Quality-Adjusted Life Months (QALMs), and Quality-Adjusted Life Years (QALYs) for short- and long-term models, respectively.

Results: In the short-term model, DPYD screening was a cost saving option resulting in an incremental cost of QAR −16.72, and an incremental QALMs of 0.00113 per patient over six months compared to no-screening. The long-term results showed that DPYD screening resulted in improved outcomes, with an incremental QALYs of 0.163 and an incremental total cost of QAR 7,166 per patient, leading to an ICER of 44,041 QAR/QALY. In probabilistic sensitivity analysis using a willingness-to-pay threshold of QAR 90,000/QALY, DPYD screening was preferred to no-screening in 100% of iterations.

Conclusions: DPYD screening before fluoropyrimidine-based adjuvant chemotherapy in CRC patients is a cost-effective strategy in Qatar, potentially reducing severe adverse events and improving patient outcomes. Implementing DPYD screening guides personalized treatment strategies and aligns with precision medicine goals in Qatar.

Keywords: Chemotherapy, Cost-effective, Genetic screening

152

Pharmaceutical operations and supply chain management: A review of literature

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Background: The operations and supply chain management (OSCM) of pharmaceuticals present unique challenges due to the medical, regulatory, and specific nature of the problems in this sector. Addressing these challenges and developing innovative solutions can yield substantial benefits, both for the healthcare industry and society.

Aim: This study aimed to explore the landscape of research conducted in the field of pharmaceutical operations and supply chain management (OSCM). It seeks to understand the evolution of the field, provide a comprehensive

content analysis of the existing literature, and propose a future research agenda to guide scholars and practitioners.

Methods: The Scopus database was used as the source for bibliographic records, following PRISMA guidelines for systematic reviews. This process resulted in the inclusion of 386 scientific journal articles published up to 2023. Data-mining and quantitative methods, including clustering, co-citation analysis, and multiple correspondence analysis, were employed alongside manual content analysis to gain in-depth insights into the field.

Results: The descriptive analysis provided a detailed overview of OSCM research, its sources, and subfields. Historical analysis traced the significant developments and highlighted the emergence of key subfields over time. Development analysis identified seven main research themes, each comprising multiple subthemes, while evolution analysis revealed thematic transformations across four distinct periods. The content analysis identified research gaps and proposed future directions under four domains, encompassing nine specific sub-directions.

Conclusion: This study presents a comprehensive synthesis of the pharmaceutical OSCM field, shedding light on its current state and historical progression. By identifying research gaps and suggesting future directions, it offers a roadmap for advancing this critical area of study.

Keywords: Supply chain management, Medicines, Literature review

153

Factors affecting the knowledge, attitude and practice of vitamin D supplement use among adults in the MENA region

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Background: Vitamin D, an essential fat-soluble metabolite, plays a critical role in various physiological functions and human growth and development.

Aim: This study investigated the factors influencing the knowledge, attitudes, and practices of the population regarding vitamin D supplementation.

Methods: A self-administered questionnaire was distributed via social media platforms. Statistical analysis involved frequency, percentages, and simple/multiple-linear regression analysis to assess study predictors.

Results: A total of 1340 responses were obtained from four countries: Saudi Arabia ($n = 412$, 30.8%), Egypt ($n = 400$, 29.9%), Jordan ($n = 310$, 23.1%), and Iraq ($n = 217$, 16.2%). Median knowledge scores ($\pm$ IQR) in Egypt, Jordan, and Iraq indicated a moderate level of understanding regarding vitamin D supplementation (11 ± 3 , 11 ± 2 , and 11 ± 3 , respectively), whereas Saudi Arabia exhibited lower levels of knowledge (7 ± 2). Neutral attitudes toward vitamin D were observed across all countries, while Jordan and Iraq showed good median practice scores (17 ± 5 , 16 ± 5), compared to moderate levels in Saudi Arabia and Egypt (13 ± 3 , 15 ± 5), respectively. Multi-linear regression analysis indicated that in Saudi Arabia, younger respondents exhibited higher knowledge, attitudes, and practice scores, while individuals working in the non-health sector displayed higher attitudes and practice scores (p -value = 0.06, 0.009, <0.001). Similarly, in Iraq and Jordan, those working in the non-health sector demonstrated higher knowledge scores (p -value <0.001, <0.001, respectively). Furthermore, males attained higher attitude scores in samples from Jordan, Egypt, and Iraq (p -value <0.001, 0.003, <0.001).

Conclusion: This study highlights varying levels of knowledge, attitudes, and practices regarding vitamin D supplementation across different demographic groups and countries. Addressing misconceptions and promoting awareness, particularly among younger individuals and those working outside the health sector is crucial for optimizing vitamin D supplementation practices and enhancing public health outcomes.

Keywords: Vitamin D, Knowledge, Attitude, Self-medication

154

Adapting team-based learning for clinical skills training of undergraduate medical students

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Background: Clinical skills are an indispensable part of health professionals' practice ensuring patient safety. Traditionally, it is taught through simulations and real patients using various approaches in the undergraduate medical

curriculum. The team-based learning (TBL) approach has been used to develop problem-solving, critical thinking, and teamwork skills among students.

Aim: We adapted TBL by using OSCEs in iRAT and tRAT format along with feedback from the peers as well as the facilitators for teaching clinical skills and then evaluated its impact on learning.

Methods: This was a quasi-experimental study with a cross-over design. The clinical students from medicine were assigned into two groups. The control group underwent traditional skills training while the experimental group underwent an adapted TBL learning experience with OSCEs implemented in iRAT and tRAT during clinical skills teaching. Pre and post-OSCEs were conducted and inferential statistics were calculated.

Results: A statistically significant improvement was noted in the mean post-test scores of the Experimental group on both OSCE stations. There was no significant difference in the mean pre-test scores of the experimental and control groups.

Conclusions: This study evaluates an adapted Team-Based Learning (TBL) approach for teaching clinical skills to undergraduate medical students. We found the approach learner-centred, engaging and effective in the clinical skills training of undergraduate medical students.

Keywords: Team-based learning, Education, Medical students

155

A clinical decision support system for optimizing albumin therapy

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Background: Albumin is a critical medication requiring careful consideration of patient-specific factors and adherence to guidelines. To optimize therapy and reduce adverse events, a reliable decision-support tool is essential.

Aim: This research developed and evaluated a web-based Clinical Decision Support System (CDSS) for albumin therapy, assisting healthcare providers in making accurate and timely decisions regarding the indication and dosing of albumin therapy.

Methods: The CDSS was developed using R and Shiny, based on the human albumin prescribing protocol of MOUWASAT Hospital. Functionalities include: 1) patient information input (serum albumin level, indications); 2) protocol-based decision support; 3) indication determination; 4) dosage

calculation; 5) real-time feedback; 6) guideline access. A survey with 27 healthcare professionals assessed usability and impact.

Results: A decision support tool for albumin indication was successfully developed to assist healthcare professionals in evaluating the appropriateness of albumin prescriptions. The tool allows healthcare professionals to enter patient-specific information, including clinical parameters and laboratory values such as serum albumin level. Based on a built-in algorithm and customized guidelines, the application generates real-time recommendations on whether albumin administration is indicated. To validate the functionality and usability of the application, 27 healthcare professionals tested the tool and provided satisfaction scores. The evaluation results showed that: The tool demonstrated 63% accuracy in dosage calculations and indication recommendations, aligning with established national guidelines. Pharmacists reported 96.3% improved efficiency in decision-making and reduced time spent on albumin evaluations. Users reported 85.2% satisfaction, with high ratings for ease of navigation, straightforward data input, and clarity of clinical recommendations.

Conclusion: The CDSS aims to optimize albumin therapy by providing accurate, timely guidance, and empowering evidence-based decisions. Positive survey feedback indicates strong potential to improve patient outcomes, reduce costs, and enhance care quality. A pilot test is planned to evaluate its impact on prescribing practices, patient outcomes, and cost.

Keywords: Clinical decision support system, Albumin, Algorithm

156

NephroXpert: An intelligent platform for personalized estimation of glomerular filtration rate (GFR)

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Background: Accurate evaluation of kidney function is essential for delivering high-quality healthcare, with glomerular filtration rate (GFR) serving as a critical metric for assessing renal performance.

However, commonly used equations for estimating GFR such as Cockcroft-Gault (CG), Modification of Diet in Renal Disease (MDRD), Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI), and Cystatin C-based equation yield varying results influenced by patient-specific factors like

age, weight, gender, race, serum creatinine, and cystatin C levels. Selecting the most appropriate equation remains a challenge in clinical practice due to these variations, impacting clinical decision-making.

Aim: This study aimed to develop a more precise and reliable approach to GFR estimation by integrating intelligent algorithms that optimize equation selection based on individual patient characteristics, minimizing errors, enhancing clinical decision-making, and improving patient outcomes.

Methods: We developed our intelligent algorithm based on the recommendations of the National Kidney Foundation. The algorithm was implemented using the R programming language in conjunction with OpenAI technologies like ChatGPT.

To validate the platform's functionality and accuracy, we manually tested multiple patient cases, comparing the algorithm's equation selection and GFR calculations against the National Kidney Foundation's guidelines.

Results: The platform allows users to input patient data, including age, weight, gender, race, serum creatinine, and cystatin C. The algorithm analyzes these inputs to select the most appropriate equation and calculates the patient's GFR accordingly. Additionally, it provides a rationale for the selection, enhancing transparency in clinical decision-making.

Conclusion: The platform demonstrated its ability to accurately select patient-specific GFR equations, reducing errors and supporting standardized renal function assessments. These results highlight its potential to enhance clinical decision-making and patient care. Future directions include integrating more personalized functions for renal assessments and testing the platform in real-life clinical environments to further validate its effectiveness and improve its practical applicability.

Keywords: Intelligent algorithms, Kidney function, Estimation, ChatGPT

157

Exploring cultural humility in simulation debriefing from the perspective of debriefer

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Background: Cultural humility is a "process of self-reflection to understand personal and systemic bias and privilege" that may contribute to health disparities. There has been an increasing interest in the integration of cultural humility within simulation debriefing which is a complex endeavor, likely

challenged by the diverse cultural backgrounds, lack of training for debriefers, power dynamics, implicit bias and resources constraint. Simulation-based education lacks the focus on cultural humility, indicates the need for further research and educational change.

Aim: The study aimed at exploring cultural humility in simulation debriefing from the perspective of debriefers.

Methods: A qualitative phenomenological study was conducted to explore the lived experiences of debriefers at ITQAN simulation centre in Qatar. Participants included simulation debriefers with at least two years of experience as a debriefer in simulation. A semi-structured interview guide was developed using Foronda's Cultural Humility Scale. Data were transcribed verbatim and analyzed using thematic approach.

Results: Participants identified the relevance of cultural humility in multicultural settings, context in which cultural humility is needed. Participants also reported various personal, systemic and educational design-related challenges and enablers to the application of cultural humility in simulation debriefing. Promoting a safe and equitable learning environment increases learners' engagement, encourages culturally appropriate communication approaches, raises awareness of cultural, ethical issues, and enhances skills & strategies necessary for effective debriefing.

Conclusion: Study recommended cultural humility training to enhance debriefers' skills providing culturally appropriate debriefing experiences. Reporting cultural humility assists simulation leaders and policy makers in multi-cultural educational settings e.g. Qatar, setting best practices towards valuing cultural diversity among learners

Keywords: Simulation debriefing, Cultural humility, Health professions education

160

Evaluation of Isotretinoin management practices in primary care in Qatar

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Background: Isotretinoin is a commonly prescribed treatment for severe acne, but its use requires stringent monitoring due to significant adverse effects and teratogenic risks.

Aim: This study evaluated prescribing patterns and adherence to safety protocols in primary care settings within the Primary Health Care Corporation (PHCC) in Qatar.

Methods: A retrospective review of patient records was conducted using the Cerner electronic health record system at PHCC facilities. Data collected included isotretinoin prescriptions (dosage, frequency, and duration), adverse events, laboratory investigations (liver function test, CBC, and lipid profile), documentation of informed consent, and pregnancy test requests for women of childbearing age. The data were collected for the period between 2016 and 2024.

Results: A total of 9,252 patient records were reviewed, including 2,194 Qatari patients. The age of patients ranged from 12 to 77 years. Around 96% (N=8,879) of isotretinoin prescriptions were indicated for acne. Urine pregnancy tests were conducted for 246 patients. Informed consent was documented in 54.1% (N=3,247) of cases.

Conclusion: The study highlights critical gaps in the management of isotretinoin in PHCC primary care settings. There is a clear need for better documentation practices and stronger adherence to clinical guidelines to ensure patient safety and minimize the risks associated with isotretinoin therapy.

Keywords: Isotretinoin, Primary care, Adherence

164

Enhancing pharmacist competency in immunisation: A micro-credential approach through active and experiential learning in Malaysia

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Background: Pharmacists are increasingly involved in immunisation services globally, necessitating specialized training to ensure the safe and effective delivery of vaccines. In Malaysia, the Certified Training Programme in Immunisation for Pharmacists (CTPIP) addresses this need by providing comprehensive training through a series of micro-credential modules.

Aim: To enhance pharmacists' competency in immunisation practices through active and experiential learning, aligning with Sustainable Development Goals (SDG) 3 (Good Health and Well-being) and SDG 4 (Quality Education).

Methods: The CTPIP consists of eight micro-credential modules available on the OpenLearning platform, culminating in an experiential learning workshop. The modules incorporate interactive discussions, case studies, forums, and group activities to enhance critical thinking, problem-solving, and decision-making skills. Participants engage in simulated immunisation scenarios, practice vaccine administration techniques, and receive basic life support (CPR) training. Post-course assessments and feedback surveys evaluate the impact on pharmacist competency and patient care.

Results: By the end of 2024, about 443 participants had enrolled in the CTPIP course, with a group of people completing the hands-on session. Preliminary results indicate significant improvements in pharmacists' knowledge of immunisation guidelines, confidence in managing adverse events, ability to administer vaccines safely and efficiently, and provision of immunisation counselling. The programme generated approximately RM 22000.00 (USD 4900.00) in income, with an expected annual income of RM 92,800 (USD 20751.00) based on projected enrolments.

Conclusion: The CTPIP effectively reskills and upskills pharmacists through active and experiential learning, providing a flexible platform for professional certification and competency enhancement. This approach is crucial for advancing the pharmacist profession and improving public health outcomes in Malaysia.

Keywords: Pharmacist, Micro-credential, Active learning, Experiential learning, Immunisation

165

Prescribing skills assessment utilization for pharmacy curriculum improvement: An experience from Gulf Medical University

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Background: The Prescribing Skills Assessment (PSA), developed by the British Pharmacological Society, evaluates prescribing competencies across eight domains of patient care. Initially created for newly qualified physicians, Gulf Medical University (GMU) – College of Pharmacy (CoP) has adapted the PSA to enhance the clinical training of pharmacy students, aiming to improve their competency in prescribing and decision-making.

Aim: This study aimed to evaluate the impact of PSA integration on improving pharmacy students' clinical skills in pharmacotherapy, laboratory data interpretation, and prescribing. Additionally, this study aims to explore how PSA can enhance curriculum design and delivery to meet the evolving roles of pharmacists.

Methods: Between 2020 and 2024, GMU CoP integrated the PSA across its pharmacy curriculum. This included incorporating PSA-based calculation and laboratory data questions into the Pharmacotherapy series and using PSA as an assessment tool in clinical rotations. Students participated in assignments and team-based learning activities designed around PSA content. Data on student performance was collected and analyzed to assess improvements in key clinical competencies, including prescribing accuracy and clinical reasoning.

Results: The findings indicate significant improvements in students' prescribing skills and their ability to interpret clinical data. PSA-based assignments and activities helped students apply theoretical knowledge to practical, real-world scenarios, leading to enhanced prescribing competency. The students demonstrated increased accuracy in medication selection and dosing calculations, as well as improved handling of complex patient cases.

Conclusion: The PSA's integration into GMU's pharmacy curriculum has proven beneficial in enhancing the clinical skills of pharmacy students. By addressing key areas including pharmacotherapy and clinical decision-making, the PSA has helped bridge the gap between theoretical knowledge and clinical practice. The PSA also helped the CoP to identify areas of weakness that are being addressed in the didactic and experiential education components of the curriculum.

Keywords: Education, Prescribing skills, Pharmacy education

166

Impact of pharmacist-led intravenous to oral medications conversion at an intensive care unit: A single-center, prospective, interventional study

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Background: Switching from intravenous (IV) to oral (PO) therapy at the appropriate time improves medication use and leads to cost savings. Studies in various countries have shown benefits such as increased appropriate oral therapy conversion, shorter IV therapy duration, reduced ICU stays,

and significant cost savings. However, there is limited prospective data from the UAE.

Aim: This study aimed to evaluate the impact of a pharmacist-led IV to PO switch service in critically ill patients using antibiotics, proton pump inhibitors (PPIs), and acetaminophen.

Methods: A single-center, prospective, interventional study was conducted in a UAE tertiary care hospital from January to August 2024. Newly admitted ICU patients' charts were reviewed, and clinical pharmacists made IV to PO switch recommendations to ICU physicians. A previous retrospective study at the same hospital, assessing IV to PO conversion practices, served as the pre-intervention comparison.

Results: A total of 338 patients were analyzed, with 118 from the retrospective study and 220 from the prospective study. After implementing the pharmacist-led service, 74.5% of patients met the switch criteria and were converted to oral therapy, compared to 11% in the pre-intervention group ($p < 0.0001$). The duration of IV therapy decreased significantly for antibiotics (4.44 vs. 7 days), acetaminophen (2.92 vs. 4.5 days), and PPIs (2.01 vs. 5.5 days) ($p < 0.0001$). The intervention also resulted in cost savings of approximately 46,000 AED (\$12,523.82) over 8 months.

Conclusion: The pharmacist-led IV-to-PO switch service significantly improved conversion rates, reduced IV therapy duration, and achieved cost savings. This approach could be expanded to other hospital departments and implemented across the UAE.

Keywords: Pharmacist-led, Route of administration, Intensive care unit

168

Medication use evaluation of erythropoietin in non-dialysis chronic kidney disease patients

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Background: Chronic kidney disease (CKD) patients are known to develop anaemia as one of the disease complications. Erythropoietin plays a key role in the treatment of anaemia by stimulating the production of red blood cells. Iron supplementation is also essential in the erythropoiesis process which makes both erythropoietin and iron therapy critical factors in treating CKD-associated anaemia, with careful consideration of necessary lab tests.

Aim: The aim of this medication use evaluation was to assess the appropriateness of laboratory testing and iron correction before erythropoietin administration.

Method: This is a single-center, retrospective chart review medication use evaluation, which included non-dialysis CKD patients admitted to the hospital between March 2023–February 2024 who received erythropoietin. We evaluated laboratory assessments (iron panel, ferritin, TSAT, folic acid, vitamin B12, folate levels, and creatinine) for patients who were either newly initiated on or had previously been taking erythropoietin. The evaluation included the assessment of adjunct therapy, such as iron supplementation.

Results: A total of 28 patients were included in the study. Nine patients were newly initiated on erythropoietin, while 19 patients had previously received erythropoietin. All patients had laboratory tests for haemoglobin and MCV, 96.4% had creatinine tests, 60.7% had iron panels, 42.8% had ferritin tests, 7% had folic acid tests, and 17.8% had vitamin B12 levels measured. Additionally, 82.1% of patients did not receive iron supplementation before erythropoietin administration. For the patients newly initiated on erythropoietin, 88.8% did not have a haemoglobin test one week after initiation. Among the patients who were previously receiving erythropoietin, 12.5% had iron panels tested monthly, and 23.5% had ferritin and TSAT levels tested at least every three months

Conclusion: Most patients did not receive iron therapy before erythropoietin administration. Additionally, pre-treatment iron status and necessary lab monitoring are important for achieving optimal patient outcomes.

Keywords: Erythropoietin, Medication use evaluation, Anemia, CKD

170

Perceptions of pharmacy students and clinical preceptors on feedback during experiential learning

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Background: Pharmacy students and preceptors in Qatar face challenges in providing and receiving effective feedback during experiential learning. This study addressed the gap by exploring the specific facilitators and barriers to feedback in this context.

Aim: The study aim is to examine students' perceptions of feedback and its impact on their experiential learning.

Methods: An exploratory qualitative methodology was employed using semi-structured interviews (face-to-face and virtual) conducted in Qatar from July 2023 to January 2024. A purposive sample of 12 pharmacy students and 12 preceptors were included. The settings included preceptors from the community and HMC hospital settings. The data were transcribed and analyzed using thematic analysis.

Results: From the analysis, four key themes and nine subthemes emerged: facets of perceptions, effective feedback strategies, perceived barriers to feedback, and facilitators of effective feedback. Preceptors emphasized their responsibilities as educators and mentors, guiding the students to apply their knowledge in a practical training setting. Similarly, students consistently recognized preceptors as essential guides throughout their experiential learning journey.

Conclusion: The study found that both pharmacy students and preceptors recognized the importance of timely, individualized, and private feedback, emphasizing its role in fostering a constructive learning environment and professional growth. Overall, the results have implications for designing and implementing preceptor training programs and feedback processes within pharmacy education and healthcare practice sites. By addressing these implications, colleges of pharmacy and healthcare organizations can create enriching experiential learning experiences for students.

Keywords: Feedback, Pharmacy education, Pharmacy student, Pharmacy resident, Preceptors, Experiential learning

174

Risk of liver injury among patients exposed to direct oral anticoagulants: A systematic review and network meta-analysis

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Background: Although direct oral anticoagulants (DOACs) have been increasingly used for multiple indications owing to their predictable pharmacokinetics and fixed dosing regimen as compared to Vitamin K Antagonists (VKAs), safety alerts have been issued highlighting their risk of drug

induced liver injury (DILI). However, data on the effects of DOACs on the liver is sparse, with most studies reporting conflicting results when compared to VKAs and other DOACs.

Aim: To ascertain the exact risks of liver injury among patients exposed to the various DOAC analogues compared to placebo and VKAs.

Methods: We carried out a systematic review and network meta-analysis of Randomized Controlled Trials (RCTs) and observational studies involving patients without pre-existing liver disease who were treated with DOACs for any indication that reported liver injury outcomes. Eligible studies were identified through a comprehensive search of relevant databases (PubMed, Cochrane Library, and ClinicalTrials.gov) from 2009 until January 2024. Direct comparisons of the liver safety outcomes between DOAC analogues were made using Generalized Pairwise Modelling (GPM) with a random effects model. All analyses were done using MetaXL with statistical significance set at $p < 0.05$.

Results: Twelve direct estimates were generated based on head-to-head comparisons across 23 studies ($n = 719,931$ patients), including 18 RCTs and 5 observational studies. Pooled analyses revealed that DOACs, particularly apixaban was associated with a 45% reduction in risk of DILI compared with warfarin (OR 0.55, 95%CI 0.35–0.85). Indirect estimates showed no significant differences in odds of DILI risks among DOACs, except for Edoxaban, which showed a 70% higher risk of DILI compared to Apixaban (OR 1.70, 95% CI 1.02–2.82).

Conclusion: DOACs, particularly apixaban, were associated with a reduced risk of DILI compared to warfarin. These findings provide both reassurance as well as valuable guidance to clinicians regarding the choice of anticoagulation in patients at high risk of liver injury.

Keywords: DOACs, Anticoagulation, Warfarin

175

Knowledge and practice of paediatric pharmacy: The approach of community pharmacists in Nigeria

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Background: Community pharmacies in Nigeria often serve as the first point of healthcare contact, including for paediatric patients whose caregivers seek assistance for various health concerns.

Aim: This study evaluated community pharmacists' knowledge and practices related to paediatric pharmacy.

Methods: A cross-sectional study was conducted in August 2023 among community pharmacists in Nigeria. Data were collected using a 40-item instrument divided into three sections: pharmacists' demographics, knowledge of paediatric pharmacy, and practice of paediatric pharmacy. Knowledge scores were summarized as means and standard deviations, categorized as poor, average, or good. Practice was reported as proportions, and associations between knowledge levels and socio-demographic variables were analyzed with a significance level set at $p < 0.05$. Ethical approval was obtained for the study.

Results: Of the 381 participating community pharmacists, 313 (82.2%) held only a Bachelor of Pharmacy degree, and most (239, 62.7%) were aged 31–50 years. The mean knowledge score was 8.6 ± 2.1 . More than half (204, 53.3%) of pharmacists reported not consulting paediatric-specific drug information sources, while 347 (91%) stocked paediatric-appropriate dosage forms in about a quarter of their inventory. Additionally, 289 (68%) did not engage in extemporaneous compounding for paediatric patients. Commonly treated conditions included malaria (353, 92.7%), fever (334, 87.7%), catarrh (333, 87.4%), and cough (325, 84.3%). Nearly all respondents (379, 99.5%) expressed interest in further training in paediatric pharmacy. Knowledge levels were significantly associated with possessing additional qualifications.

Conclusion: Despite their extensive experience, community pharmacists demonstrated moderate knowledge of paediatric pharmacy. Establishing targeted training programs is essential to enhance paediatric care in community pharmacy practice.

Keywords: Community pharmacists, Knowledge, Paediatric

176

Effectiveness and safety of Brentuximab Vedotin containing regimen versus standard chemotherapy in relapsed or refractory Hodgkin Lymphoma: A multicenter cohort study

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Background: Hodgkin Lymphoma (HL) constitutes 3.4% of all cancers in Saudi Arabia and presents significant challenges, with relapse rates reaching 40% in advanced stages. Brentuximab vedotin (BV), a novel agent targeting CD-30, has demonstrated potential in improving clinical outcomes. Despite the increasing adoption of BV-based regimens for treating relapsed/refractory Hodgkin Lymphoma (rrHL), there is a notable lack of direct comparative studies evaluating the efficacy of these regimens against standard chemotherapy.

Aim: This study aims to fill this gap by assessing the real-world effectiveness and safety profiles of BV-based regimens compared to standard chemotherapy in patients with rrHL in Saudi Arabia.

Methods: This retrospective multicenter cohort study included 93 adult rrHL patients in Saudi Arabia between 2016 and 2021. Patients were grouped into BV-containing regimens ($n = 61$) and control regimens ($n = 32$). Data collection adhered to ethical guidelines, and outcomes assessed included overall response rate (ORR), complete response (CR), progression-free survival (PFS), and post-chemotherapy complications.

Results: BV-containing regimens exhibited a higher ORR (82.5% vs. 76.9%), though this was not statistically significant (adjusted odds ratio (aOR) 12.25, 95% CI 0.99–151.79, $p = 0.05$). CR rates were comparable between groups (61.5% in the control group vs. 59.6% in the BV group; OR 1.37, 95% CI 0.19–9.83, $p = 0.76$). A trend toward prolonged PFS was observed with BV (22.4 months vs. 11.1 months), but this was not statistically significant (Beta coefficient 0.38, 95% CI -0.13 – 0.89 , $p = 0.14$). Post-chemotherapy complications were similar, with higher incidences of hepatotoxicity and acute kidney injury reported in the control group.

Conclusion: BV-containing regimens show potential benefits in response rates and PFS in rrHL patients. However, these findings warrant validation through long-term studies with larger sample sizes to optimize treatment strategies.

Keywords: Hodgkin Lymphoma, Refractory, Brentuximab vedotin, Saudi Arabia

178

Effectiveness of an online mental health educational programme to improve mental health literacy among the youth

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Background: Mental health issues among the youth population have become a growing concern globally. Mental health literacy (MHL) is crucial for early recognition and effective management of mental health issues. However, university students have limited knowledge, contributing to delayed help-seeking behaviours and increased stigma. A targeted programme is needed to address the challenges and improve MHL among young people.

Aim: To assess the effectiveness of a newly developed online youth-centred mental health educational programme.

Methods: A pre–post-study design was employed. University students were invited to participate in a 3-hour mental health educational programme. The programme was developed and validated by a multidisciplinary team (i.e. pharmacists, educators, counsellors, social workers and a psychologist). Participants' knowledge of MHL was assessed before the programme (pre-test), post-test 1 (immediate), and post-test 2 (2 weeks later). The intervention was delivered via a workshop conducted on the Webex platform and the assessments were administered through Google Form.

Results: A total of 46 undergraduate students from Universiti Sains Malaysia consented to the programme, and completed pre-test and post-test 1 while 23 completed the post-test 2 assessments. After the workshop, participants demonstrated significant improvements in their MHL and ability to manage mental health challenges. The mean total score for MHL increased from 9.9 (pre-test) to 11.7 (post-test 1) and 11.9 (post-test 2). Similarly, the understanding of managing mental health challenges mean score increased from 8.6 (pre-test) to 9.9 (post-tests 1 and 2). The intervention led to a statistically significant enhancement in the participants' knowledge, with p -values of <0.001 between pre-test and post-test 1 for MHL and managing mental health challenges.

Conclusions: The mental health educational programme effectively improved mental health literacy and management skills among university students. Findings highlighted the importance of targeted educational interventions in enhancing mental well-being.

Keywords: Mental health, Health education, Youth, Malaysia, Extended pharmacist's role

180

Venous thromboembolism after moderna COVID-19 vaccination: A case report

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Background: Since the beginning of the COVID-19 outbreak in December 2019, several vaccines have been produced to prevent the disease. Moderna COVID-19 vaccine was approved in December 2020 after it has been proven safe and highly effective in preventing symptomatic COVID-19.⁽¹⁾ Later, several cases reported the incidence of thrombosis and thrombocytopenia after the vaccine, which raised some concerns about its safety.

Aim: To report a case of a patient who developed extensive pulmonary embolism (PE) and deep vein thrombosis (DVT) after receiving the Moderna vaccine.

Method: A 32-year-old Bangladeshi male, not known to have chronic diseases, presented to the hospital with a productive cough, subjective fever and occasional hemoptysis for 18 days which started 1 day after receiving his second dose of Moderna vaccine. The patient presented in critical condition with hypoxia, tachypnea, and tachycardia. Given the history of recent vaccination, with the presenting symptoms and elevated D-dimer level of 4.59 mg/L, PE was suspected, so urgent computed tomography pulmonary angiogram was done which was impressive of acute PE. Furthermore, lower limb ultrasound Doppler showed findings suggestive of acute DVT; though, serial complete blood counts showed no evidence of thrombocytopenia neither at presentation nor after that.

Results: Vaccine-induced thrombosis and thrombocytopenia has been reported previously in healthy individuals after receiving adenovirus-based vaccines. It was mostly occurring 5–30 days after vaccine administration. A previous case report has described thrombosis and thrombocytopenia developed after m-RNA vaccine (Moderna) vaccination, however, our case report thrombosis without thrombocytopenia after Moderna

vaccination. Yet, it is difficult to establish a direct association between this serious thrombotic event and exposure to Moderna vaccine from few case reports among millions of vaccine doses administered globally.

Conclusion: This case described the possible link between Moderna vaccine and venous thromboembolism; yet, further evidence is needed to confirm this association.

Keywords: Vaccine, Covid-19, Thromboembolism

182

Pharmacists and TB adherence: Bridging gaps in directly observed therapy

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Background: Tuberculosis (TB) remains a critical public health issue in regions like Malaysia, where challenges such as treatment non-adherence and diminished quality of life persist despite established DOTS programs. Pharmacist-led educational interventions, aimed at enhancing adherence through tailored patient counselling, offer a potential solution.

Aim: This study aimed to evaluate the effectiveness of a pharmacist-led educational intervention in improving treatment adherence among tuberculosis patients enrolled in the DOTS programme.

Method: A randomized controlled study at Hospital Pulau Pinang assessed the impact of pharmacist-led counselling, guided by the health belief model, on treatment adherence. Participants ($n = 206$) were randomized into intervention (pharmacist counseling) and control (standard nursing care) groups. The MARS-5 scale measured adherence at baseline and at two, four, and six months. Final sample size was 186, considering attrition, with sequential sampling used to reach this target.

Results: Among the participants, 68.4% were male, and 38.8% were aged 45–64 years. The intervention group showed a significant 2.4-point improvement in MARS-5 adherence scores ($p = 0.001$, $\eta^2 = 0.21$) from baseline to follow-up. At baseline, no significant differences were noted in most health belief domains, except perceived barriers, which were higher in the intervention group ($p = 0.01$). By the first follow-up, significant improvements were observed in perceived susceptibility ($p = 0.001$, $\eta^2 = 0.21$), perceived barriers

($p = 0.012$, $\eta^2 = 0.11$), and cues to action ($p = 0.01$, $\eta^2 = 0.09$). These gains were further enhanced in perceived susceptibility ($p = 0.001$, $\eta^2 = 0.5$) and severity ($p = 0.002$, $\eta^2 = 0.9$) by the last follow-up.

Conclusion: Pharmacist-led interventions significantly improved TB treatment adherence, demonstrating the effectiveness of tailored counselling based on the health belief model, with a marked 2.4-point increase in MARS-5 scores.

Keywords: Pharmacist-led, Tuberculosis, Adherence, MARS-5

183

Perspectives of pharmacy students towards academic programs accreditation: A focus on awareness, perception and attitude

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Background: The accreditation of pharmacy programmes is essential for maintaining educational quality in academic institutions, entailing a systematic evaluation process.

Aim: This study examines the perspectives of pharmacy students (i.e. awareness, perception and attitude) towards academic program accreditation.

Methods: A cross-sectional study surveyed Doctor of Pharmacy (PharmD) students at King Saud University, Riyadh, Saudi Arabia, towards academic program accreditation. Data from 112 participants were collected *via* a structured questionnaire covering demographics, awareness evaluation, and perception and attitude assessment. The data was tabulated and analyzed using the SPSS software.

Results: The sample included both male (56.30%) and female (43.70%) students from various academic years. Comprehensive awareness of the accreditation process was demonstrated by 38.40% of participants, with a significant variation across academic years ($p < 0.001$). Additionally, 54.50% of participants had a general awareness of the concept of accreditation, although 74.10% lacked a formal orientation from their institutions. Over half (53.60%) knew their program's accreditation status, but many were unclear about accreditation requirements. Only 38.40% were aware of other accredited pharmacy programs. In terms of perception and attitudes, the study revealed a generally positive outlook on accreditation. A meaningful proportion of male students (53.60%) compared to female students (42.90%) viewed joining an accredited

programme as important, with this aspect differed significantly across academic levels ($p < 0.05$). The study found a generally positive outlook on accreditation, and academic level differences in perceptions about its importance and role in post-graduate training eligibility.

Conclusions: The present study revealed a significant improved trend of students' awareness of accreditation by academic level, demonstrating that awareness level increases with each academic year. Educational strategies deepening students' awareness about the accreditation process are needed.

Keywords: Pharmacy education; Academic programs, Accreditation, Students' perspectives

185

Evaluation of an integrated in silico modelling module in an undergraduate pharmacy technology program: Enhancing competency in biopharmaceutics and pharmacokinetics

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Background: Pharmacokinetics and biopharmaceutics are essential components of pharmaceutical curricula, crucial for understanding drug design, development, and clinical applications. Integrating in silico modelling into these disciplines provides an innovative platform to bridge theoretical concepts with practical applications.

Aim: This study evaluated the effectiveness of an integrated module that utilizes advanced software tools including: GastroPlus®, ADMET Predictor®, and DDDPlus® from Simulation Plus, Inc., within the third year Biopharmaceutics and Pharmacokinetics course in the pharmacy technology bachelor's program in Qatar.

Methods: The module was implemented in the Biopharmaceutics and Pharmacokinetics course. Its effectiveness was assessed through: a) Student performance scores across three assessment levels; b) Alignment with the seven principles of good practice in undergraduate education; c) Mapping module components to identify learning outcomes to evaluate the development of core competencies and professional soft skills.

Results: Student performance consistently yielded high scores, with averages of 90.7%, 97.1%, and 91% for the overall grade, the project score, and the poster presentation, respectively ($n = 79$), reflecting strong alignment with

institutional objectives and significant knowledge acquisition. The module adhered to the seven principles of good undergraduate education, offering significant benefits to learners. The module provided a robust platform for synthesizing biopharmaceutical and pharmacokinetic knowledge, enhancing students' professional identity and nurturing practical competencies. It also fostered soft skills such as real-world data analysis and effective knowledge dissemination, aligning with workplace demands. Moreover, students developed a deeper appreciation of the distinct objectives of pharmacokinetics and pharmacodynamics aspects of the course.

Conclusion: The integration of in silico modelling into the Biopharmaceutics and Pharmacokinetics course effectively bridged academic learning with real-world applications. It enhanced students' understanding of core concepts, cultivates critical soft skills, and prepares them for professional careers in pharmaceutical and health sector. The module serves as a model for innovative, practical, and student-centred education in pharmacy technology.

Keywords: Technology education, In silico modelling, Biopharmaceutics, Pharmacokinetics

186

Anticoagulation in patients with provoked venous thromboembolism: A real-world analysis of clinical characteristics, management strategies and outcomes from an outpatient tertiary anticoagulation clinic in Qatar

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Background: Patients with venous thromboembolism (VTE) are categorized into provoked or unprovoked episodes depending on the aetiology of their disease. Such categorization has an important prognostic value and aids in determining treatment duration.

Aim: This study aims to describe the clinical characteristics, treatment approaches, and outcomes of patients diagnosed with provoked venous thromboembolism at the anticoagulation clinic of Hamad General Hospital. This covered the period from the clinic's establishment in May 2016 to December 2024.

Methods: This cross-sectional descriptive study aims to examine the characteristics of patients receiving anticoagulant therapy for provoked venous thromboembolism (VTE) episodes. It outlines the provoking factors, types of anticoagulant medications used, treatment duration, thrombophilia investigation, the persistence or progression of the underlying aetiology, and the reasons for extending treatment beyond the initially planned duration. Additionally, it assesses any adverse drug reactions.

Results: A preliminary analysis of the data revealed that the most common diagnosis among patients with provoked events was deep vein thrombosis (44.4%), followed by pulmonary embolism or a combination of pulmonary embolism and deep vein thrombosis. Limited mobility due to travel or acute illness was the most frequently reported provoking factor, affecting 44% of patients, with surgery being the second most common factor (22%). Thrombophilia work-up was conducted in 33% of cases, yielding positive results based on clinical presentation or a significant family history of blood clotting disorders. However, in only 11% of patients, the findings from thrombophilia testing led to a change in the management plan, resulting in lifelong anticoagulation.

Conclusion: The findings of this study provide real-world data on patients attending Qatar's largest anticoagulation clinic, serving as a foundation for generating hypotheses for future anticoagulation research.

Keywords: Anticoagulation, Thromboembolism, Patient safety

187

Bamlanivimab use in patients with COVID-19 infection in Qatar: A retrospective observational study

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Background: COVID-19 pandemic has led to a global public health crisis. As research continues to explore effective treatments for COVID-19, monoclonal antibodies have emerged as a potential therapeutic option. Bamlanivimab is a monoclonal antibody that was shown to reduce the severity and hospitalization rate among COVID-19 patients.

Aim: The objective of this study was to evaluate the appropriateness, safety, and effectiveness of Bamlanivimab use in patients with mid-moderate COVID-19 infection in Qatar.

Methods: A retrospective observational study was conducted on all COVID-19 patients who received Bamlanivimab in Qatar during the COVID-19 pandemic. A standardized assessment tool with pre-determined criteria was developed to evaluate the indications, outcomes, and safety measures. Demographic and clinical data were collected from patients' electronic medical records. The primary outcome was to determine the percentage of patients with all-cause hospitalization/emergency department (ED) visits up to 30 days after treatment. A descriptive data analysis was undertaken.

Results: A total of 20 COVID-19 patients received the drug and were included in the study. Most patients were male ($n = 14$, 70%), with a median age of 60 years [IQR = 28.75]. All patients have at least one appropriate indication for Bamlanivimab, with diabetes mellitus being the most common risk factor for progression to severe disease or hospitalization ($n = 12$, 60%). Seven patients (35%) required hospitalization or ED visit within 1 month after Bamlanivimab and only three patients (15%) clinically worsened, required mechanical ventilation, and intensive care unit admission due to COVID-19 infection. Two patients (10%) died within 90 days after treatment with Bamlanivimab. None of the patients discontinued the infusion due to adverse events or hypersensitivity reactions.

Conclusion: The indication for treatment with Bamlanivimab was appropriate in most COVID-19 patients in Qatar with questionable effectiveness in preventing hospitalization or ED visits due to worsening COVID-19 disease.

Keywords: Covid-19, Hospitalization, Bamlanivimab

188

Gentamicin appropriateness and therapeutic drug monitoring at a diverse multicultural middle eastern tertiary hospital setting: A retrospective evaluation and quality audit

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Background: The use of gentamicin in the treatment of infectious diseases requires frequent monitoring to attain the best treatment outcomes.

Aim: The objective of this study was to evaluate the appropriateness of therapeutic drug monitoring (TDM) for gentamicin at a tertiary care hospital in Qatar.

Methods: A one-year retrospective chart review of all gentamicin TDM records was conducted. Evidence-based criteria were applied to evaluate the appropriateness of gentamicin TDM in terms of indication, sampling times, and post-analytical actions.

Results: Out of 59 captured gentamicin TDM records, 58 gentamicin samples were eligible for evaluation. Overall, gentamicin TDM appropriateness was achieved in 50% (n = 29) of the evaluated records. However, 12% (n = 7) of gentamicin drug concentrations were below the assay quantification limits or were not sampled appropriately. Inappropriate post-analytical actions (22.4%, n = 13) and inappropriate sampling times (44.8%, n = 26) were recorded. Most of the gentamicin blood samples (n = 43; 74.2%) were taken appropriately at steady state, but the samples were not appropriately timed in relation to the dosing schedule in 31% (n = 18) of the cases. Although 27.6% (n = 16) of gentamicin concentrations were non-therapeutic, continuing gentamicin dosing without adjustment was the most frequent post-analytical action (69.8%, n = 37). Gentamicin dose regimen continuation, dose regimen decreases, and dose regimen discontinuation were inappropriately applied in 27% (n = 10), 25% (n = 2) and 14% (n = 1) of the times respectively.

Conclusion: Suboptimal gentamicin TDM practices exist in relation to sampling time and post-analytical actions. Studies exploring setting-specific reasons behind inappropriate TDM practices and methods of its optimization are needed.

Keywords: Antibiotics chart review, Therapeutic, Monitoring

189

Effect of Pioglitazone on clinical outcomes and inflammatory response in patients with type 2 diabetes and COVID-19 infection

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Background: COVID-19 infection has emerged as a rapidly spreading communicable disease across the globe. Recent studies showed that people with type 2 diabetes (T2D) are at a higher risk of mortalities from COVID-19, mainly due to the chronic inflammatory state associated with it. Therefore, it was hypothesized that pioglitazone may improve COVID-19 outcomes in patients with T2D.

Aim: The objective of this study was to evaluate the efficacy of pioglitazone in improving the clinical outcomes of patients with T2D admitted with moderate-severe COVID-19.

Methods: This was a randomized, double-blinded, multicenter trial on patients with T2D admitted with COVID-19 to hospitals in Qatar and Kuwait. Patients were randomized to receive either pioglitazone 45mg orally daily for 10 days then 30mg daily for 18 days or a matching placebo. The study had two primary outcomes: (1) the incidence of a clinical composite outcome of a) requirement for mechanical ventilation, b) death, c) myocardial damage indicated by Troponin > 3x upper limit of normal range; and (2) an increase in C-Reactive Protein (CRP). Blood samples were collected at baseline and repeated weekly until day 28 or discharge.

Results: A total of 343 patients, pioglitazone (n = 188) and placebo (n = 155), were enrolled in the study. Twenty-eight participants met the primary composite outcome. There was no significant difference in the incidence of primary outcome between subjects receiving placebo (n = 14) and pioglitazone (n = 14). Cox Hazard ratio for patients receiving pioglitazone was 1.10 ($p = \text{NS}$). Total of 4 patients died in the placebo group vs. 3 in the pioglitazone group ($P = 0.91$). No statistically significant difference was observed between the two groups in plasma CRP level at the end of the follow-up.

Conclusion: Diabetes treatment with pioglitazone did not provide additional benefit to patients admitted with moderate to severe COVID-19 infection.

Keywords: Diabetes, Covid-19, Antidiabetics, Pioglitazone

190

Excellence in medication reconciliation during COVID-19 pandemic

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Background: Effective medication reconciliation (MedRec) promotes patients' safety, while poor compliance with MedRec may result in adverse event and medication errors. In Communicable Disease Center (CDC), majority of COVID-19 admissions were at risk of having drug interactions and medication errors from the concomitant use of COVID-19 medications. Therefore, it was crucial to conduct proper MedRec to prevent these harmful events in this vulnerable population. A review of practices at the CDC in December 2020 showed a median MedRec compliance rate of 86%, indicating room for improvement.

Aim: This quality improvement project aimed to enhance MedRec compliance and reduce medication-related harm in this high-risk population.

Methods: The current practice of MedRec in CDC was reviewed and analysed to identify areas for improvement. A process map for inpatients was prepared and the Institute for Healthcare Improvement (IHI) model for improvement was utilized, where multiple measures were developed to monitor the project's effectiveness. A physician satisfaction survey was distributed to physicians as a balance measure. The multidisciplinary task force team managed to test and implement multiple interventions with various change concepts

Results: Median compliance increased from 86% to 90% by December 2021 and it further increased to 95% by the end of December 2022. During the pandemic, there was only one reported medication error that is due to missed or improperly conducted MedRec. It was about one patient who missed his insulin dose and caused hyperglycemia. Peer review was done for this case and found that the medication reconciliation was incomplete for this patient. For the balance measure, most physicians ($n = 27/33$, 82%) agreed that MedRec process does not have a negative impact on their tasks.

Conclusion: CDC has successfully ensured patient safety and consistently delivered high-quality care during and beyond the COVID-19 pandemic by enhancing MedRec practices.

Keywords: Medication reconciliation, Compliance, Satisfaction

191

Development of a comprehensive curriculum for sports pharmacy educational course

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Background: Sport pharmacy field is gaining recognition for its important role in enhancing athlete performance and treating their disease conditions while promoting clean sport practices. There are many existing courses provided curricula with valuable insights. However, there is a lack of unified, comprehensive course framework tailored to all healthcare providers who are interested in sport pharmacy.

Aim: This review aimed to analyse global sports pharmacy courses and conferences, joining their objectives into an integrated curriculum for sports pharmacy course.

Methods: To identify available sports pharmacy courses and curricula worldwide, a narrative review using exploratory search was conducted utilizing publicly accessible online platforms, including medical databases (e.g. PubMed), general search engines (e.g. Google), university websites, educational online platforms, and conference pages. The search focused on retrieving course titles, methods of delivery, objectives, and targeted audiences. The review also extended to related conferences and their objectives. This approach ensured the inclusion of a wide range of publicly available material, guaranteeing a comprehensive and globally inclusive framework for the course.

Results: A total of 8 courses and 4 conferences worldwide were analyzed. The framework included eight modules. It addresses foundational knowledge, traditional and novel methods of doping techniques, detection strategies, sports pharmacotherapy, the roles of healthcare providers in sports medicine, and evidence-based nutrition and supplementation, providing a comprehensive framework for sports pharmacy education and athlete care.

Conclusion: This project focused on creating a comprehensive sports pharmacy curriculum by integrating global educational resources. The proposed framework aims to empower healthcare providers with the knowledge and skills necessary to optimize athlete care while promoting clean sport practices.

Keywords: Sports pharmacy, Curriculum, Undergraduate

192

From manual to digital: Revolutionizing medication home delivery at Primary Health Care Corporation (PHCC) in Qatar

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Background: PHCC's medication home delivery service ensures patients receive medications without visiting health centers. However, manual processes cause inefficiencies, errors, and high medication return rates. Digitalizing the service enhances efficiency and patient satisfaction.

Aim: This study aims to develop and implement an electronic system using Microsoft Power Apps – SharePoint to digitalize all steps of the medication home delivery process, reducing medication returns, minimizing medication errors, and enhancing service efficiency.

Methods: An electronic system was developed via Microsoft Power Apps-SharePoint to streamline the medication home delivery process. Multidisciplinary teams from pharmacy, drug supply, ICT, and finance departments developed, tested, and piloted the system before implementation across 31 health centres. The system includes modules for patient registration, contact validation, address verification, QPost barcode scanning, phone consultations, billing, delivery confirmation, return management, printing stickers with patient details and tracking information, and sending satisfaction survey links via SMS post-delivery.

Results: PHCC delivered 75,675 medication orders to 69,685 patients, achieving a 98% success rate. The return rate dropped from 5% to 2% after implementing the electronic system. The top five medical conditions managed were diabetes mellitus, hypertension, vitamin D deficiency, dyslipidemia, and viral upper respiratory tract infection. Consequently, the top medications delivered were Ergocalciferol, metformin, rosuvastatin, levothyroxine, and paracetamol. Real-time dashboards enabled tracking of key performance indicators, including delivery success rates and reasons for returns.

Conclusion: The digitalization of the medication home delivery service at PHCC has significantly improved operational efficiency and reduced medication return rates. This study enlightens us on the immense value of digital tools in enhancing healthcare delivery services.

Keywords: Digitalized delivery, Primary care

193

Foundations of communicating effectively in healthcare (Fakeh) module: A transformative approach for interprofessional education

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Background: Effective communication is vital in healthcare for delivering quality patient care, team coordination, and operational efficiency. To address this need, the "Foundations of Communicating Effectively in Healthcare (FAKEH)" module was developed.

Aim: This innovative module comprises a comprehensive training course, designed to provide healthcare professionals for effective communication within a healthcare context.

Methods: The FAKEH module uses a blended learning approach, combining online and self-study components with interactive, face-to-face sessions. The e-learning segment includes six didactic topics covering communication concepts, supplemented by reflective tasks to analyse personal experiences. Group activities such as infographic creation and scenario-based exercises enhance written and verbal communication skills. The module also includes an Objective Structured Pharmacy Examination (OSPHE) to address ethical patient care dilemmas and difficult conversations.

Results: A pilot implementation involving 103 final-year students from the School of Pharmaceutical Sciences at Universiti Sains Malaysia revealed a significant improvement in communication skills awareness after module completion. Most students (76%) rated their awareness of communication skills between 8 and 10 after the module, compared to 0–6 before.

Students praised the module's objectives, content, and interactive approach. Feedback highlighted the benefits such as, "We learned crucial communication skills with the online modules," and "This course simulated real-life situations as a pharmacist."

Role-play sessions were particularly impactful, allowing students to reflect on their behavior and identify areas for improvement. One student noted, "During role plays, I realized I can be intimidating. Now, I practice staying calm" and another, "The role-play part was a 'wow' moment because we got to experience handling patients and other healthcare providers in different scenarios."

Conclusion: The FAKEH module significantly improved students' communication awareness and skills which provides a sustainable model for healthcare curricula, supporting interprofessional education and effective communication in medical practice.

Keywords: Module, Interprofessional Collaboration, Communication skill

Enhancing oncology pharmacy education: A gamified simulation module for improved student engagement and competency

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Background: Gamification in education is an innovative approach to enhance learning experiences, particularly in complex fields like oncology pharmacy.

Aim: This study explored the implementation of a gamified oncology pharmacy simulation designed for pharmacy students.

Methods: The simulation integrates role-play elements, with participants assuming the roles of pharmacist, patient, evaluator, and videographer. Each scenario is based on a pre-selected case, providing a structured yet dynamic environment for students to apply theoretical knowledge in practical, real-world situations. The patient role is enriched with behavioural cards, ensuring a diverse range of responses and challenges for the pharmacist. Evaluators, equipped with detailed assessment forms, provide structured feedback on both clinical and communication skills. Pharmacists also self-assess using these forms, fostering self-reflection and continuous improvement. The videographer captures the interactions, allowing for post-simulation review and analysis.

Results: This multi-faceted approach aims to enhance critical thinking, clinical decision-making, and interpersonal skills among pharmacy students. Preliminary findings indicate that the gamified simulation not only increases engagement but also improves competency in managing oncology patients.

Conclusion: The gamified OPS module shows promise in enhancing student engagement and competency in oncology pharmacy. Further research is needed to optimize the design and assess long-term impacts on professional practice.

Keywords: Gamification, Oncology pharmacy, Competency, Role-play

196

Transforming healthcare: Elevating interprofessional collaboration and pharmaceutical care through the collaboration (IPC) course

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Background: Effective interprofessional collaboration (IPC) is essential for improving healthcare outcomes. Recognizing the need for enhanced IPC skills among healthcare professionals, Universiti Sains Malaysia (USM) and Pharos University in Alexandria (PUA) developed the IPC Course.

Aim: This initiative aimed to enhance interprofessional collaboration and pharmaceutical care by equipping participants with the necessary skills to manage conflicts and identify pharmaceutical care issues (PCI) and drug-related problems (DRP).

Methods: The IPC Course includes both theoretical and practical components. The theoretical section consists of six Massive Open Online Course (MOOC) modules covering IPC concepts, reinforced through assessment tests and learning activities. The practical component involves a one-day Pharmacists Workup Drug Therapy (PWDT) application workshop (SLT: 16.5 hours) and a two-day IPC workshop (SLT: 36 hours) featuring case-based learning and role-play simulations.

Results: Participants demonstrated improved understanding of pharmaceutical care principles, effective use of the PWDT tool in clinical practice, and enhanced ability to recognize PCI and DRP. They also gained skills in conflict management and interprofessional collaboration. Certification was achieved by passing a final quiz with a score of over 90%.

Conclusion: The IPC Course successfully enhanced interprofessional collaboration and pharmaceutical care skills among pharmacy students and practicing pharmacists. This aligns with Sustainable Development Goal 4: Quality Education, contributing to improved healthcare outcomes.

Keywords: Interprofessional Collaboration, Pharmaceutical Care, Healthcare Education, Massive Open Online Course

Exploring the current practice of drug information service provision among community pharmacists in Qatar: A cross-sectional study

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Background: Community pharmacists are considered as the first point of contact for the public. As highlighted by American Society of Health-System Pharmacists (ASHP), drug information services are among the fundamental professional responsibilities of the pharmacists. This means that community pharmacists are responsible for addressing the drug information queries of patients and healthcare providers.

Aim: The study investigated the current practices, preparedness, barriers, and professional development needs of the community pharmacists in Qatar drug information services provision.

Methods: A descriptive cross-sectional online 48 item questionnaire-based survey constructed by experts in drug information and survey administration, was administered between 26 April 2023 and 14 May 2023. Questions related to current practice required responses on 5-point Likert scale, ranging from “always” to “never”. Questions related to level of preparedness, barriers and recommendations required responses on 5-point Likert scale, ranging from “strongly agree” to “strongly disagree”. Questions related to competency required responses on 3-point Likert scale, ranging from “competent” to “not competent”. The questionnaire was evaluated for face and content validity by experts in survey administration.

Results: Out of 236 responses received, only 211 completed the survey in entirety and were included in the analysis, resulting in a usable response rate of 89.4%. Most pharmacists responded that they “always” received drug information questions from patients/clients (50.0%). Majority of the respondents were well prepared to answer drug information questions, while also admitting that they need further training in the provision of drug information services. Time was one of the common barriers identified (50.5%). There was no significant relation between demographics and competency to deliver drug information services.

Conclusion: The community pharmacists in Qatar are prepared to deliver drug information services and are willing to take additional training to improve their drug information provision capabilities.

Keywords: Drug information, Community pharmacists, Qatar, Preparedness

199

Optimizing patient care at Hamad Bin Khalifah Medical City (HBKMC) using advanced transport automation

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Background: Many healthcare institutes employ pneumatic tube systems to deliver drugs between units. While pneumatic tubes may be easy, quick, and cost-effective, they are not suitable for all types of drugs. The admission of more patients has made it more challenging to administer medications effectively. This issue has been fixed by a pneumatic tube syst?? (system?).

Aim: This study aimed to evaluate the impact of the pneumatic tube system on reducing turnaround time, improving drug administration, and assessing nurses' satisfaction through a structured questionnaire.

Methods: The investigation was conducted in a sizable tertiary care hospital where medication was provided via a pneumatic tube system. Data was gathered between March and May of 2024 to compare the distribution of drugs by pneumatic tube technology to human-based transportation. Also, find out through a questionnaire how satisfied nurses are with the pneumatic tube system.

Result: The study discovered that whereas drug distribution using a pneumatic tube method takes an average of 2 min for the three institutions, each cycle completed by human power takes approximately 37 min on average due to distance, elevator issues, and personnel shortages. Data show that when compared to the pneumatic drug delivery technology, human resources make much fewer deliveries each hour. It is that most nurses are quite satisfied with the pneumatic tube system.

Conclusion: A useful substitute for using human strength to administer drugs is a pneumatic tube system. From each round, we set aside these thirty-five minutes, so it is used for other crucial activities. This results in the delivery of medications to the designated wards in a matter of

seconds, reducing the need for human power while simultaneously achieving high-quality patient care.

Keywords: Pneumatic tube, Automation, Hospital pharmacy

201

Genetic determinants of vitamin D deficiency in Middle Easterners with a relevant characterization into the novel *SDR42E1* gene

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Background: Vitamin D deficiency is highly prevalent in Middle Eastern populations, yet most genetic studies primarily focus on European cohorts. Interestingly, genomic research has identified a nonsense variant in the *SDR42E1* gene as a potential contributor to vitamin D deficiency, but its role in vitamin D regulation remains understood.

Aim: This study aimed to identify genetic factors influencing vitamin D levels in Qatari and Lebanese populations and develop a regional vitamin D polygenic risk score. Additionally, it aims to elucidate the *SDR42E1* gene's role in vitamin D regulation and its integration into pharmacogenomic approaches for personalized healthcare.

Methods: We conducted the first genome-wide association study (GWAS) on Middle Eastern populations to identify genetic determinants of vitamin D levels, analyzing 6,047 Qatari (IRB: QF-QGP-RES-ACC-00075) and 199 Lebanese (IRB: IM-GEHF-20) subjects. Functional and structural characterization of *SDR42E1* was performed using CRISPR/Cas9 genome-editing in cutaneous HaCat and intestinal HCT116 human cells, integrated with multi-omics approaches.

Results: Our analysis identified a novel variant, rs2298850 ($P = 1.71 \times 10^{-08}$), in the *GC* gene among Qataris and two suggestive variants, rs141064014 in *MGAM* ($P = 4.40 \times 10^{-06}$) and rs7036592 in *PHF2* ($P = 8.43 \times 10^{-06}$), in Lebanese. European-derived polygenic risk scores for vitamin D levels exhibited lower predictive performance in Middle Easterners, yet effectively predicted vitamin D deficiency. Functional studies of *SDR42E1* revealed its regulatory role in vitamin D pathways, influencing genes, like *ALPP*, *ABCC1*, and *SLC7A5*, alongside pathways related to steroid biosynthesis and cancer prognosis in *SDR42E1* knockout models.

Conclusion: Our research emphasized the importance of population-specific genetic determinants in advancing pharmacogenomics and precision medicine. The novel *SDR42E1* gene plays a pivotal role in vitamin D deficiency, providing a foundation for personalized interventions and innovations in pharmacy education and clinical practice.

Keywords: Vitamin D deficiency, *SDR42E1*, Pharmacogenomics, Personalized medicine

202

Budget impact analysis of subcutaneous versus intravenous trastuzumab for HER2-positive breast cancer in Jordan

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Background: In Jordan, the introduction of biosimilar versions of intravenous (IV) trastuzumab has prompted concerns regarding cost-efficiency of the patented subcutaneous (SC) trastuzumab.

Aim: This study aimed to undertake a budget impact analysis to evaluate the economic implications of adopting SC trastuzumab versus IV trastuzumab for treating HER2-positive breast cancer.

Methods: A prevalence-based budget impact model (BIM) was created during one year to coincide with annual national budget planning and pharmaceutical procurement in Jordan. The investigation covered the public prices for trastuzumab therapies as well as any direct medical expenditures associated with therapy delivery in Jordan. The base case analysis calculated the annual cost per patient for each trastuzumab treatment available in Jordan. In the scenario study, the net monetary budget (NMB) impact was computed for two environments: one without SC trastuzumab and one with SC trastuzumab, with a gradual replacement from 10% to 100% (complete replacement). Sensitivity studies were conducted to examine the NMB impact using 95% confidence ranges for estimated prevalence and incidence.

Results: The anticipated number of eligible patients receiving trastuzumab treatment is 566 (95%CI = 498-633). Increased expenses that differ depending on the treatment plan are typically linked to the introduction of trastuzumab

SC. In general, incremental NMB at an average of JOD11,209 per patient-year is linked to the conversion from IV trastuzumab to SC trastuzumab. With 566 people expected to be diagnosed and treated by 2024, the additional expense of replacing all patients with trastuzumab SC might total JOD 5,160,613 (95%CI = JOD 4,543,779-JOD 5,777,446).

Conclusions: The adoption of SC trastuzumab in Jordan should take into account patient preferences, the price elasticity of SC trastuzumab, the price advantage of IV biosimilars, and possible savings in healthcare resources with the SC formulation over the IV formulations.

Keywords: Budget impact, Breast cancer, Jordan, Trastuzumab

203

Pharmacological management of severe behavioural disturbances in cognitively impaired elderly hospital inpatients in a teaching hospital in Australia

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Background: The management of severe behavioural disturbances in cognitively impaired elderly hospital inpatients can be quite challenging due to the high vulnerability of this patient group. The pharmacological options, antipsychotics, recommended can increase risk of falls, cardiovascular events and/or mortality. Ensuring the most appropriate pharmacological option, in terms of safety and efficacy, is prescribed is a priority.

Aim: The aim was to identify if the pharmacological management options of severe behavioural disturbances, in cognitively impaired elderly hospital inpatients, in the updated procedure are: 1) more aligned with “standard” resources and 2) more appropriate for this patient group.

Methods: Australian resources in this procedure, identified by observing the most commonly accessed and relied on resources, by hospital prescribers, for this specific patient group, are labelled as “standard”: 1) Australian Medicine’s handbook Aged care Companion 2) Therapeutic Guidelines (eTG) and 3) Cognitive Impairment management toolkit. A direct comparison of pharmacological recommendations in the outdated and updated local procedures, in addition to the “standard” resources, occurred to identify alignment and appropriateness.

Results: The updated procedure showed significantly improved alignment with all three standard Australian resources. Key changes included clearer

delineation of first-line pharmacological options, with a shift away from agents associated with higher risks in older adults (e.g. high-potency typical antipsychotics). The updated procedure now prioritizes low-dose, short-term use of atypical antipsychotics such as risperidone, consistent with the recommendations in the AMH Aged Care Companion and eTG.

Conclusions: The procedure was updated appropriately to ensure safe and effective pharmacological options are recommended for the management of severe behavioural disturbances in this patient group. Further research will need to be conducted to assess compliance of prescribers to this procedure.

Keywords: Behavioural disturbance, Cognitive impairment, Antipsychotic

204

Clinical features and outcomes of newly diagnosed Hodgkin lymphoma patients in Saudi Arabia: A multicenter cohort study

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Background: Hodgkin Lymphoma (HL) is a prominent B-cell neoplasm affecting global and Saudi populations. Given limited large-scale studies on

HL in Saudi Arabia, this retrospective multicenter cohort study reviewed the characteristics and clinical outcomes of HL patients in Saudi Arabia.

Aim: To evaluate clinical features, treatment responses, and survival outcomes of HL patients in Saudi Arabia.

Methods: Adult patients aged more than 18 years, newly diagnosed with classical HL (cHL) at three tertiary care hospitals between 2016 and 2021, were included. Primary outcomes were histopathology, risk category, disease stage, and risk factors. Secondary outcomes included complete response (CR) rates post-initial chemotherapy, post-chemotherapy complications, time to relapse/refractory disease, progression-free survival (PFS), and overall survival (OS).

Results: Of 765 screened patients, 498 with cHL were included (median age: 30 years). Most had nodular sclerosis (80.7%) and stage IV disease (44.8%). ABVD chemotherapy was predominant (84.9%), with a 90.4% overall response rate and 85.2% achieving CR. After a median follow-up of 41.3 months, the OS rate was 94.9%, and median PFS was not reached. Among 89 patients (17.8%) with relapsed/refractory HL, 56 underwent hematopoietic stem cell transplantation as second-line therapy. Post-second-line therapy, the overall response rate was 80%. Transplanted patients had significantly longer median OS than non-transplanted patients, though PFS differences were not significant.

Conclusion: This national study highlights HL characteristics and outcomes in Saudi Arabia. ABVD chemotherapy achieved high response and remission rates with favourable survival outcomes. Hematopoietic stem cell transplantation improved OS for relapsed/refractory patients, underscoring its importance in enhancing outcomes.

Keywords: Hodgkin Lymphoma, Saudi Arabia, Refractory, Cytogenetics

207

Factors associated with prescribing initial antidiabetic drugs among patients with Type 2 diabetes mellitus in Scotland, 2010–2020: A retrospective population-based study

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Background: Clinical guidelines emphasize a patient-centered approach in selecting antihyperglycemic drugs (ADDs) for managing type 2 diabetes mellitus (T2DM), considering demographic, clinical, and socioeconomic factors. However, limited guidance on prioritizing these factors contributes to variability in prescribing practices. Understanding key determinants of ADD prescribing can enhance clinical practices, refine guidelines, and inform healthcare policies.

Aim: The study aimed to explore factors influencing ADD prescribing at initiation among adults with T2DM in Scotland.

Method: Using national data from Scotland (2010–2020), this retrospective cohort study analyzed patients with T2DM initiating ADD therapy. Inclusion criteria required patients to be new ADD users, without prior prescriptions in the preceding year. The study assessed demographic (e.g. age, sex), clinical (e.g. comorbidities), socioeconomic, and prescriber-related factors. Descriptive statistics and logistic regression were employed with statistical significance set at $p < 0.05$.

Results: The study included 145,909 patients (median-age: 61-years; 57.9% male). Demographic factors, especially age and sex, significantly influenced prescribing patterns. Elderly patients (≥ 65 years) were more likely to receive Dipeptidyl-peptidase-4-inhibitors (DPP4-I) or sulfonylurea (SU) and less likely to be prescribed sodium-glucose-transporter-2-inhibitors (SGLT2-I). Female patients were less likely to receive SU than males (Adjusted OR[95% CI]: 0.89[0.85–0.93]).

Clinical factors are only partially aligned with guidelines. Surprisingly, cardiovascular/renal conditions, such as ischemic heart disease or low eGFR ($< 60 \text{ ml/min/1.73 m}^2$), were not significantly associated with prescribing newer drugs like glucagon-like-peptide-1-receptor agonists and SGLT2-I. Additionally, patients with HbA1c $\geq 9\%$ were more likely to receive insulin or SU, while overweight individuals had lower odds of being prescribed DPP4-I, insulin, or SGLT2-I.

Conclusion: This study highlights significant associations of sociodemographic and clinical factors with ADD prescribing. Clinical factors related to cardiovascular/renal diseases often deviated from guideline recommendations, particularly for newer ADDs. This finding underscores the need for improved prescriber education to enhance adherence to evidence-based practice and to optimize patient outcomes.

Keywords: Diabetes, Antihyperglycemic, Evidence-based practices

The neonatal outcomes of 12-hourly betamethasone regimen compared to 24-hourly betamethasone dosing regimen for pregnant patients

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Background: Preterm delivery is associated with neonatal complications such as: respiratory distress syndrome (RDS) and neonatal mortality. Antenatal betamethasone is shown to reduce these complications and is typically delivered as a 24 mg course administered intramuscularly every 24 hours in two divided 12 mg doses. However, some practitioners prefer to prescribe it every 12 hours to assure that patients receive the full regimen before delivery. No robust evidence supports the new frequency regimen.

Aim: To compare the incidence of RDS, RDS severity, and mortality in neonates of mothers who received 24 mg of betamethasone at 12-hour or 24-hour intervals.

Methods: This retrospective cohort study was conducted between 2019 and 2022 in Doha, Qatar. A total of 168 patients who received betamethasone between 23 0/7 and 36 0/7 weeks of gestation were included. Only patients who delivered prematurely within 7 days of receiving the betamethasone regimen were included. Frequency and severity of RDS and neonatal mortality were compared between 12-hour and 24-hour cohorts.

Results: There was no significant difference in RDS rate ($p = 0.569$) or mortality ($p = 0.349$) between the cohorts. Stratification by gestational age at delivery was performed to account for varying RDS risk; no differences in RDS between the two dosing groups were observed following stratification ($p = 0.122$). No statistically significant difference in RDS severity was noted.

Conclusion: An accelerated 12-hourly betamethasone dosing regimen did not improve neonatal RDS, or mortality outcomes compared to a 24-hourly regimen.

Keywords: Betamethasone, Retrospective cohort, Neonates

211

Multidisciplinary interventions for male infertility: An exploratory study combining nutritional supplementation and traditional malay massage therapy to boost sperm quality

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Background: Male infertility is an alarming reproductive health concern and is frequently associated with aberrant semen parameters.

Aim: This study explored the effectiveness of combining nutritional supplements (Profertil®), a standard treatment for aberrant semen parameters, and traditional Malay massage on sperm quality, notably motility, concentration and count.

Methods: Seven (7) men diagnosed with abnormal semen analysis by a reproductive consultant underwent a 10-week treatment regimen of daily Profertil® supplementation combined with weekly Malay massage therapy (MMT) sessions. The sartorius muscle, femoral vasculature, inguinal region, belly, pubic area, lower back, and buttocks were the main targets during the MMT, which was massaged by a certified masseuse. Each session lasted 60 min, with a total of 10 sessions. Semen samples were collected before, during and after the intervention to assess sperm motility, count, and concentration.

Results: Results showed an improvement in sperm motility following the intervention, whereas sperm count and concentration were variable. This exhibits that combining nutritional supplements (Profertil®) with MMT may enhance sperm motility and potentially increase the probability of conception, even when sperm count and concentration remain inconsistent.

Conclusions: These findings highlight the potential role of traditional Malay massage in improving sperm quality in men with aberrant semen parameters. Further research is required for investigating the long-term effectiveness of this combined treatment and the mechanisms underlying its impact on sperm quality.

Keywords: Male infertility, Malay massage, Sperm motility, Semen quality improvement

212

Development and evaluation of a fall prevention flipchart as a pharmacist-provided counselling material

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Background: Medications are one of the modifiable risk factors contributing to fall. Those are commonly known as “Fall-Risk Increasing Drugs” (FRIDs) and are common among older people (OP). However, older people and their family caregivers (FCGs) are largely unaware of FRIDs and their risks in increasing the risk of fall (ROF).

Aim: The fall prevention flip chart was created and assessed to educate OPs and FCGs about FRIDs. Current existing literatures have shown lack of published and validated pharmacist-provided counselling materials worldwide that specifically included FRIDs.

Methods: The fall prevention flip chart adapted the Centres for Disease Control and Prevention (CDC) Development of STEADI-Rx framework incorporating suitable design for OP. The flip chart was content validated by a panel of 6 reviewers through the Patient Education Materials Assessment Tool (PEMAT). Evaluation of the flip chart, was done among 50 OPs and 50 FCGs in Felda Purun, a rural area located in Bera District, Pahang. Perceived knowledge of the participants were assessed prior to and after completing the flip chart counselling session through set of questionnaires.

Result: The fall prevention flip chart intervention led to significant improvements in knowledge ($p < 0.001$) among both OP and FCG groups with both groups achieving high post-test scores across all measured domains.

Conclusion: The fall prevention flip chart is a vital, user-friendly tool that effectively educates OP and FCGs in Felda Purun. This reduces falls and improving health outcomes, while also serving as a key resource for public health policy and community safety efforts. This flip chart also serves an important commitment towards OP and people living in the rural area equivalent to the Sustainable Development Goal of the United Nation, which to ensure that no one is left behind.

Keywords: Fall prevention, Pharmacist, Counseling

213

Cost analysis of ceftriaxone vs. moxifloxacin as first-line empirical antibiotics for community-acquired pneumonia in hospitalized adults

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Background: Community-acquired pneumonia (CAP) remains a major healthcare burden due to its significant cost and impact on patient morbidity and mortality. With the increasing prevalence of CAP, hospitals are continually exploring cost-effective treatment strategies. While antibiotics like ceftriaxone and moxifloxacin are commonly used as first-line empirical therapies, a pharmacoeconomic analysis comparing the two has not been conducted.

Aim: The primary objective of this study was to evaluate the clinical efficacy and economic impact of ceftriaxone compared to moxifloxacin as a first-line empirical antibiotic for the treatment of CAP in hospitalized adult patients.

Methods: This retrospective observational study utilized medical records and billing data from Liaquat National Hospital, covering the period from September 2024 to December 2024. Data collection occurred between June and July 2024. Adult inpatients aged 18 years or older, diagnosed with CAP and treated with either moxifloxacin or ceftriaxone as a first-line empirical antibiotic, were included. Key variables included patient demographics, antibiotic effectiveness, length of hospital stay, and direct medical costs.

Results: The study found that moxifloxacin has significantly higher antibiotic effectiveness rate compared ceftriaxone. Additionally, patients treated with moxifloxacin had a shorter length of stay and lower total direct medical costs compared to those receiving ceftriaxone.

Conclusion: Moxifloxacin demonstrates superior antibiotic effectiveness, reduces the length of hospital stay, and results in cost savings when used as a first-line empirical treatment for CAP in hospitalized adult patients.

Keywords: Moxifloxacin, Antibiotic, Pharmacoeconomics, CAP

Comparative efficacy of subcutaneous compared to intravenous biologics for inflammatory bowel disease: A systematic review and meta-analysis

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Background: The comparative efficacy and safety of subcutaneous versus intravenous administration of biologics for achieving remission in inflammatory bowel disease (IBD) remain uncertain. Evaluating these two delivery routes is essential to support clinicians in making well-informed decisions in collaboration with their patients.

Aim: This systematic review and meta-analysis synthesized all existing evidence to compare the efficacy and safety of subcutaneous versus intravenous biologics in achieving remission for moderate-to-severe IBD.

Methods: In this systematic review and meta-analysis, we searched Cochrane, PubMed, SCOPUS, CINAHL, and preprint archives for trials that compared the efficacy and safety of subcutaneous to intravenous biologics for the induction, , and endoscopic remission of IBD. Meta-analysis was carried out using the bias adjusted inverse variance heterogeneity (quality-effects) model with subgroup analysis for Crohn's disease (CD) and Ulcerative Colitis (UC), heterogeneity using I^2 , and publication bias using funnel and Doi plots.

Results: A total of 15 randomized controlled trials were included. Subcutaneous, compared to intravenous administration, was less efficacious for induction of remission, overall (OR 0.68, 95%CI 0.35-1.31), worse in UC (OR 0.35, 95%CI 0.07-1.79), and showed similar efficacy in CD (OR 0.97, 95%CI 0.73-1.30, $I^2 = 0\%$). For maintenance of remission, subcutaneous biologics were almost similar to intravenous biologics, overall (OR 0.97, 95%CI 0.63-1.49, $I^2 = 47.1\%$), with less efficacy in UC (OR 0.82, 95%CI 0.54-1.23, $I^2 = 52\%$), but superior efficacy in CD (OR 1.81, 95%CI 1.09-3.01, $I^2 = 0\%$). For endoscopic remission, Subcutaneous administration was less effective than intravenous overall (OR 0.84, 95%CI 0.56-1.26), worse in UC (OR 0.72, 95%CI 0.52-0.99, $I^2 = 34\%$), but more effective in CD (OR 1.90, 95%CI 0.85-4.26, $I^2 = 0\%$). Subcutaneous compared to intravenous biologics, showed nearly similar odds for

adverse events (OR 1.05, 95%CI 0.84-1.31, $I^2 = 30.2\%$) and serious adverse events (OR 1.09, 95%CI 0.83-1.42, $I^2 = 0.0\%$).

Conclusion: Subcutaneous administration demonstrates lower efficacy in inducing remission but achieves comparable efficacy and safety to intravenous administration in maintaining remission in IBD. Subcutaneous injection shows greater efficacy and safety in CD compared to UC.

Keywords: Systematic review, IBD, Biologicals, Intravenous

218

Competency-based curricula in the eastern mediterranean region schools of pharmacy: A framework-informed mixed method study

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Background: Competency-based education (CBE) has been implemented across healthcare disciplines, including pharmacy to address the limitations of the traditional teaching system. While CBE implementation in pharmacy education has been studied globally, research examining its adoption across pharmacy schools in the Eastern Mediterranean Region (EMR) remains limited.

Aim: Explore the nature and extent of CBE incorporation within pharmacy schools' curricula across the EMR.

Methods: The study applied a sequential explanatory mixed-methods approach, combining an online pre-tested survey for the quantitative phase followed by semi-structured online interviews for qualitative insights. The study was guided by the International Pharmaceutical Federation Global Competency Framework (FIP GbCF).

Results: A total of 38 responses were collected from the pharmacy schools in the EMR (response rate 47%) and 18 participants participated in the online interview. Most pharmacy schools were accredited (82%) and (63%) of which implement/incorporate CBE in their curricula. Analysis of the FIP GbCF competencies incorporation revealed that medicines information and advice and

Research theme

Teaching and Learning in Pharmacy

communication skills were the most prevalent competencies (92%), followed by professional and ethical practice (88%). The least incorporated competencies included health promotion and workplace management (42%) followed by human resource management (38%). Inductive thematic analysis identified key challenges for the implementation of CBE in some pharmacy schools including internal resistance, apprehensive leadership and faculty overload. Schools implementing CBE highlighted faculty mindset adaptation and assessment restructuring as ongoing challenges.

Conclusion: Most accredited pharmacy schools in the EMR adopted CBE, yet considerable gaps remain in implementing many of the of FIP GbCF competencies. The study reveals that pharmacy schools face significant challenges in CBE implementation. This is regardless of their current adoption status, suggesting opportunities for enhancement of pharmacy education across the region.

Keywords: Pharmacy education, Competency-based education, Accreditation, Framework, Eastern Mediterranean Region

219

Strengthening medication adherence through social support: A qualitative study of schizophrenia outpatients in Indonesia

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Background: Schizophrenia is a stigmatized mental illness requiring sustained social support for therapeutic success. Patients rely heavily on caregivers, who experience burdens due to the chronic nature of the disease. Understanding the forms of social support that influence medication adherence is crucial.

Aim: This study explored social support experienced by people with schizophrenia (PWS) that promotes medication adherence.

Methods: A qualitative study was conducted in 2024 in rural Yogyakarta, Indonesia. Semi-structured interviews were held with 20 participants, including PWS (n = 14), caregivers (n = 3), and community health workers (n = 3). Deductive thematic analysis was used to analyse the data. Ethical approval was obtained from the Medical and Health Research Ethics Committee

(MHREC), Faculty of Medicine, Universitas Gadjah Mada, under reference number: KE/FK/1580/EC/2023.

Results: Four main themes of social support emerged: (1) advocacy and education, (2) professional and formal support, (3) social relationships, and (4) community program interventions. Community-led efforts, spearheaded by a non-governmental organization (NGO), included advocacy, education, and community-based rehabilitation programs in collaboration with local public health centers. These efforts improved social relationships between PWS, families, and the community, fostering better medication adherence. Barriers to effective social support were identified and this include (1) prejudice, (2) stigma, (3) discrimination, and (4) challenges in implementing interventions. Prejudice, stigma, and discrimination were more noticeable during the initial stages of NGO intervention but gradually decreased over time. Programmatic challenges included insufficient government involvement and limited availability of male community health workers.

Conclusion: Social support is critical for medication adherence in schizophrenia. While NGO initiatives have reduced stigma and enhanced community-based rehabilitation, structural and cultural barriers persist. Greater government involvement and targeted interventions are needed to optimize social support systems.

Keywords: Schizophrenia, Medication adherence, Social support

221

Efficacy and safety of therapeutic alternatives of the management of hypertriglyceridemia-induced acute pancreatitis: A systematic review

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Background: The optimal management strategy for hypertriglyceridemia-induced acute pancreatitis (HTGAP) remains uncertain, despite widespread use of treatment modalities including insulin therapy, heparin infusion, and plasmapheresis. Current evidence supporting the effectiveness and safety of these interventions is limited, leading to ongoing debate about their role in HTGAP treatment.

Aim: To assess the efficacy and safety of insulin infusion, plasmapheresis, hemofiltration, and heparin in the management of HTGAP.

Method: Following PRISMA guidelines for systematic reviews, we conducted a comprehensive search of the following electronic databases from their inception until October 2024: PubMed, EMBASE, Science Direct, Cochrane library, ProQuest and Scopus. The studies were included if they were human-based quantitative study design involving a comparison group. The quality assessment tools applied were Jadad scale and CASP tool.

Results: Fourteen eligible studies were included in this review, the majority of which were cohort studies ($n = 12$) with ($n = 10$) followed a retrospective observational study design. Most studies were conducted in China ($n = 9$). Plasmapheresis significantly reduced triglyceride levels in six studies, while hemofiltration showed similar effectiveness in two studies. Plasmapheresis also demonstrated a statistically significant reduction in length of hospital stay across four studies. Majority of the studies ($n = 12$) reported no statistically significant effect on mortality. Safety reporting was notably limited, with only four studies documenting treatment-related adverse events.

Conclusion: There is some evidence to suggest that plasmapheresis and hemofiltration can have desirable effect in the management of HTGAP. However, it is challenging to draw a definitive conclusion about the efficacy and safety of HTGAP therapeutic interventions due to heterogeneity of the reported data and lack of well-designed studies. There is a need for high-quality research with robust methodological design to establish an informed consensus regarding the efficacy and safety of the available treatment strategies.

Keywords: Hypertriglyceridemia, Hyperlipidemia, Acute pancreatitis, Therapeutic management

222

Assessing quality improvement knowledge, skills, and attitudes among pharmacy professionals in Qatar

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Background: Optimizing patient care and medication safety requires a strong foundation in quality improvement (QI) principles and practices. Pharmacists play a crucial role in QI initiatives, but a gap exists between awareness

of QI principles and their practical application.¹⁻² Limited data on the awareness and preparedness of hospital pharmacists in Qatar highlights a need for assessment.

Aim: This study investigates pharmacy professionals' understanding of QI principles, their perceived competency in implementing QI initiatives, and their overall attitudes towards QI.

Methods: A cross-sectional web-based questionnaire using the Beliefs, Attitudes, Skills, and Confidence in Quality Improvement (BASiC-QI) scale was distributed to pharmacy professionals electronically from Hamad Medical corporate pharmacy office by email.

Results: Out of 158 respondents, the majority (63.7%) had received formal QI training, with 58.1% rating it as "very effective" and 33.3% as "moderately effective," despite this a significant 86% still expressed a need for further training. Most respondents expressed strong agreement with statements indicating their interest in QI (34.3%) and understanding of its role in strengthening healthcare systems (47.5%). Furthermore, an understanding of core QI concepts was reported, including theory (39.1% agree), systems thinking (42.1% agree), and the link between change and improvement (38.4% agree). However, when it came to specific skills, confidence levels varied. The percentage reporting "Extremely confident" remained consistently low across all skills (4.8-11.2%). Of note was the relatively high percentage of respondents who reported "Not confident whatsoever" in skills related to using the Model for Improvement (15.5%) and using PDSA cycles (15.3%) in quality projects.

Conclusions: Pharmacy professionals in Qatar demonstrate a positive attitude towards QI and possess knowledge of core concepts. However, targeted training and support in areas like the Model for Improvement, leadership, and evaluation are needed to enhance the effectiveness of quality improvement initiatives within the pharmacy profession.

Keywords: Attitude; Pharmacy, Knowledge, Quality improvement

223

Evaluating knowledge of high-alert medications among nurses, pharmacists, and clinicians to improve medication safety

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Background: Medication errors are unintentional errors that can happen at any stage of the drug treatment process, including prescribing, dispensing, storing, preparing, or administering medications. High alert medications are those that carry the greatest risk of causing severe harm to patients if used improperly. Nurses are tasked with administering these high-risk medications, and any errors in this process can lead to serious clinical consequences.

Aim: This study aims to evaluate hospital staff's knowledge of high-alert medications, focusing on their use, storage, dilution, and compatibility with other drugs.

Methodology: A cross-sectional survey was conducted to evaluate the knowledge and practices concerning high-alert medications (HAM), focusing specifically on electrolytes included in the HAM list. The survey aimed to gather insights knowledge from three professional groups – clinicians, nurses, and pharmacists. A sample of 60 participants was selected, with 20 individuals from each of the three groups. The survey was complemented by an analysis of high-alert medication consumption patterns over a 6-month period (June 2024–Nov 2024), providing a comprehensive view of both knowledge gaps and medication usage related to HAMs.

Result: The study assessed 60 participants and revealed varying levels of knowledge regarding high-alert medications. Pharmacists demonstrated the highest level of knowledge, with 80% of their responses being correct, followed by nurses at 75%, and clinicians at 70%. Notably, all participants (100%) showed an understanding of the composition, characteristics, and drug content percentages of high-alert medications. However, there was a significant gap in knowledge regarding the compatibility, storage, and labelling of these medications.

Conclusion: The findings reveal that pharmacists possess the most comprehensive understanding of high-alert medications underscoring a significant knowledge gap among other healthcare professionals, particularly clinicians and nurses.

Keywords: High alert medication, Medication error, Patient safety, Nurses, Questionnaire

224

Collaborative models between pharmacists and community health workers to enhance medication adherence: A scoping review

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Background: Medication adherence remains a significant challenge in managing chronic diseases, especially among vulnerable populations. This scoping review explores collaborative models between pharmacists and community health workers (CHWs) to enhance medication adherence.

Aim: The primary objective is to identify collaboration models involving pharmacists and CHWs that are designed to improve medication adherence.

Methods: A comprehensive literature search was conducted across five databases (PubMed, Scopus, Lens, Central, and Web of Science) for studies published between 2014 and 2024. The review included original research with both quantitative and qualitative designs focusing on interventions involving pharmacist-CHW collaborations. Following duplicate removal, three reviewers screened the results using Rayyan, extracted data onto an Excel spreadsheet, and conducted quality appraisals.

Results: Fifteen studies met the inclusion criteria: 7 quantitative (5 randomized controlled trials, 1 quasi-experimental, and 1 cohort study), 6 qualitative (5 interviews or focus groups, 1 case study), and 2 used mixed-methods design. Most (13 of 15) were conducted in the United States. Three collaboration models emerged: (1) CHWs provided education and cultural support while pharmacists managed medication therapy; (2) pharmacists offered remote consultations with in-person CHW support; and (3) CHWs served as cultural liaisons. Interventions were delivered through direct patient interactions, telehealth, or home visits. Barriers included communication gaps, workflow disruptions, and cultural misunderstandings. Enablers included cultural competence, trust-building, and CHW integration into healthcare teams. Collaborative interventions improved medication adherence, reduced hospital readmissions, and enhanced outcomes such as blood pressure control and glycaemic management. Patient satisfaction was consistently high due to culturally sensitive care.

Conclusion: This review highlights the potential of pharmacist-CHW collaborations in addressing medication adherence challenges, particularly in underserved communities. The success of these models lies in their integration of clinical expertise with cultural competence. However, sustainable implementation requires careful consideration of resource allocation and seamless integration into healthcare systems

Keywords: Collaboration, Pharmacist, Communication

225

AI meets human expertise: Comparison between clinical pharmacist interventions and artificial intelligence at a tertiary care hospital in Pakistan

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Background: Artificial Intelligence refers to the set of technologies that allows computer to perform a wide range of advanced functions including problem-solving, analysing data, making recommendations and more. Artificial Intelligence revolutionizes medication therapy management by improving patient care by analyzing vast amount of data to identify potential drug interactions, adverse effects, and dose optimization. It has been shown to be effective in other sectors of clinical medicine.

Aim: To compare and evaluate the interventions made by clinical pharmacist versus artificial intelligence (AI) at a tertiary care hospital setting in Pakistan.

Methods: This prospective study compared the interventions made by clinical pharmacists and artificial intelligence using the advanced version (ChatGPT 4.0) at a tertiary care hospital setting. The study spanned six months (June to November 2024) with 60 randomly selected patients. Data was collected from clinical pharmacist notes and detailed patient medical record. Patients with incomplete medical record were excluded from the study. Patient data including medications, lab results, medical history and current diagnosis was input into ChatGPT 4.0 to generate recommendations based on three basic intervention criteria: drug–drug interaction, renal adjusted doses of antibiotics and electrolyte imbalance management. To ensure validity and reliability three clinical pharmacist independently reviewed and validated the results using different references.

Result: ChatGPT 4.0 effectively solved 60 provided cases and successfully identified drug–drug interactions and electrolyte imbalance management but it did not accurately recommend renal adjusted doses of antibiotics which are critical in ICU set-up. Can you elaborate the results a bit more, its very brief

Conclusion: Although there are some restrictions and possible biases that open the door for advancements, our research indicates that ChatGPT has the potential to completely transform the management of drug therapy.

Keywords: Artificial Intelligence, Clinical Pharmacist, Interventions, Technology, Patient Care

227

Effectiveness and safety assessment of calcium channel blockers compared to beta blockers in patients with angina: An observational study

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Background: Angina pectoris is a prevalent sign of myocardial ischemia in people who have obstructive coronary artery disease. Angina causes 25% of the total deaths in developed and 80% in middle- and low-income countries. The prevalence of angina in Pakistan is 26.9% and it is one of the leading causes of death however females are more vulnerable for precipitating angina (18.30% women and 16.60% men) in Pakistan.

Aim: To combat the rising prevalence of ischemic heart disease and related problems in changing socioeconomic situations in Pakistan requires comprehensive public health policies that include preventive interventions and conveniently accessible healthcare facilities. European and US guidelines recommend that stable angina should be routinely treated with either a β -blocker (BB) or calcium channel blocker (CCB).

Methods: This study was performed in two tertiary care hospitals that provide cardiac care to approximately 50–100 patients in a month, the age ranges from 20 to 75 years. The study was approved by the Institutional Bio-ethical Research Committee of University of Karachi, with reference No. (IBC KU-317/2023) of Informed consent was taken prior from all enrolled patients since the study did not involve any intervention.

Results: In the current study, we found most of the angina patients' average age was 54.12 years which reflects a population within a middle-aged range, with a fairly equal distribution between males (55.45%) and females (44.54%). In Pakistan, these findings are in accordance with who reported the same incidence age of angina in Pakistani population. Elaborate the results a bit more

Conclusions: Combinations (CCB+BB) are more effective in angina treatment as BB and CCB alone. However, the ADRs related to CCB are less frequent over long-term use than BB.

Keywords: Angina, Beta-blockers, Calcium channel blockers, Myocardial ischemia, ADRs

Pharmacogenomics education and knowledge assessment in healthcare curricula: A scoping review of the middle eastern and North African Region

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Background: Pharmacogenomics (PGx) is an essential component of precision medicine, yet its integration into clinical practice remains slow, primarily due to knowledge gaps among healthcare professionals (HCPs). The lack of education has been identified as the most influential barrier to the successful implementation of PGx into practice, particularly in the Middle East and North Africa (MENA) region, where numerous studies have reported limited PGx education in their curricula.

Aim: To explore the current state of PGx education in healthcare curricula and its impact on the knowledge of students and HCPs in the MENA region.

Methods: A scoping review conducted in accordance with the PRISMA guidelines. A systematic search of PubMed, Embase, and CINAHL databases was conducted up to June 2023. Article screening and data extraction were performed independently by five reviewers. Eligible studies were analyzed for the availability of PGx education, its curricular placement, modes of delivery, levels of PGx knowledge among participants, and perceived barriers to PGx implementation.

Results: Seven cross-sectional studies were included, with the majority (57%) conducted in the Gulf region. PGx education was predominantly offered to undergraduate students (71%), delivered entirely through didactic lectures (100%), with standalone courses reported in only 29% of studies. Knowledge levels among participants ranged from low (43%) to moderate (43%), assessed either via objective scales or subjective self-assessments. Key barriers to PGx implementation included knowledge gaps, economic constraints, and challenges in achieving system sustainability.

Conclusion: PGx education in the MENA region was found to be insufficient within the healthcare undergraduate curricula, which is reflected on the low to moderate knowledge levels observed among students and HCPs. To support the implementation of PGx in clinical practice, more universities need to incorporate PGx in their curricula and to develop standardized

assessment tools. There is also a need to evaluate knowledge acquisition across HCPs using objective measures.

Keywords: Pharmacogenomics, Cross-sectional, Scoping review

230

Management of hand-foot syndrome (HFS) associated with capecitabine therapy for colorectal & breast cancer: Protocol of a randomised controlled clinical trial

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Background: Cancer therapies, particularly chemotherapy regimens like capecitabine, an oral chemotherapeutic drug frequently used to treat colorectal and breast cancers are often associated with a range of side effects, one of the most common being hand-foot syndrome (HFS), also known as palmar-plantar erythrodysesthesia. Despite the significant impact of this side effect on patient well-being, there are limited treatment options like use of NSAIDs that can effectively prevent or alleviate HFS without compromising the efficacy of the chemotherapy regimen.

Aim: The aim for this study is to compare the effectiveness of topical diclofenac versus Vaseline in preventing hand-foot syndrome in patients undergoing capecitabine therapy for colorectal and breast cancer.

Methods: A randomized, controlled and single-blind will be conducted at Liaquat National Hospital, Karachi, Pakistan. The prospective, parallel-group superiority trial further follow stratification based on age, gender, and type of cancer. Data collection allows patients through four to eight cycles of capecitabine therapy, with the aim to compare the effectiveness of the treatments in reducing HFS categorized on NCI-CTCAE grading scale.

Results: The primary outcome is to assess the prevention of Hand-Foot Syndrome (HFS), with secondary objectives including time to onset of HFS, patient-reported outcomes, adherence, capecitabine dose adjustments, and safety.

Conclusion: Ultimately, the results of this study could provide valuable insights into evidence-based, cost-effective solutions the use of these topical treatments, offering clinicians and patient's with viable options for

managing skin toxicity associated with chemotherapy, potentially improving patient comfort, treatment adherence, and overall quality of life.

Keywords: Hand-foot syndrome, Capecitabine, Randomised controlled trial

231

Enhance medication home delivery service process and improve staff satisfaction using a medication quantity per pack list during Covid-19 pandemic

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Background: During the Covid-19 pandemic, all medications for home delivery were dispensed from two centralized locations. Pharmacy staff involved in the service had varied experience and familiarity with medication pack quantities. This led to frequent errors in quantity entry during prescription processing, requiring pharmacists to cancel fills, adjust quantities, and reprint bills – a time-consuming and inefficient process, especially with high patient volumes.

Aim: To develop and implement a medication quantity per pack list to reduce pharmacist entry errors, improve workflow efficiency, and enhance staff satisfaction during the medication home delivery process.

Methods: The project followed multiple Plan-Do-Study-Act (PDSA) cycles: Cycle 1: Consolidated pack quantities for all medications into a single list. Cycle 2: Organized the list alphabetically to enhance usability. Cycle 3: Modified the list layout and design based on feedback from pharmacists to improve clarity and accessibility.

Results: The majority of pharmacists reported using the list multiple times per day. All respondents expressed high satisfaction, noting that the list significantly reduced entry errors and eliminated the need to cancel and reprocess prescriptions. The tool saved time and effort, reduced stress, and improved overall productivity, especially during peak hours by enabling faster prescription processing and minimizing workflow interruptions.

Conclusion: The implementation of a standardized medication quantity per pack list effectively reduced entry errors, improved efficiency, and enhanced pharmacist satisfaction. By preventing job duplication and streamlining the home delivery service process, this simple yet impactful

tool contributed to better time management and reduced stress among pharmacy staff.

Keywords: Medication home delivery, Covid-19, PDSA cycle

235

Cost-effectiveness analysis of fetal fibronectin testing in pregnant women

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Background: The foetal fibronectin (fFN) test is used in clinical practice to predict true preterm delivery, and its appropriate use can potentially reduce unnecessary medical interventions and healthcare costs in women experiencing preterm labour.

Aim: This study aimed to evaluate the cost-effectiveness of fFN testing versus no-fFN testing in identifying true preterm labour among women exhibiting preterm labour symptoms and at risk for premature birth in Qatar.

Methods: This retrospective cohort-based cost-effectiveness analysis was conducted at the Women's Wellness and Research Center (WWRC) of Hamad Medical Corporation (HMC) in Qatar. A conventional decision-tree model was designed to compare the use of the fFN testing against no-fFN testing. The primary outcome was the incremental cost-effectiveness ratio (ICER), which represented the added cost associated with an additional successful delivery within 48 hours of hospital admission. Clinical data were collected from the medical records of HMC, and the cost of direct medical resource data was locally obtained, reflecting the Qatari hospital perspective. Sensitivity analyses confirmed the study results against input uncertainties.

Results: A total of 519 women were analyzed, reporting significant differences in favor of the fFN study arm, with 94% delivery within 48 hours rate, versus 80%, $P < 0.0001$), and a cost saving of QAR6,470 (US\$1,777) per successful delivery within 48 hours, irrespective of the presence of adverse drug events, also in favor of the fFN testing. Additionally, the fFN group showed a lower rate of rehospitalization and shorter hospitalization duration

($P<0.001$). Neonates in the fFN group had higher birth weights (2920g vs. 2255g, $P<0.001$) and experienced a lower frequency of complications.

Conclusion: The findings seem to support the integration of fFN testing as a standard practice in managing preterm labour. This strategy contributes to improved maternal and neonatal health, at a reduced healthcare cost.

Keywords: Fetal fibronectin, Cost effectiveness, Premature births

236

Exploring healthcare professional students' perceptions of disaster management: A mixed-methods study

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Background: The increasing frequency of disasters highlights the role of healthcare professional students (HCPS) as future responders and potential volunteers during crises. However, disaster preparedness research often focuses narrowly on specific disciplines, such as nursing and medicine, and overlooks countries with low disaster risk.

Aim: This study explored HCPS perceptions in Qatar, examining their knowledge, attitudes, and readiness for disaster response, while identifying barriers and facilitators to enhance preparedness.

Methods: An explanatory mixed-methods approach was used. Quantitative data were collected via a self-reported questionnaire aligned with the Disaster Management Framework and administered to HCPS from Qatar University Health Cluster. Qualitative data were gathered through six online focus groups (FGs) conducted via Microsoft Teams and analyzed using thematic analysis, guided by the Haddon Matrix framework.

Results: Out of 1,392 invited participants, 455 HCPS responded (response rate = 32.7%), and 35 joined FGs. HCPS demonstrated fair knowledge (Median [IQR] = 44 [8]), good attitudes (Median [IQR] = 39 [5]), and good readiness to practice (Median [IQR] = 42 [6]). Healthcare Professional Students emphasized the importance of training, practical simulations, and mental health support. Facilitators for student preparedness included institutional backing and social media for communication, while barriers included limited knowledge, psychological readiness challenges, and time constraints due to academic demands, such as coursework, exams, and clinical training.

Conclusions: Despite positive attitudes, HCPS face knowledge gaps and readiness barriers. Interdisciplinary training integrating practical and psychological skills is crucial to developing a well-prepared healthcare workforce, particularly in regions considered at low disaster risk.

Keywords: Disaster management, Healthcare professional students, Preparedness, Response

237

Prevalence and patterns of inappropriate antibiotics prescribing in dialysis patients: A systematic review

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Background: Optimizing antibiotic use among dialysis patients is challenging, owing to altered pharmacokinetics and pharmacodynamics parameters as well as the need for frequent dosing adjustments. Inappropriate use is associated with treatment failure and the emergence of antimicrobial resistance which necessitates understanding the patterns of antibiotic prescribing.

Aim: To synthesize the current rates and patterns of antibiotic prescribing among dialysis patients and assess the prevalence and types of inappropriate antibiotic use.

Methods: This is a systematic review conducted in accordance with the PRISMA guidelines. Four electronic databases (PubMed, EMBASE, SCOPUS, CINAHL) were searched until October 2024. Studies published in the English language on antibiotic prescribing rates, patterns, or appropriateness in dialysis patients were included. Screening, selection and data extraction were performed by two independent reviewers. Data on antibiotic types, prescribing rates, indications, appropriateness tools, and inappropriateness details were extracted and synthesized qualitatively.

Results: A total of 13 studies were included covering 20,065 dialysis patients across both inpatient (61.5%) and outpatient (53.8%) settings. Most studies (61.5%) were from the USA, and hemodialysis (84.6%) was the most common dialysis modality. 54,040 antibiotics were prescribed, with prescribing rates ranging from 16% to 75.5%. Bloodstream and respiratory tract infections were the most common infections treated with antibiotics. Vancomycin, piperacillin/tazobactam, amoxicillin/clavulanate, and ciprofloxacin were the most frequently prescribed antibiotics. Eleven studies (84.6%) evaluated antibiotic appropriateness, with international guidelines being the most common

tool. Inappropriate antibiotic use ranged from 10% to 65.7%, mainly related to inappropriate dosing and indication, with under-dosing and unindicated broad-spectrum use reported. Vancomycin, meropenem, and amoxicillin/clavulanate were the most antibiotics associated with inappropriate use.

Conclusion: Antibiotic prescribing in dialysis patients varies significantly, with prevalent inappropriate prescribing. Antimicrobial stewardship is recommended to improve prescribing practices and reduce the emergence of resistance.

Keywords: Renal replacement therapy, Antibiotics, Prescribing patterns

238

Therapeutic drug monitoring of aminoglycosides in paediatrics: A scoping review

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Background: Aminoglycosides are commonly used in paediatrics and require therapeutic drug monitoring (TDM) to optimize dosing. There is variability in TDM practices and a lack of guidelines for extended-interval aminoglycoside dosing (EIAD) TDM and dose adjustment in paediatrics.

Aim: To identify and explore TDM guidelines recommendations for EIAD in paediatrics, globally.

Methods: We searched MEDLINE, EMBASE, Web of Science, and Google Scholar from their inception to December 2024. Additional sources included guideline registries, namely, Guideline Central and Guidelines International Network and websites of infectious disease and/or paediatric organizations. Search results were imported to EndNote version 20 where duplicates were removed. We included guidelines that provided recommendations for the use of EIAD for treatment of infectious diseases in paediatrics. Guidelines recommending single-dose or non-parenteral aminoglycosides were excluded. Title/abstract and full-text screening were done by two reviewers using Rayyan. Data were extracted using a standardized data extraction sheet and analysed using descriptive statistics.

Results: As preliminary results, a total of 38 guidelines discussing the use of EIAD in paediatrics have been included in this review. The type of documents included were mainly clinical practice guidelines (58%) and consensus statements (32%).

Most guidelines (55%) did not provide any recommendations for monitoring of aminoglycosides. Around 20% of guidelines recommended serum concentration monitoring without further details. An additional 24% of guidelines discussed peak and/or trough targets. One guideline recommended preserving maximum antibiotic concentration to minimum inhibitory concentration ratio to optimize efficacy. Another guideline provided detailed recommendations for TDM of gentamicin in neonates. Only one guideline provided recommendations for dose adjustment however, it was not specific to pediatrics.

Conclusions: There is limited and inconsistent guidance on EIAD TDM and dose adjustment in paediatrics. Practice guidelines to inform and standardize TDM practices are highly needed.

Keywords: Aminoglycosides, Extended-interval dosing, Guidelines, Pediatrics, Therapeutic drug monitoring

240

Vitamin D status of gastrostomy-fed children with neurological impairment in paediatric long-term units

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Background: Vitamin D is crucial for calcium and phosphate metabolism, and essential for bone health, particularly in children. Gastrostomy-fed children with neurological impairment face challenges in absorption, necessitating individualized monitoring and dosing adjustments. Sustaining adequate 25-hydroxyvitamin D (25OHD) levels is vital for their overall health.

Aim: This study aimed to optimize vitamin D levels in gastrostomy-fed children at paediatric long-term clinics (Al-Maha units) through a targeted intervention program.

Methods: A retrospective study design was used to collect baseline data from 70 children (aged 17 months to 17 years) residing in long-term paediatric units in Doha, Qatar. Participants received consistent, commercially prepared enteral feeds via gastrostomy. Serum 25OHD levels were measured per hospital protocols, with adequacy defined as levels >20 ng/ml, insufficiency as 12–20 ng/ml, and deficiency as <12 ng/ml.

Results: Among the 74 initially recruited participants, 70 were analyzed after excluding four deceased. About 87% of participants had adequate serum 25OHD levels, 12.86% had insufficient levels, and none had deficient levels. The mean vitamin D intake from medications was 1120 units compared to 415.86 units from diet alone. Only 46% of subjects were on anti-epileptics. Furthermore, only 6% received mega doses of vitamin D.

Conclusions: The nutritional completeness of gastrostomy feeds effectively prevented vitamin D deficiency in most children with neurological impairment. A targeted intervention program is recommended to ensure optimal vitamin D levels, particularly for gastrostomy-fed children.

Keywords: Gastrostomy, Vitamin D, Children

241

Medications used by stroke patients for secondary prevention in the first year: A medication profile study

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Background: When considered different from other cardiovascular diseases, stroke ranks fifth among all causes of death. Poor adherence or discontinuation of secondary prevention medications results in poor health outcomes.

Aim: This study aimed to reveal the change in the medication used by stroke patients for secondary prevention in one year.

Methods: This study was conducted as a retrospective data analysis of another study in which we evaluated the effects of pharmaceutical care on stroke patients at a hospital in Türkiye. The antithrombotic, antihypertensive, antidiabetic and antihyperlipidemic medications used by the patients for secondary prevention of stroke when they were admitted to the hospital with a diagnosis of stroke and 1 year later were recorded.

Results: A total of 193 patients (67.4% male) were included in the study. The mean age of the patients was 60 years. The majority of the patients were diagnosed with ischemic stroke. Most common comorbidities were hypertension, dyslipidemia and diabetes. About seventy-seven percent were using 5 or more medications. Initially, 33.2% of the patients were using aspirin, 4.1% were using clopidogrel, 5.7% were using aspirin and clopidogrel, 3.1% were

using warfarin, and 4.7% were using Non-vitamin K oral anticoagulants. At the end of 1 year, these values were 26.6%, 28%, 17.1%, 5.7% and 22.3%, respectively. At the beginning, 68.9% of the patients were using antihypertensives and 27.5% were using antidiabetics, and these values were 79.3% and 35.8%, respectively, at the end of 1 year. While 11.4% of the patients were using statins at the beginning, this value was 67.4% at the end of 1 year. All these increases compared to baseline in the number of prescribed medications were found to be significant ($p < 0.001$).

Conclusion: The use of secondary prevention medications significantly increased one year after stroke, indicating improved adherence to treatment guidelines. Continued follow-up and pharmaceutical care are essential to support long-term management in stroke patients.

Keywords: Stroke, Cardiovascular, Hospitalization, Secondary prevention

243

Academic resilience of the Health Sector undergraduate students at Qatar University

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Background: Academic resilience (AR) is crucial for university students, enabling them to overcome academic setbacks, cope with pressure, and maintain a positive attitude to achieve their educational goals. While previous studies have focused on AR in specific health profession undergraduate students, our study evaluated AR among Health Sector undergraduate students.

Aim: We used the ARS30 scale to assess the academic resilience of Qatar University (QU) Health Sector undergraduate students.

Methods: We conducted a cross-sectional survey among undergraduate Health Sector students at Qatar University (QU) between May 2023 and October 2023. The validated ARS30 scale, including a case vintage with a 5-point Likert scale to negative behavioral responses, besides the demographic questions, was built into an online survey and emailed to QU health sector students. Descriptive and inferential statistics were performed to explore the ARS30 scores and its three sub-scales: perseverance, reflecting and adaptive help-seeking, and negative affect and emotional response.

Results: Two hundred thirty college students from the College of Medicine (34%), College of Pharmacy (31%), Health Sciences (24%), and Dental Medicine (11%) responded to the survey. The mean age was 21 years; 82% of

the students were females, and 66% were juniors in the Medicine and Pharmacy program. The findings showed moderate AR among health sector students with a mean ARS30 score of 113 (range 30–150). A weak positive correlation is statistically significant between GPA and ARS30 (p -value 0.036). ARS30 subscales were insignificant except for the negative affect and emotional response subscale, which was significantly higher in males than females (p -value 0.028).

Conclusions: Our study provides insights into the AR level of QU Health Sector undergraduate students. Being a female, a junior student, or a pharmacy student has been correlated with higher AR scores. These findings can embrace future implications for advancing health professions students' education, performance, and well-being.

Keywords: Academic resilience, Pharmacy education

244

Evaluating pharmaceutical education situation in the Eastern Mediterranean Region using the international pharmaceutical federation's development goals of academic capacity and quality assurance

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Background: Pharmacy workforce intelligence involves the development, implementation, and evaluation of effective strategies and tools to ensure the availability and quality of the pharmacy workforce.

Aim: This study aimed to evaluate the status of Academic-Capacity (AC) and Quality-Assurance (QA) of pharmacy education in the Eastern-Mediterranean-Region (EMR) using the International-Pharmaceutical-Federation (FIP)'s Development-Goals and their associated mechanisms as a framework.

Methodology: An explanatory sequential mixed-methods approach was used. The quantitative phase involved administering a newly developed and validated questionnaire to pharmacy leaders of all pharmacy schools in the EMR. Both descriptive and inferential statistics were applied as

appropriate. The qualitative phase involved the conduct of semi-structured interviews with pharmacy leaders, and data were thematically analyzed.

Results: Of 112 pharmacy leaders in EMR, 61 participated in the survey (54.5% response rate) and 14 participated in the interviews. Planning students' enrollment capacity based on local workforce needs was reported by 51% of the schools. Capacity-building programs for teacher-practitioners were provided by 47% of the programs. Collaboration with stakeholders in designing and/or QA of the curricular content was reported by 71% of the participants. Most programs (67%) stated that they adopt/adapt global standards for QA in pharmacy education. At the country level, 70% of the programs adhered to standards of practice for pharmacists. The main gaps identified through the interviews were the lack of adequate density distribution in all career pathways and countries, the need for improving preceptors' training programs, developing the national standards for pharmacy education/practice, and forming national pharmacy professional bodies.

Conclusion: Although some mechanisms associated with the AC and QA goals are achieved and maintained adequately in the pharmacy programs in the region, most of them require improvements. Factors identified that might impact the distribution of the pharmacy workforce in their career paths such as job opportunities require further investigation.

Keywords: Education, Academic capacity, Quality assurance

245

Eliminating Omnicell stock discrepancies in inpatient clinical units across Mental Health Service (MHS)

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Background: Omnicell is a securely automated supply chain medication dispensing system that provides fast and easy access to medications at the point of care. The inpatient pharmacy is responsible for storing, managing, and securing medication inventory in clinical units' Omnicell's. Nurse compliance with the proper workflow upon removing any medication from the Omnicell significantly impacts the stability and preservation of the medication inventory.

Aim: This quality improvement project was aimed at eliminating and maintaining Omnicell stock discrepancies in inpatient clinical units across MHS from to less than 5% per week per each cabinet by end of December 2024.

Method: A quality improvement project was conducted in all inpatient units to eliminate Omnicell stock discrepancies using four Plan-Do-Study-Act (PDSA) framework after identifying a multidisciplinary team Baseline audit for a month identified discrepancies. Reasons for numerous discrepancies were identified using cause and effect diagram. Interventions included Individual staff training and system optimization. Weekly audits and monthly statistical analysis for outcomes measures utilizing run charts were conducted. Follow up meeting with head nurses were also organized. Staff Satisfaction and time consumption were set as balance measures.

Results: The monthly discrepancies got reduced to 0.25% across all MHS inpatient units by end of the year.

Conclusion: A systematic approach was successfully implemented to maintain a discrepancy rate below 5%, aligning with HMC's maximum accepted variance per store. Despite the time investment required for the project, staff expressed high levels of satisfaction and recognized the initiative as a cornerstone for achieving optimal workflow efficiency.

Keywords: Omnicell, Inpatient, Mental health

246

Antibiotic prophylaxis for recurrent urinary tract infections in adults: Overview of current practice

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Background: Recurrent urinary tract infections (rUTIs) contribute significantly to patient morbidity and are linked to increased utilization of healthcare services. RUTIs are defined as ≥ 2 UTIs in 6 months, or ≥ 3 in 12 months. Both international and Qatar local guidelines recommend antimicrobial prophylaxis for rUTI patients. Despite the morbidity of rUTIs and concerns over frequent antibiotics use, it's unclear how well clinical practice follows current guidelines recommendations.

Aim: The aim of this audit is to assess current practice in prescribing antibiotic prophylaxis for rUTI and to evaluate the appropriateness of the prescribed antimicrobials for rUTI in Al Wakra Hospital.

Method: A retrospective evaluation was conducted from January till August 2023. A list of patients who had at least two positive urine cultures in the internal medicine wards during the review period was generated from the

local electronic medical records. The primary focus was on therapeutic decisions including the appropriateness of prescribed prophylaxis in terms of dose, duration, frequency, based on culture sensitivity. Secondly, on follow-up and referral to infectious disease specialty.

Results: A total of 90 patients met the clinical criteria for recurrent UTIs, with 56% being female. Prophylactic antibiotics were initiated in 13% (n = 12) of patients, with 92% (n = 11) receiving an appropriate choice, and all receiving the correct dose and frequency. Prophylaxis was indicated but not prescribed in 21% (n = 16) of patients. Among them, 24% were ineligible due to the presence of a catheter, while no documented reason was found for withholding prophylaxis in the remaining eligible cases. Infectious disease specialists were involved in the care of 75% of the patients.

Conclusion: This study highlights gaps in the prevention of rUTIs practices. Enhancing guideline adherence through providing targeted education for healthcare providers is essential to optimizing stewardship practices and patient care.

Keywords: Urinary tract infection, Antibiotics, Prophylaxes

247

Psychotropic polypharmacy in women of child-bearing age: Prevalence and influencing factors at outpatient mental health clinics in Qatar

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Background: There is a rising trend in psychotropic prescribing among women, often leading to polypharmacy and off-label use.

Aim: Identifying such trends can address potential health risks such as adverse drug effects and contribute to tailoring recommendations to enhance rationality in prescribing for women with childbearing age potential.

Method: This was a quantitative retrospective chart review of patient medical records for women of childbearing age patients who attended an outpatient appointment at mental health services (MHS) between Jun 2021 to Jun 2022. Demographic information and clinical variables, medication-related details were collected. IBM SPSS® Statistics was used for data analysis. A linear

regression and a binary logistic regression model were constructed. Odds ratios (ORs) and 95% CIs were calculated and an alpha value of 0.05 was used.

Results: Four sixty female patients were included. Around one-third (34.3%, $n = 158$) had a somatic comorbidity. The most common disorder was depressive disorder (36.5%, $n = 168$), and anxiety disorders (24.6%, $n = 13$). Most patients (68%, $n = 227$) had a full response. Only 28.2% ($n = 130$) were on psychotropic polypharmacy. The total number of prescribed psychotropic drugs at the last clinic visit was higher in older patients ($B = 0.013$; $p = 0.020$; $r = 0.137$), patients who were unemployed ($B = -0.150$; $p = 0.036$; $r = -0.124$), those with a longer duration of illness ($B = 0.416$; $p = 0.007$; $r = 0.158$). A full response was not found to be associated with any of the studied variables.

Conclusion: The prevalence of polypharmacy was not higher than international figures. Some of the factors linked to the use of polypharmacy are modifiable. Continuous promotion of rational approaches to women's health in psychiatry is advised.

Keywords: Polypharmacy, Psychotropic, Mental health

249

Incidence of hyponatremia in older adults taking antidepressants in Qatar: A retrospective study

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Background: Hyponatremia is a common electrolyte disturbance with potentially serious health consequences, particularly in older adults. Antidepressant use has been linked to an increased risk of hyponatremia in this population. This is often attributed to the syndrome of inappropriate antidiuretic hormone secretion (SIADH), a well-documented mechanism of antidepressant-induced hyponatremia.

Aims: The project aimed at determining the incidence of hyponatremia among older adult patients receiving antidepressant within mental health services in Qatar.

Methods: A quantitative retrospective chart review was conducted for older adult patients treated with antidepressants for mental health disorders from June 1, 2022, to June 1, 2023. Data collected included the incidence of hyponatremia, demographic details, and clinical characteristics. Statistical analysis

was performed using SPSS® Statistics, with binary logistic regression applied to identify sociodemographic and clinical predictors of hyponatremia. An alpha level of 0.05 was considered significant.

Results: Among 245 randomly selected patients, the incidence of hyponatremia over a one-year period was 24.5% (60 patients). Mirtazapine, used by 30.2% ($n = 74$) of patients, showed the highest association with hyponatremia, affecting 35.0% ($n = 21$) of its users. Key predictors of hyponatremia included chronic kidney disease (CKD) ($p = 0.028$, OR: 2.55, 95% CI [1.1, 5.8]) and concurrent use of medications known to induce hyponatremia ($p = 0.006$, OR: 3.047, 95% CI [1.3, 6.7]). Hyponatremia onset within the first month of treatment was observed in 6.5% of cases.

Conclusions: This study revealed an incidence rate consistent with the upper range reported in the literature (0.5% to 33%). Healthcare providers should prioritize proactive sodium level assessments, particularly during the initial months of antidepressant therapy and among patients with predisposing factors such as CKD or concurrent diuretic use.

Keywords: Hyponatremia, Electrolytes, Antidepressant

250

Exploring the desirable attributes of clinical preceptors: A scoping review of the available instruments

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Background: Experiential learning is an important component of health professional education programs because it provides students an opportunity to apply their knowledge through practice before becoming practitioners. Preceptors (i.e. clinical educators) play a vital role in developing students' professional practice skills and competencies during clinical training. Ensuring the quality and preparedness of preceptors is a key aspect in the students' experiential learning experience. In Gulf region, specifically Qatar, there is a knowledge gap regarding the desired attributes of pharmacy preceptors, which highlights the opportunity to explore more in this topic.

Aim: The review aim was to identify the available tools in literature to explore desirable attributes of pharmacy preceptors as clinical educators.

Methods: Four databases were utilized to identify relevant original research articles published in English: CINAHL, ERIC, ProQuest, and PubMed. Key concepts were preceptorship, attributes, pharmacy, and tools. Quantitative and mixed-methods studies were included. Included articles were summarized according to design, setting, population and outcomes. Also, the validity of the used instruments in these studies was reported.

Results: Out of twenty-three articles that reached the full-text screening, six studies were included in the review. Exclusion reasons were grouped into three groups: different setting, different outcomes, and different study design. All identified tools from the pharmacy profession needed some modifications to help exploring the desirable attributes of pharmacy preceptors as clinical educators. Some studies utilized frameworks which considered general to meet this study aim. Others included sections which were not applicable for the study aim (i.e. assessing preceptors' training programs).

Conclusion: There was lack of well-designed and validated tools in pharmacy profession that can be used to explore the desirable attributes of pharmacy preceptors as clinical educators. This necessitates further research to develop and validate new tools to ultimately understand the perception of pharmacy preceptors about the desirable attributes to be a clinical educator.

Keywords: Experiential learning, Health professional education

251

Epidemiology and clinical outcomes of nosocomial bloodstream infections in ECMO patients in Qatar: A seven-year retrospective analysis

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Background: Extracorporeal Membrane Oxygenation (ECMO) is a vital procedure for patients experiencing acute respiratory or cardiac failure. However, its invasive nature presents a considerable risk for nosocomial bloodstream infections (BSIs). BSIs complicate patient management and correlate with unfavorable outcomes.

Aim: This is the first analysis of BSI prevalence, pathogen profile, and clinical outcomes among ECMO patients in Qatar and the Middle East.

Methods: The study examined 172 ECMO patients admitted to the ICU of Hamad General Hospital between 2014 and 2021. Data collected included patient demographics, comorbidities, ECMO indications, BSI occurrence, pathogen profiles, and clinical outcomes.

Results: Among ECMO patients, 49 patients (28.5%) experienced BSIs, yielding an infection density rate of 16.7 infections per 1,000 ECMO-days over a cumulative total of 4,485 ECMO-days. The mean age of patients was 41.17 years, predominantly male (77.9%). Diabetes (26.2%) and hypertension (23.8%) were the most prevalent comorbidities. The main indications for ECMO were viral infections (46.5%), primarily SARS-CoV-2 (30.7%). *Klebsiella pneumoniae* was the predominant BSI pathogen, accounting for 12% of BSIs, followed by *Candida auris* and *Escherichia coli* at 8% each, and *Pseudomonas aeruginosa* and *Staphylococcus epidermidis* at 6.7% each.

BSI patients had an elevated prevalence of comorbidities (55.1% vs. 44.3%). Diabetes was much more common among BSI patients (32.7% vs. 17.2%). BSI patients had a higher mortality rate post-ICU discharge (55.1% vs. 38.2%), with an elevated mortality rate during ECMO (42.9% vs. 32.5%). The occurrence of BSI correlated with significantly extended hospital (95.27 vs. 56.71 days), ICU (75.86 vs. 31.49 days), and ECMO stays (52.55 vs. 16.34 days).

Conclusions: BSIs in ECMO patients significantly increase the clinical burden, prolonging hospital stay and complicating management. Despite similar baseline severity scores, BSIs correlate with adverse outcomes, highlighting the urgent need for strict infection control, diligent monitoring, and enhanced antimicrobial stewardship in ECMO management.

Keywords: Cardiac failure, Epidemiology, Extracorporeal Membrane Oxygenation

252

Evaluating the impact of Trigger Tool implementation on enhancing adverse drug reaction reporting in the Intensive Care Units

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Background: Adverse drug reactions (ADRs) can present as unexpected, unintended, or excessive responses to medications, often requiring interventions like discontinuing the drug, modifying therapy, adjusting dosages, or even hospital admission. The World Health Organization (WHO) has highlighted key priority areas that must be addressed nationally or internationally, including ADR detection, reporting, assessment, communication, and impact.

Various initiatives have been introduced to enhance and simplify the reporting of ADRs. Most of these strategies have traditionally concentrated on voluntary reporting and error tracking. The Institute for Healthcare Improvement (IHI) developed the Global Trigger Tool (GTT), which allows for a retrospective review of “triggers” (or clues) to identify possible adverse events.

Aim: This project aimed to evaluate the impact of implementing a modified IHI trigger tool on ADR reporting in the Intensive Care Units (ICU) at Al-Wakra Hospital (AWH).

Methods: A list of detected triggers had been auto generated from the electronic medical records and emailed to the assigned team. Weekly, one reviewer revised the extracted triggers and prompted the ICU pharmacists to report the true ADRs detected by the tool, if not already reported.

This work assessed the tool’s Positive Predictive Value (PPV), the monthly number of ADRs reported per 1000 ICU days, and the percentage of spontaneously reported ADRs

Results: Between February and September 2024, the tool demonstrated an average PPV of 25.5% when including all triggers. The number of ADRs reported per 1,000 ICU days varied monthly, increasing from an initial average of 2 reports per month to a new average of 20 reports per month. The percentage of spontaneously reported ADRs varied considerably, fluctuating between 6% and 50%.

Conclusion: While ADR reporting improved from baseline, continued reminders and education interventions are essential. The involvement of the other multidisciplinary team members further enhances the reporting and monitoring efforts.

Keywords: Adverse drug reactions, Intensive care unit

254

Retrospective analysis of nosocomial bloodstream infections in COVID-19 patients undergoing ECMO therapy in Qatar

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Background: The COVID-19 pandemic markedly heightened the demand for extracorporeal membrane oxygenation (ECMO) therapy in critically ill patients experiencing severe respiratory failure. Although ECMO can be life-saving, its application is associated with extended ICU stays, increasing the likelihood of nosocomial infections and adverse clinical outcomes.

Aim: This study is the first investigation in Qatar and the Middle East on the epidemiology of nosocomial bloodstream infections (BSIs) in COVID-19 patients undergoing ECMO.

Methods: Data from 54 COVID-19 patients on ECMO (admitted April 2020–November 2021) were analyzed to calculate infection rates and infection density. Pathogen identification was conducted, and baseline parameters were compared between individuals with and without BSIs.

Results: The patient cohort primarily comprised males (85.2%) aged 29 to 65 years, with notable comorbidities including diabetes (66.7%) and hypertension (68.5%). Out of 54 patients, 25 (46.3%) experienced BSIs, yielding an infection density of 22.7 BSIs per 1000 ECMO days throughout a cumulative total of 2022 ECMO days. The primary BSI pathogens comprised *Klebsiella pneumoniae* (15.22%), *Escherichia coli* (13.04%), and notable fungal agents such as *Candida auris* (10.87%). Patients with BSIs demonstrated a markedly extended hospital stay (116.12 days vs. 61.17 days, $p = 0.0116$) and ICU stay (87 days vs. 39.31 days, $p = 0.0132$). Nonetheless, total death rates were comparable in the two groups (48% with BSIs versus 37.9% without BSIs), suggesting that additional factors, such as initial disease severity (APACHE2 mean of 26.3 among deceased patients), may substantially affect outcomes. Comorbidities, including diabetes (48%) and obesity (72%), were more frequent in individuals with BSIs, indicating increased vulnerability to infections.

Conclusions: These results underscore the considerable burden of nosocomial bloodstream infections in COVID-19 ECMO patients, highlighting the

necessity for improved infection control protocols and customized antimicrobial stewardship initiatives in Qatar and in the wider Middle East.

Keywords: Bloodstream infections, Extracorporeal Membrane Oxygenation, COVID-19

260

Safety of SGLT2 inhibitors peri-percutaneous coronary intervention: A national experience

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Background: Sodium-glucose cotransporter-2 inhibitors (SGLT2i) have cardio-renal protection benefits regardless of diabetes status. However, the outcomes of uninterrupted SGLT2i use peri-percutaneous coronary intervention (PCI) among patients with acute coronary syndrome (ACS) remains uncertain.

Aim: Thus study aimed to evaluate the safety of SGLT2i use peri-PCI.

Methods: We conducted a retrospective observational cohort study that included all patients admitted with ACS who underwent revascularization with PCI at Qatar's main tertiary cardiology center between 1/01/2018 and 31/12/2020. The use of SGLT2i peri-PCI was defined as the established use of SGLT2i before the index admission or early initiation of SGLT2i during the index admission within 24 hours of contrast exposure. The study population was divided into two groups: (1) SGLT2i users and (2) SGLT2i non-users. The primary outcome was the risk of contrast-induced nephropathy (CIN) with SGLT2i use. A propensity score-matched model (1:1) was used to adjust for baseline characteristics between the study groups. Logistic regression analysis was used with a *P*-value <0.05, indicating statistical significance.

Results: We included 3421 ACS patients who met the eligibility criteria (88% male, mean age 54 ± 10 years, and 59% Asian), with 95 patients classified as SGLT2i users and 3326 as SGLT2i non-users. Using 1:1 propensity score matching, 95 patients were included per arm with an absolute standardized difference < 0.1 for all baseline characteristics. The use of SGLT2i peri-PCI was not associated with an increased risk of CIN [before propensity score matching: 4 (4.2%) versus 146 (4.4%), aOR = 1.31 (0.45-3.78), *P*-value 0.622; after

propensity score matching: 4 (4.2%) versus 5 (5.3%), OR = 0.79 (0.21-3.04), *P*-value 0.733].

Conclusion: Our findings suggest that the uninterrupted use of SGLT2i among patients with ACS undergoing PCI is safe without increasing the risk of CIN, warranting continuation of SGLT2i use or early initiation during ACS.

Keywords: SGLT2, ACS, PCI

261

Impact of sodium glucose cotransporter-2 inhibitors on post-operative atrial fibrillation in cardiothoracic surgery: A multicenter cohort study

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Background: Sodium-glucose cotransporter-2 (SGLT2) inhibitors are considered cornerstone therapy in patients with type 2 diabetes, chronic kidney disease, and heart failure and were recently proposed to be protective against atrial fibrillation (AF). However, the impact of SGLT2 inhibitors on post-operative AF, a major concern after cardiothoracic surgery, has not yet been evaluated.

Aim: Therefore, in this multicentre study, we assessed the effectiveness of SGLT2 inhibitors in preventing post-operative AF.

Methods: We conducted a multicentre retrospective observational cohort study that included all patients undergoing cardiothoracic surgery, including coronary artery bypass graft (CABG), valve replacement, and valve repair over six years (1/06/2017-1/06/2023). Patients who were receiving SGLT2 inhibitors for a minimum of one week before surgery, regardless of diabetes status, were included in the SGLT2 inhibitor users' arm and then matched into non-SGLT2 inhibitor users using propensity score-matching model (1:1) for baseline demographics. The primary outcome was confirmed AF after surgery regardless of frequency, duration, or intervention used for cardioversion within 30 days of surgery. Multivariate Cox proportional hazard regression analysis was used to determine the primary outcome. *P*-value <0.05 indicated statistical significance.

Results: We included 2240 patients who met the eligibility criteria (78% male, mean age 62±10). Using 1:1 propensity score matching for baseline demographics, including age and gender, 1123 patients were included in the

SGLT2 inhibitor users arm, and 1117 in the non-SGLT2 inhibitor users arm with an absolute standardized difference <0.1 . Around 79% of patients underwent CABG. SGLT2 inhibitors use significantly reduced post-operative AF within 30 days of surgery compared to SGLT2 inhibitor non-users (11.2% vs. 13%, aHR 0.75, 95% CI 0.61-0.91, $P = 0.004$).

Conclusion: Using real-world data, SGLT2 inhibitors were found to be protective against post-operative AF, which may widen the spectrum of indications for them.

Keywords: SGLT2 inhibitors, Atrial fibrillation

262

Sodium glucose cotransporter-2 inhibitors post acute coronary syndrome induced new heart failure

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Background: Patients with acute coronary syndrome (ACS) have an unmet need for further risk reduction in major adverse cardiovascular events (MACE) and heart failure (HF) development despite guideline-directed medical therapy. As sodium glucose cotransporter-2 inhibitors (SGLT2i) are cardioprotective, in this study, we assessed the effectiveness of SGLT2i post-ACS with new-onset HF.

Aim: To evaluate the effectiveness of sodium glucose cotransporter-2 inhibitors (SGLT2i) in reducing major adverse cardiovascular events and heart failure outcomes in diabetic patients with acute coronary syndrome complicated by new-onset heart failure.

Methods: We conducted a retrospective cohort study that included all diabetic patients admitted to the main tertiary cardiology center in Qatar between 1/06/2017 and 1/06/2021 with ACS complicated by new HF defined as new clinical HF requiring diuretics during the index admission, or ejection fraction (EF) $<40\%$. We divided the study population into two groups: (1) SGLT2i users and (2) non-SGLT2i users. Primary outcomes were evaluated at 30 and 360 days, including a composite of ACS, HF hospitalization, and all-cause mortality. A propensity score-matched model (1:1) was used to adjust for baseline characteristics between the study groups. Cox proportional hazard regression analysis was used to evaluate the outcomes. P -value <0.05 indicated statistical significance.

Results: We included 465 ACS patients who met the eligibility criteria (93% male, mean age 55 ± 10 years, and 72% Asian). Using 1:1 propensity score matching, 78 patients were included per arm with an absolute standardized difference < 0.1 for all baseline characteristics. The use of SGLT2i postischemia-induced new HF resulted in lower composite outcomes at 30 and 360 days [30-day: 2.6% vs. 11.5%, HR = 0.20 (0.04-0.94), $P = 0.041$; 360-Day: 14.1% vs. 23.1, HR = 0.46 (0.22-0.99), $P = 0.046$].

Conclusions: Our findings suggest cardioprotective effects of SGLT2i post-ACS induced new HF, which widens the SGLT2i spectrum of indications and may encourage cardiologists to prescribe SGLT2i for this population.

Keywords: ACS, SGLT2i, MACE

264

Examining the perception of undergraduate pharmacy students towards their Leadership Competencies: A mixed-methods study

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Background: Leadership development among pharmacy students is essential to meeting the evolving demands of the healthcare sector. However, research on pharmacy students' perceptions of their leadership competencies and development opportunities remains limited.

Aim: This study explores the perceptions of students at the College of Pharmacy at Qatar University regarding their leadership knowledge, skills, attitudes, and the available opportunities to support leadership development using an explanatory sequential mixed-methods approach.

Methods: The quantitative phase of the mixed methods involved developing, validating, and administering a questionnaire to all undergraduate students, with responses analyzed using SPSS®. In the qualitative phase, four focus groups (FGs) were conducted virtually using a topic guide guided by the Medical Leadership Competency Framework. The FGs were transcribed verbatim and analyzed thematically using NVivo®.

Results: The results revealed a fair level of leadership knowledge and skills, and a good attitude toward leadership. The results suggested leadership development was supported through curricular and extracurricular

opportunities, including interprofessional education (IPE), clinical rotations, and the integration of leadership concepts within various courses. Students recommended expanding IPE to clinical rotations, incorporating standalone leadership courses, and international exchange programs.

Conclusions: These findings emphasize the importance of a structured comprehensive approach to leadership development in pharmacy education.

Keywords: Leadership, Pharmacy students, Education

265

Inflammatory biomarkers as predictors for unlocking antidepressant efficacy: Assessing predictive value and risk stratification in major depressive disorder in a prospective cohort study

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Background: Major depressive disorder (MDD) is characterized by significant heterogeneity in treatment response, with inflammation hypothesized to play a role in its pathophysiology. Peripheral inflammatory biomarkers, such as the neutrophil-to-lymphocyte ratio (NLR) and C-reactive protein (CRP), may predict antidepressant efficacy, but evidence remains limited.

Aim: This study aimed to investigate the association between baseline inflammatory biomarkers, their changes, and antidepressant treatment outcomes in patients with MDD.

Methods: A prospective cohort study was conducted in Qatar, recruiting 123 MDD outpatients aged 18–64 years. Participants underwent baseline assessments of inflammatory biomarkers, including NLR, CRP, monocyte-to-lymphocyte ratio (MLR), and platelet-to-lymphocyte ratio (PLR). Clinical severity was measured using the Zung Self-Rating Depression Scale (ZSRD) at baseline and 12 weeks post-treatment. Statistical analyses, including multiple regression models and the Random Forest machine learning model, were performed to identify predictors of antidepressant response.

Results: Improvement in depressive symptoms was significantly associated with female sex, higher mean corpuscular volume (MCV), lower absolute

neutrophil count (ANC), and higher eosinophil counts. However, changes in biomarkers such as NLR, MLR, PLR, and CRP did not predict treatment response. Machine learning models identified folate levels, PLR, and change in cholesterol level as potential predictors but demonstrated limited specificity.

Conclusion: Baseline inflammatory biomarkers, including ANC and eosinophil count, may help stratify treatment outcomes in MDD. Changes in biomarkers post-treatment were not associated with symptom improvement, highlighting the complexity of the inflammation-depression relationship. Future research should explore advanced biomarkers and refine precision psychiatry approaches.

Keywords: Major depressive disorder, Inflammatory biomarkers, Anti-depressant response, Neutrophil-to-lymphocyte ratio, C-reactive protein, Precision psychiatry

268

Comparing dual oral agents plus insulin vs. triple oral agents in uncontrolled Type II diabetes: A pilot study

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Background: Type II diabetes mellitus (T2DM) patients often do not achieve glycaemic control with oral hypoglycaemic agents (OHAs). There are two main approaches to address this challenge: transitioning to a triple OHA regimen or adding Insulin to the existing dual OHA regimen.

Aim: This study aimed to compare the efficacy of adding Insulin to dual OHAs (Sitagliptin + Metformin) against adding a third OHA to Sitagliptin + Metformin in achieving glycaemic control among patients with uncontrolled T2DM.

Method: A pre-post study was conducted between 21 September 2023 and 21 December 2023 at Services Hospital Peshawar, Pakistan. Patients with uncontrolled T2DM with >7% HbA1c were divided into group 1 (Sitagliptin + Metformin plus a third OHA), and group 2 (Sitagliptin + Metformin plus pre-mixed Insulin 70/30). Glycemic control based on HbA1c values, fasting and random blood sugar levels, lipid profile, and body weight were evaluated after 3 months of therapy. The pre- and post-effect was compared by using a paired t-test.

Results: The study included $n = 80$ patients with T2DM. Between groups 1 and 2, no significant difference was found in HbA1c values (9.1 vs. 9, with $p = 0.724$). However, BMI, cholesterol, and LDL significantly decreased in group 1 compared to group 2 ($p < 0.001$ vs. $p = 0.131$, $p = 0.023$ vs. $p = 0.896$, and $p = 0.003$ vs. $p = 0.395$, respectively). Additionally, the incidence of hypoglycaemic episodes was significantly lower in group 1 (7.5%) than in group 2 (47.5%, $p = 0.004$). No significant difference was observed between the triple OHA and dual OHA plus Insulin regimens in achieving glycemic control.

Conclusion: The triple OHA regimen improved BMI, cholesterol, and LDL levels, and reduced hypoglycemic episodes more effectively than dual OHA plus Insulin, despite similar HbA1c outcomes, suggesting it may be preferable for uncontrolled T2DM.

Keywords: Insulin, Sitagliptin, Metformin

269

Development and implementation of a clinical pharmacy programme in Syria

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Background: The role of pharmacists in patient care has expanded significantly in recent years, highlighting the need for advanced training at both undergraduate and postgraduate levels to ensure the delivery of high-quality care. Although the local Ministry of Health in Syria has recently introduced a clinical pharmacy residency programme, a formal curriculum to support this specialized field remains unavailable.

Aim: This initiative aimed to develop and implement a clinical pharmacy diploma program to equip practicing pharmacists in Syria with essential skills and knowledge to enhance patient outcomes.

Methods: We designed a clinical pharmacy diploma program based on the latest international guidelines and best practices in the field. The program comprises nine modules that address core topics in clinical pharmacy. Pharmacists holding a valid practice license were eligible to enroll in the program, which includes didactic lectures, formative assessments, and interactive case-based learning. Upon completing the modules, enrolled pharmacists will participate in supervised hospital rotations to gain practical experience.

Results: The program has been accredited by the Syrian Students Union and the Syrian Pharmacists Association. Since September 2024, a total of 265

pharmacists have enrolled in the program. The modules cover specific therapeutic areas such as chronic disease management, infectious diseases, and critical care pharmacotherapy. These are tailored to qualify practicing pharmacists to care for patients in diverse clinical settings. The first cohort is expected to complete their final examinations in July 2025.

Conclusion: The newly developed clinical pharmacy diploma programme is expected to address a critical gap in the Syrian healthcare system by enhancing the capacity of pharmacists. Continuous evaluation and feedback will be necessary to ensure further improvements to the programme.

Keywords: Clinical pharmacy diploma, Pharmacist training, Education, Pharmacotherapy

270

Spatial mapping of antimicrobial resistance patterns using global information system in selected cities of Punjab, Pakistan

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Background: Pakistan is among the countries which are facing the issue of antimicrobial resistance, however, limited data or surveillance systems are available to give any idea or statistics about the extent or the patterns of the resistance of different species of bacteria against antibiotics.

Aim: To apply GIS-based spatial analysis to identify and map antimicrobial resistance patterns across key cities in Punjab, Pakistan, providing insights to support evidence-based decision-making in antibiotic use and policy planning.

Methods: A retrospective study was conducted to evaluate the patterns of the resistance prevailing in the Punjab province of Pakistan for which tertiary care hospitals and laboratories having the facilities of culture sensitivity tests were approached for the collection of the culture sensitivity reports in four cities of Punjab, including Faisalabad, Lahore, Multan, and Rawalpindi. The resistance pattern was spatially mapped using GIS mapping software ArcGIS®Pro.

Results: The antibiotic resistance patterns of *Acinetobacter baumannii*, *Klebsiella pneumoniae*, *Proteus mirabilis*, *Pseudomonas aeruginosa* (Gram negative bacteria), *Staphylococcus aureus* and *Streptococcus pneumoniae* (Gram positive bacteria) were determined towards the antibiotics tested against them

The results showed that a very high resistance in the bacterial isolates was present among the antibiotics in use, which included Amikacin, Ampicillin, Cefoperazone/Salbactam, Ceftriaxone, Ciprofloxacin, Doxycycline, Gentamycin, and Trimethoprim/Sulfamethoxazole. The antibiotics which were found sensitive included Colistin, Linezolid, Meropenem, Polymyxin-B, and Teicoplanin.

Conclusion: GIS-based spatial mapping of antimicrobial resistance provided valuable insights into regional variations in bacterial resistance patterns across Punjab. This information can support evidence-based decision-making by healthcare authorities, enabling more targeted antibiotic procurement and effective antimicrobial stewardship tailored to the resistance profiles of specific cities.

Keywords: Resistance, Spatial mapping, Antimicrobial, GIS

272

A retrospective cross-sectional comparative study identifying the risk factors associated with recurrent venous thrombosis among oncology patients

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Background: Oncology patients are 6–7 times more likely at the risk of venous thromboembolism (VTE). VTE is among the leading causes of mortality among oncology patients. Use of anticoagulants is common among cancer patients; however, it results in undesirable events.

Aim: The aim of this study was to evaluate the incidence of recurrence of VTE by comparison of subcutaneous (LMWH) and direct oral anticoagulants (DOCA), and to identify risk factors associated in these cancer patients along with cumulative incidence of bleeding and recurrence of VTE.

Method: A retrospective cross-sectional design was used, and data were accessed from hospital information systems (HIS). From March 1, 2024, to May 28, 2024, eligible patient profiles were gathered for all adult's cancer patients aged 18 years and above, both genders, diagnosed with VTE during their treatment for cancer under treatment at Shaukat Khanum Memorial Cancer Hospital, Peshawar.

Results: A total of 207 patients' data was retrieved from health information system (HIS), majority of the patients were female (59.9%), aged more than 50 years of age, most of the patients were having gastric cancer (35.7%) followed by duodenal (30.4%) and esophageal cancer (25.1%) and (35.7%) of the patients were on anticoagulant therapy for duration of 6 months. Upon statistical analysis it was revealed that recurrence of VTE was independent of type and routes of administration of difference anticoagulants ($p = 0.187$). A Log Rank (Mantel–Cox) analysis was showed no difference between the bleeding time and recurrence of VTE with type of anticoagulants only. Multivariate logistic regression analysis revealed no significant association between various clinical factors and VTE recurrence.

Conclusion: Our study findings showed, no significant difference between oral versus subcutaneous anticoagulants for recurrence of VTE in cancer patients. However, the literature supports oral anticoagulant are associated with higher risk of bleeding.

Keywords: Direct oral anticoagulant, Low molecular weight heparin, Venous thromboembolism, Bleeding

273

Theoretical framework on the modular-based course learning outcome assessment: The need to change the pharmaceutical education

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Aim: This study aimed to evaluate the feasibility of the integration of modular-based teaching and Course Learning Outcome (CLO)-driven assessments within the Doctor of Pharmacy (PharmD) or Bachelor of Pharmacy (BPharm) curriculum, emphasizing its role in enhancing student learning, academic progression, and comprehensive program evaluation.

Method: A theoretical framework was developed through extensive benchmarking and surveys among students ($n = 100$) and faculty ($n = 10$) to evaluate the effectiveness of modular-based CLO assessments.

Results: Indicated that 91% of students strongly endorsed the modular approach, citing reduced academic stress and improved self-awareness, professionalism, and punctuality. Similarly, 95% of faculty members believed that modular assessments enhanced teaching practices and supported students' academic success. Faculty also agreed (90%) that this approach fosters

focused evaluations, ensures alignment between CLOs and Program Learning Outcomes (PLOs), and positively influences exit exams and APPE preparedness. The study identified redundancies in traditional assessments, emphasizing the need for CLO-focused evaluations to improve clarity and efficiency. Pilot data revealed that students scored higher (55-75%) in continuous assessments targeting all CLOs compared to final comprehensive exams (50-60%). These findings support the need for tailored reassessments focusing on specific CLO deficiencies. This framework promotes a competency-driven approach through diverse teaching methods, such as Problem-Based Learning (PBL), simulation, and interprofessional education. Modular assessments, aligned with Bloom's taxonomy and Miller's pyramid, ensure both formative and summative evaluation of clinical and theoretical competencies.

Conclusion: Modular teaching and CLO-focused assessments enhance learning outcomes, minimize redundancy, and provide clarity for competency-based programmatic evaluations. These findings advocate for broader adoption across pharmacy education.

Keywords: Modular-based teaching, Course learning objectives

274

The efficacy and safety of the FDA-approved phosphoinositide 3-kinase (PI3K) inhibitors in cancer: A systematic review of meta-analyses

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Background: Phosphoinositide 3-kinase inhibitors (PI3Kis) represent a targeted therapeutic class with growing use in oncology due to their role in regulating tumor cell growth, survival, and metabolism. Five PI3K inhibitors – Idelalisib, Copanlisib, Duvelisib, Alpelisib, and Umbralisib – have received FDA approval for use in various cancers. Despite their approval, there remains variability in their clinical efficacy and safety profiles across different cancer types.

Aim: This systematic review aims to synthesize the existing evidence about the efficacy and safety of the five Phosphoinositide 3-kinases inhibitors (PI3Kis) (Idelalisib, Copanlisib, Duvelisib, Alpelisib, Umbralisib) that have

received Food and Drug Administration (FDA) approval for treatment in cancer patients.

Methods: Several databases, such as PubMed, Embase, and Cochrane library, were searched using targeted keywords. Only meta-analyses (MAs) that evaluated the PI3Kis as monotherapy or adjunctive therapy in adult patients with any type of cancer were included. Data extraction focused on efficacy and safety outcomes reported in eligible studies. The Risk of Bias In Systematic Reviews (ROBIS) tool and the Assessment of Multiple Systematic Reviews-2 (AMSTAR-2) were used to evaluate the risk of bias and quality of methods of the included MAs.

Results: MAs involving 12,535 patients were included. Two MAs focused on Alpelisib in breast cancer BC patients, demonstrating its superiority in improving progression free survival (PFS) among patients with PI3KCA-mutated BC. Two MAs evaluated Idelalisib combined with standard immunotherapy in relapsed/refractory chronic lymphocytic leukemia (R/R CLL), showing promising efficacy but it demonstrated some severe hematological adverse events. Two MAs were conducted in relapsed/refractory B-cell non-Hodgkin lymphoma (R/R B-NHL) patients; one of them studied Copanlisib monotherapy or in combination with other agents, while the other studied Duvelisib as monotherapy. Lastly, three MAs addressed only safety outcomes.

Conclusion: Alpelisib showed favorable efficacy outcomes in breast cancer treatment regardless of the patient's mutation status while maintaining a tolerable adverse event profile. It was challenging to find a meaningful difference between the remaining PI3Kis that are included in this review. It has been clarified, that Idelalisib and Duvelisib have a role in the treatment of R/R CLL and R/R NHL whether alone or in combination with immunotherapy.

Keywords: Pi3k inhibitors, PIK3CA mutation, Breast cancer, Chronic lymphocytic leukemia, Lymphoma

277

Characteristics of psychotropic medications adverse drug reactions: A retrospective descriptive analysis

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Background: Spontaneous reporting systems for Adverse Drug Reactions (ADRs) play a key role in detecting new, rare, and severe ADRs in clinical practice. However, there is limited research on ADRs associated with psychotropic medications in Qatar.

Aim: To investigate psychotropic-related ADR characteristics and risk factors.

Methods: A retrospective chart review analyzed ADR reports linked to psychotropic medications submitted to Hamad Medical Corporation's (HMC) Spontaneous Reporting System database from 2018 to 2022. Data on clinical descriptions, suspected agents, severity, seriousness, preventability, and causality were assessed. Associations between patient demographics, comorbidities, and ADR types were also explored.

Results: In 908 ADR reports, the majority involved male patients (543, 59.83%) and were reported in 2020 (256, 28.2%). As per the reporters' assessment, approximately two-thirds of the ADRs (67.7%) were moderate in severity, and 64.5% of patients recovered or were in the process of recovery at the time of reporting. Notably, 65.6% of the ADRs were non-preventable. Regarding causality, 56.8% were probable, while 3.9% were definite. Extrapyramidal symptoms were the most frequently reported ADRs (46.5%), followed by gastrointestinal (13.3%), hyperprolactinemia (13.2%), and cardiac (3.7%) side effects. The most common offending agents were haloperidol (17.2%), olanzapine (13%), risperidone (12.6%), and clozapine (9.6%). Pharmacists reported 87% (790) of reports.

Multiple associations were discovered between patient characteristics and certain ADR types. Gender and weight were associated with the highest number of ADR types.

Conclusion: Although the recovery rate from ADRs associated with psychotropic medications is relatively high, the notable prevalence of non-preventable ADRs highlights the necessity of adopting a comprehensive approach to enhance the safety and efficacy of these medications. This approach should encompass advancements in regulatory policies, the adoption of holistic treatment strategies, and the implementation of personalized medicine.

Keywords: Adverse drug reactions, Psychotropic medications, Mental health